

The mode of the corner-to-corner first-passage time in two and three dimensions: rigorous asymptotics

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Companion note R3

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Note. This document is the companion note R3 of the article “Mean and most probable first-passage times of lazy random walks in reflecting hypercubic lattices” (paper.pdf, Section 7; Supplementary Material, Section S8), in the repository <https://github.com/zhouyi-xiaoxiao/master-dissertation>. In the repository, its scripts are in `code/msc_rigorous_asymptotics/` and its data in `data/msc_rigorous_asymptotics/`; paths written `scripts/...` and `data/...` below refer to these folders; the paths in the `% src` comments of the $\text{\LaTeX}$ source are relative to the root of the repository, and the run logs they cite are in `data/msc_rigorous_*/logs/`. Other run logs, the code of the internal checks and working notes mentioned below are not part of the repository; `notes/VERIFICATION.md` summarises how this note was checked.

Abstract

For the lazy nearest-neighbour walk on $\{1, \dots, N\}^d$ ($d = 2, 3$) with move probability q , reflecting walls and an absorbing target at one corner, started at the opposite corner, we replace the formal two-pole asymptotics of the first-passage *mode* t^* by theorems with explicit error terms. The proof rests on (i) an exact decomposition $T \stackrel{d}{=} U + T_\pi$ of the first-passage time into an independent “spreading time” U with distribution function $u = v^d$ (v the distribution function of a sum of $N - 1$ independent exponentials) and the hitting time T_π from the uniform start; (ii) strict unimodality of the continuous-time density (a direct proof), and for the discrete walk either unimodality ($q \leq \frac{1}{2}$) or a four-region argument that needs no unimodality (all $q < 1$, with constants uniform in q); (iii) an exact identity for the derivative of the density in which the two slowest poles appear with explicit coefficients and every remainder term has a sign and an explicit exponential bound; (iv) two-sided bounds for the pole data in terms of the lattice sum $X_s = \mu \mathbb{E}_\pi T$; (v) sharp two-sided bounds for X_s itself; and (vi) a ball-arithmetic certificate (Arb) for the resulting inequalities, uniform in $N \geq N_0$.

0 Results and status

Notation: $x = \pi/(2N)$, $\mu = \frac{2q}{d} \sin^2 x$ (slowest relaxation rate), $X = \mu \mathbb{E}T$, $\tau^* = \mu t^*$, closed form $\tau_{\text{cf}} = X \ln(2dX)/(X + 2d - 1)$, $c_3 = \frac{\pi^2}{2} \int_0^\infty (e^{-z}(I_0 + I_1)(z))^3 dz$.

label	statement	status
Prop. 1.5, Rem. 1.6	$g = \pi_a \varphi + \varphi * h_{\text{ac}}; T \stackrel{d}{=} U + T_\pi$; exact pole expansion (continuous and discrete time, all $q \in (0, 1]$)	Theorem (proved)
Thm 1.11	continuous-time density strictly unimodal for every $d \geq 1$, $N \geq 2$ except $(d, N) = (1, 2)$ (where μT is exponential and there is no interior mode); $g' = \varphi B$ with B nonincreasing	Theorem (proved)
Prop. 2.1, Thm 2.3	exact identity for g' ; window theorem $\Phi_- \leq g' \leq \Phi_+$	Theorem (proved)

label	statement	status
Lemmas 3.1–3.5	two-sided bounds for $\nu_0, \rho_0, m_0, \nu_1, \rho_1$ and the near poles in terms of X_s	Theorem (proved)
Props 4.5, 4.6	$2\pi \ln N - 2.32 \leq X_s \leq 2\pi \ln N - 1.60$ ($d = 2$); $c_3 N - 3\pi - 5.85/N \leq X_s \leq c_3 N - 8.38$ ($d = 3$), all $N \geq 10$	computer-assisted (certified constants $c_\kappa, J, T_d(1)$)
Thm 5.1	$-K_d^- < X(\tau^* - \tau_{cf}) < K_d^+$ for all $N \geq N_0$, e.g. $(K^-, K^+) = (4.242, 7.486)$ for $d = 2, N \geq 20$; $(1.564, 3.663)$ for $d = 3, N \geq 20$	computer-assisted proof (ball arithmetic)
Thm 6.1	$d = 2, N \geq 20$: $\tau^* = \ln \ln N + \ln 8\pi + R_N$, $-\frac{4.6+3 \ln(8\pi \ln N+2.4)}{2\pi \ln N - 0.24} \leq R_N < 0$	computer-assisted (consequence of Thm 5.1, Cor. 4.7)
Thm 6.2	$d = 3, N \geq 20$: $\tau^* = \ln N + \ln(6c_3) + R_N$, $-\frac{9.2+5 \ln(6c_3 N)}{c_3 N - 7.63} \leq R_N < 0, \ln(6c_3) = 3.75282 \dots$	computer-assisted (consequence of Thm 5.1, Cor. 4.7)
Thm 6.3	two-sided bounds $c_1 N^2 \ell_d(N) \leq t^* \leq c_2 N^2 \ell_d(N)$, $N \geq 20$, explicit c_1, c_2	computer-assisted (consequence of Thms 6.1, 6.2)
Prop. 7.1	discrete-time identity for the forward difference of the PMF, all $q \in (0, 1]$	Theorem (proved)
Thm 7.4	discrete-time PMF unimodal for $q \leq \frac{1}{2}, d \geq 2, N \geq 3$ (log-concavity of $u_{t+1} - u_t$)	Theorem; uses the theorem of Walkup and Liggett that ultra log-concave sequences are closed under convolution
Thm 7.8	discrete time, all $q \in (0, \frac{1}{2}]$: $t_{cf} - \frac{K^-}{X_\mu} < t^* < t_{cf} + \frac{K^+}{X_\mu} + 2$, e.g. $(K^-, K^+) = (4.479, 7.542)$ ($d = 2$), $(2.661, 3.761)$ ($d = 3$) for $N \geq 20$	computer-assisted proof
Cor. 7.9	discrete time, $q \leq \frac{1}{2}$: the laws of Thms 6.1–6.3 with slightly larger constants	computer-assisted (consequence of Thm 7.8)
Lemmas 7.10–7.12, Thm 7.13	all $q < 1$ (no unimodality): pairing lemma for the one-dimensional factors, bound for the poles with negative multiplier, four-region localisation of the maximisers	Theorem (proved)
Thm 7.14	discrete time, $q \leq q_{\max} \in \{0.8, 0.9, 0.95, 0.99\}$: the same window for all $N \geq N_0$; e.g. $q \leq 0.8$: $(K^-, K^+) = (2.850, 6.613)$ for $d = 2, N \geq 37$; $(5.189, 6.634)$ for $d = 3, N \geq 13$	computer-assisted proof
Cor. 7.15	discrete time, $q \leq 0.99$: $t^* = \frac{4N^2}{\pi^2 q} [\ln \ln N + \ln 8\pi + O(\frac{\ln \ln N}{\ln N})]$ resp. $\frac{6N^2}{\pi^2 q} [\ln N + \ln 6c_3 + O(\frac{\ln N}{N})]$ with explicit constants	computer-assisted (consequence of Thm 7.14)
Lemma 7.16	under (H), the q -dependent constants of Lemma 7.11 are bounded by functions of N alone	Theorem (proved)
Thm 7.17	discrete time, every $q \in (0, 1)$ and every $N \geq \max(N_*, N_1(q))$, $N_* = 59$ ($d = 2$), 12 ($d = 3$), $N_1(q) = \min\{N \geq 4 : \sin \frac{\pi}{N} \leq \frac{1-q}{q}\}$: the same window with constants uniform in q , e.g. $(2.082, 6.061)$ ($d = 2$), $(6.379, 7.183)$ ($d = 3$)	computer-assisted proof
Cor. 7.18	the laws of Cor. 7.15 for every $q < 1$ and $N \geq \max(N_*, N_1(q))$	computer-assisted (consequence of Thm 7.17)
Rem. 7.19	$N < N_1(q)$ (where (H) fails) and some $N < N_*$: not covered; $q = 1$: the PMF is not unimodal in general, laws conjectured	open; conjecture
Rem. 5.3(iii)	$X(\tau^* - \tau_{cf}) \rightarrow K_d^\infty, K_2^\infty \approx 3.74, K_3^\infty \approx -0.22$	Numerical observation

"Computer-assisted" means: the statement is reduced by complete proofs to finitely many explicit inequalities, which are verified with rigorous ball arithmetic (Arb through python-flint); the scripts and certificate files are listed in the Appendix. All other numerical work in the scripts is sanity checking and is not used in proofs.

In real time the continuous-time laws read ($N \geq 20$)

$$\begin{aligned} d = 2 : t^* &= \frac{4N^2}{\pi^2 q} [\ln \ln N + \ln 8\pi + R_N] (1 + \rho_N), \\ d = 3 : t^* &= \frac{6N^2}{\pi^2 q} [\ln N + 3.75282 \dots + R_N] (1 + \rho_N), \quad 0 \leq \rho_N \leq \frac{0.83}{N^2}. \end{aligned}$$

1 Setting, spectral structure, and unimodality

Throughout, $d \geq 1$ and $N \geq 3$ are integers, $q \in (0, 1]$, and

$$x := \frac{\pi}{2N}, \quad \mu := \frac{q}{d} \left(1 - \cos \frac{\pi}{N}\right) = \frac{2q}{d} \sin^2 x.$$

The main results concern $d \in \{2, 3\}$; in $d = 3$ we assume $N \geq 4$ from Section 2 on. Lemmas 1.1–1.4, 1.7 and 1.10, Proposition 1.5 and Theorem 1.11 also hold for $N = 2$, with the same proofs (they do not use ε_2); Lemmas 1.8 and 1.9 need $N \geq 3$.

1.1 The walk

The state space is $\{1, \dots, N\}^d$. In one step the walk attempts, with probability $q/(2d)$ each, a move $n \rightarrow n \pm e_i$ ($i = 1, \dots, d$) and stays put with probability $1 - q$; an attempted move that leaves the box is cancelled. Let A be the symmetric $N \times N$ matrix

$$(Af)(n) = f(n) - \frac{1}{2}f(n-1) - \frac{1}{2}f(n+1), \quad \text{with the conventions } f(0) := f(1), f(N+1) := f(N),$$

and let A_i denote A acting on the i -th coordinate. The one-step transition matrix is $P = I - \frac{q}{d} \sum_{i=1}^d A_i$ (symmetric). The *continuous-time* walk has transition semigroup $p_t = \exp(-t \frac{q}{d} \sum_i A_i)$ (each allowed move has rate $q/(2d)$). The target is the corner $a = (N, \dots, N)$, the start is the opposite corner $x_0 = (1, \dots, 1)$, and T is the first time the walk is at a .

Lemma 1.1 (spectrum of A). *For $k = 0, \dots, N-1$ put $h_k(n) = \cos \frac{\pi k(2n-1)}{2N}$ and $a_k = 1 - \cos \frac{\pi k}{N} = 2 \sin^2(kx)$. Then $Ah_k = a_k h_k$, the h_k are mutually orthogonal, $\sum_{n=1}^N h_k(n)^2 = N/\alpha_k$ with $\alpha_0 = 1$, $\alpha_k = 2$ ($k \geq 1$), and*

$$h_k(1) = \cos(kx), \quad h_k(N) = (-1)^k \cos(kx).$$

Proof. Extend h_k to all integers n by the same formula. The identity $\cos(\alpha + \beta) + \cos(\alpha - \beta) = 2 \cos \alpha \cos \beta$ gives $h_k(n+1) + h_k(n-1) = 2 \cos(\pi k/N) h_k(n)$ for every n . Moreover $h_k(0) = \cos(-kx) = h_k(1)$ and $h_k(N+1) = \cos(\pi k + kx) = (-1)^k \cos(kx) = \cos(\pi k - kx) = h_k(N)$, so the extension agrees with the conventions in the definition of A ; hence $(Ah_k)(n) = (1 - \cos(\pi k/N))h_k(n)$ for $1 \leq n \leq N$. The numbers $a_0 < a_1 < \dots < a_{N-1}$ are distinct and A is symmetric, so the h_k are orthogonal. For $k \geq 1$, $\sum_{n=1}^N h_k(n)^2 = \frac{N}{2} + \frac{1}{2} \operatorname{Re}(e^{-i\pi k/N} \sum_{n=1}^N e^{2\pi i k n/N}) = \frac{N}{2}$ because $1 \leq k \leq N-1$; for $k = 0$ the sum is N . The boundary values were computed above. $\square$

1.2 Scaled spectral data

All formulas are written in the *scaled time* $\tau = \mu t$. Define, for $k \in \{0, \dots, N-1\}$,

$$\varepsilon_k := \frac{a_k}{a_1} = \frac{\sin^2(kx)}{\sin^2 x}, \quad \gamma_0 := 1, \quad \gamma_k := 2 \cos^2(kx) \quad (k \geq 1),$$

and for a multi-index $k \in \mathcal{K} := \{0, \dots, N-1\}^d \setminus \{0\}$

$$\mu_k := \sum_{i=1}^d \varepsilon_{k_i}, \quad w_k := \prod_{i=1}^d \gamma_{k_i} > 0, \quad c_k := (-1)^{k_1 + \dots + k_d} w_k.$$

Note $0 = \varepsilon_0 < \varepsilon_1 = 1 < \varepsilon_2 < \dots < \varepsilon_{N-1}$ (because $kx \leq \pi/2$), $\varepsilon_2 = 4 \cos^2 x \in [3, 4)$ and $\gamma_k > 0$. We use the abbreviations

$$A := \gamma_1 = 2 \cos^2 x, \quad W := dA, \quad \gamma_2 = 2 \cos^2(2x), \quad C_1 := d(d-1)A^2, \\ C_2 := d\gamma_2\varepsilon_2,$$

and the one-dimensional functions

$$\omega(\tau) := \sum_{k=0}^{N-1} \gamma_k e^{-\varepsilon_k \tau}, \\ v(\tau) := \sum_{k=0}^{N-1} (-1)^k \gamma_k e^{-\varepsilon_k \tau}.$$

Lemma 1.2 (propagators at the corners). *With $\pi_a := N^{-d}$ and $t = \tau/\mu$,*

$$N^d p_t(a, a) = \omega(\tau)^d = 1 + \sum_{k \in \mathcal{K}} w_k e^{-\mu_k \tau}, \\ N^d p_t(x_0, a) = v(\tau)^d =: u(\tau) = 1 + \sum_{k \in \mathcal{K}} c_k e^{-\mu_k \tau}.$$

In discrete time, $N^d P^t(a, a) = 1 + \sum_k w_k (1 - \mu\mu_k)^t$ and $N^d P^t(x_0, a) = 1 + \sum_k c_k (1 - \mu\mu_k)^t$. In particular $\omega(0) = N$, $v(0) = 0$, $1 + \sum_k w_k = N^d$ and $1 + \sum_k c_k = 0$.

Proof. By Lemma 1.1, $e^{-sA}(n, m) = \sum_k \frac{\alpha_k}{N} h_k(n) h_k(m) e^{-sa_k}$. Since the A_i commute and act on different coordinates, $p_t(n, m) = \prod_i e^{-(tq/d)A}(n_i, m_i)$. With $s = tq/d$ we have $sa_k = \varepsilon_k \tau$, $\frac{\alpha_k}{N} h_k(N)^2 = \gamma_k/N$ and $\frac{\alpha_k}{N} h_k(1) h_k(N) = (-1)^k \gamma_k/N$; expanding the products gives the two formulas. The discrete case is identical because $P = I - \frac{q}{d} \sum_i A_i$ has the eigenvalue $1 - \frac{q}{d} \sum_i a_{k_i} = 1 - \mu\mu_k$ on $\prod_i h_{k_i}(n_i)$. The last assertions are the case $\tau = 0$ (p_0 is the identity matrix). $\square$

1.3 Renewal equation and the pole expansion

Let $m_1 < m_2 < \dots < m_L$ be the *distinct* values of μ_k , $k \in \mathcal{K}$ (so $m_1 = 1$), and

$$W_i := \sum_{k: \mu_k = m_i} w_k > 0, \quad R(s) := \sum_{k \in \mathcal{K}} \frac{w_k}{s + \mu_k} = \sum_{i=1}^L \frac{W_i}{s + m_i}, \\ D(\nu) := 1 - \nu \sum_{k \in \mathcal{K}} \frac{w_k}{\mu_k - \nu}.$$

Lemma 1.3 (renewal equation). *For $s > 0$, with $\hat{u}(s) = \int_0^\infty e^{-s\tau} u(\tau) d\tau$,*

$$\mathbb{E} e^{-s\mu T} = \frac{s \hat{u}(s)}{1 + sR(s)} \quad (\text{continuous time}), \\ \sum_{t \geq 1} \mathbb{P}(T = t) z^t = \frac{\sum_{t \geq 0} z^t P^t(x_0, a)}{\sum_{t \geq 0} z^t P^t(a, a)} \quad (|z| < 1) \quad (\text{discrete time}).$$

Proof. T is a stopping time and the walk is at a at time T (on $\{T < \infty\}$). By the strong Markov property, $p_t(x_0, a) = \mathbb{E}[1_{T \leq t} p_{t-T}(a, a)]$, hence $\int_0^\infty e^{-\sigma t} p_t(x_0, a) dt = \mathbb{E}[e^{-\sigma T}] \int_0^\infty e^{-\sigma t} p_t(a, a) dt$ for $\sigma > 0$. Put $\sigma = \mu s$ and use Lemma 1.2: $\int_0^\infty e^{-s\tau} \omega(\tau)^d d\tau = \frac{1}{s} + R(s)$. The discrete identity is the same computation with $P^t(x_0, a) = \sum_{s=1}^t \mathbb{P}(T = s) P^{t-s}(a, a)$. $\square$

Lemma 1.4 (poles, residues, sum rules). (a) D has exactly L zeros, all real and simple, and they interlace the levels:

$$0 < \nu_0 < m_1 = 1 < \nu_1 < m_2 < \nu_2 < \cdots < \nu_{L-1} < m_L.$$

(b) For $s \notin \{-\nu_j\}$,

$$\hat{h}(s) := \frac{1}{1 + sR(s)} = \pi_a + \sum_{j=0}^{L-1} \frac{\rho_j}{s + \nu_j},$$

$$\rho_j = \left(\sum_{k \in \mathcal{K}} \frac{w_k \mu_k}{(\mu_k - \nu_j)^2} \right)^{-1} > 0.$$

(c) (sum rules) With $b_j := \rho_j / \nu_j$:

$$\pi_a + \sum_j b_j = 1, \quad \pi_a + \sum_j \frac{\rho_j}{\nu_j - m_i} = 0 \quad (i = 1, \dots, L),$$

$$\sum_j \rho_j = R_{\text{tot}} := \frac{d}{4N^d \sin^2 x},$$

$$\frac{1}{\rho_j} = \frac{1}{\nu_j} + \nu_j \sum_k \frac{w_k}{(\mu_k - \nu_j)^2}.$$

Proof. (a) $D'(\nu) = -\sum_i W_i m_i / (m_i - \nu)^2 < 0$ wherever defined. On $(0, m_1)$, D decreases from $D(0) = 1$ to $-\infty$; on each (m_i, m_{i+1}) it decreases from $+\infty$ to $-\infty$; on (m_L, ∞) it decreases from $+\infty$ to $1 + \sum_i W_i > 0$; for $\nu \leq 0$, $D(\nu) \geq 1$. So there is exactly one zero in $(0, m_1)$ and in each (m_i, m_{i+1}) , and none elsewhere on the real line. Now $1 + sR(s) = Q(s) / \prod_i (s + m_i)$ with Q a polynomial of degree L and leading coefficient $1 + \sum_i W_i = N^d$ (Lemma 1.2), and $Q(-\nu) = D(\nu) \prod_i (m_i - \nu)$. Thus the L numbers $-\nu_j$ are roots of Q , hence all its roots, and they are simple.

(b) $\hat{h} = \prod_i (s + m_i) / Q(s)$ is a ratio of polynomials of equal degree with simple poles, so it equals its limit $N^{-d} = \pi_a$ at infinity plus $\sum_j \rho_j / (s + \nu_j)$, where $1/\rho_j = \frac{d}{ds}(1 + sR(s))|_{s=-\nu_j} = \sum_i W_i m_i / (m_i - \nu_j)^2$.

(c) $\hat{h}(0) = 1$ gives the first rule; $\hat{h}(-m_i) = 0$ (the numerator vanishes, the denominator does not) gives the second. For the third, $1 + sR(s) = N^d - \sum_k w_k \mu_k / (s + \mu_k)$, so $\hat{h}(s) = \pi_a + \pi_a^2 \sum_k w_k \mu_k s^{-1} + O(s^{-2})$ as $s \rightarrow \infty$, i.e. $\sum_j \rho_j = \pi_a^2 \sum_k w_k \mu_k$. By Lemma 1.2 with $n = m = a$, $\pi_a \sum_k w_k \mu_k$ is the diagonal entry at a of $\mu^{-1} \frac{q}{d} \sum_i A_i$, which equals $\mu^{-1} \frac{q}{d} \cdot \frac{d}{2} = d / (4 \sin^2 x)$ ($A(N, N) = \frac{1}{2}$). The fourth: $\sum_k w_k \mu_k / (\mu_k - \nu_j)^2 = \sum_k w_k / (\mu_k - \nu_j) + \nu_j \sum_k w_k / (\mu_k - \nu_j)^2$ and $\sum_k w_k / (\mu_k - \nu_j) = 1/\nu_j$ since $D(\nu_j) = 0$. $\square$

Proposition 1.5 (density of T ; decomposition). Put $\varphi := u'$ and $h_{\text{ac}}(\tau) := \sum_j \rho_j e^{-\nu_j \tau}$. In continuous time μT has the density

$$g(\tau) = \pi_a \varphi(\tau) + \int_0^\tau \varphi(\tau - s) h_{\text{ac}}(s) ds = \sum_{j=0}^{L-1} \rho_j \hat{\varphi}(-\nu_j) e^{-\nu_j \tau},$$

$$\hat{\varphi}(s) = - \sum_{k \in \mathcal{K}} \frac{c_k \mu_k}{s + \mu_k},$$

on $\tau \geq 0$; the real-time density is $\mu g(\mu t)$ and $\int_0^\infty g = 1$. In discrete time, for $t \geq 1$,

$$\mathbb{P}(T = t) = \mu \sum_{j=0}^{L-1} \rho_j \hat{\varphi}(-\nu_j) (1 - \mu \nu_j)^{t-1}.$$

Proof. Since $u(0) = 0$ (Lemma 1.2), $s\hat{u}(s) = 1 + \sum_k c_k s / (s + \mu_k) = (1 + \sum_k c_k) - \sum_k c_k \mu_k / (s + \mu_k) = \hat{\varphi}(s)$, the Laplace transform of $\varphi = u' = -\sum_k c_k \mu_k e^{-\mu_k \tau}$. By Lemmas 1.3 and 1.4, $\mathbb{E}e^{-s\mu T} = \hat{\varphi}(s)\hat{h}(s)$, which is the Laplace transform of $\pi_a \varphi + \varphi * h_{ac}$. This function is a finite linear combination of exponentials $e^{-\mu_k \tau}$, $e^{-\nu_j \tau}$, hence integrable on $[0, \infty)$; by uniqueness of the Laplace transform of finite signed measures (apply Feller, *An Introduction to Probability Theory and Its Applications*, Vol. II, 2nd ed., Section XIII.1, Theorem 1, to the two positive measures $\text{law}(\mu T) + g^- d\tau$ and $g^+ d\tau$) it is the density of μT ; in particular $\int g = \hat{\varphi}(0)\hat{h}(0) = 1$. In the product $\hat{\varphi}\hat{h}$ the simple poles of $\hat{\varphi}$ at $-m_i$ are cancelled by the zeros of $\hat{h}$, and $\hat{\varphi}(s)\hat{h}(s) \rightarrow 0$ at infinity, so $\hat{\varphi}\hat{h} = \sum_j \rho_j \hat{\varphi}(-\nu_j)/(s + \nu_j)$; this gives the second expression for g .

Discrete time: put $s = (1 - z)/(\mu z)$ for $0 < z < 1$. By Lemma 1.2, $\sum_t z^t P^t(a, a) \cdot N^d = \frac{1}{1-z} + \sum_k \frac{w_k}{1-(1-\mu\mu_k)z} = \frac{1}{\mu z} \left(\frac{1}{s} + R(s) \right)$, and likewise $N^d \sum_t z^t P^t(x_0, a) = \frac{1}{\mu z} \hat{u}(s)$. By Lemma 1.3 the generating function of T is $\hat{\varphi}(s)\hat{h}(s) = \sum_j \rho_j \hat{\varphi}(-\nu_j) \frac{\mu z}{1-(1-\mu\nu_j)z}$, and $\frac{\mu z}{1-(1-\mu\nu_j)z} = \mu \sum_{t \geq 1} (1 - \mu\nu_j)^{t-1} z^t$. $\square$

Remark 1.6. $\hat{h}$ is the Laplace transform of the hitting time T_π of a from the uniform (stationary) start, which has an atom π_a at 0 and density h_{ac} ; and $u = v^d$ is a distribution function (Lemma 1.7). So $\mu T \stackrel{d}{=} U + \mu T_\pi$ with $U \sim u$ independent of T_π . Only the analytic content (Proposition 1.5) is used below. Write

$$X_s := R(0) = \sum_{k \in \mathcal{K}} \frac{w_k}{\mu_k}, \quad \mathbb{E}U = \int_0^\infty (1 - u) = -\sum_k \frac{c_k}{\mu_k},$$

$$X := \mu \mathbb{E}T = X_s + \mathbb{E}U,$$

$(\mathbb{E}[\mu T] = -\frac{d}{ds}(\hat{\varphi}\hat{h})(0) = -\hat{\varphi}'(0) - \hat{h}'(0) = \mathbb{E}U + R(0))$. X is the quantity " $X = \mu_1 \cdot \text{MFPT}$ " of Section 6 of the article, and the closed form is $\tau_{cf} := X \ln(2dX)/(X + 2d - 1)$ in scaled time.

1.4 The one-dimensional factor v

Lemma 1.7 (hypoexponential structure). $\int_0^\infty e^{-s\tau} v'(\tau) d\tau = \prod_{k=1}^{N-1} \frac{\varepsilon_k}{s + \varepsilon_k}$. Consequently v is the distribution function of $H = \xi_1 + \dots + \xi_{N-1}$ with independent $\xi_k \sim \text{Exp}(\varepsilon_k)$; $v' > 0$ on $(0, \infty)$, $0 \leq v \leq 1$, $v(0) = 0$, and

$$\prod_{l \neq k} \frac{\varepsilon_l}{\varepsilon_l - \varepsilon_k} = (-1)^{k+1} \gamma_k$$

$$(1 \leq k \leq N-1; \text{ product over } 1 \leq l \leq N-1).$$

Proof. Let $M = A/a_1$ (eigenvalues ε_k). By Lemma 1.1, $(s + M)^{-1}(1, N) = \sum_k \frac{\alpha_k}{N} \frac{h_k(1)h_k(N)}{s + \varepsilon_k} = \hat{v}(s)/N$. By Cramer's rule $(s + M)^{-1}(1, N) = (-1)^{N+1} \det(\widetilde{M}) / \det(s + M)$, where $\widetilde{M}$ is $s + M$ with row N and column 1 deleted. As $s + M$ is tridiagonal, $\widetilde{M}$ is lower triangular with diagonal entries equal to the super-diagonal entries $-1/(2a_1)$ of $s + M$, so $\det \widetilde{M} = (-2a_1)^{-(N-1)}$ and

$$s\hat{v}(s) = \frac{N(2a_1)^{-(N-1)}}{\prod_{k=1}^{N-1} (s + \varepsilon_k)}.$$

Since $s\hat{v}(s) \rightarrow v(\infty) = 1$ as $s \downarrow 0$, $N(2a_1)^{-(N-1)} = \prod_k \varepsilon_k$, and $s\hat{v}(s) = \hat{v}'(s)$ because $v(0) = 0$. The right-hand side is the Laplace transform of the density of H , which is positive on $(0, \infty)$; by uniqueness v' is that density. Finally $v'(\tau) = \sum_{k \geq 1} (-1)^{k+1} \gamma_k \varepsilon_k e^{-\varepsilon_k \tau}$ by definition of v , while the partial fraction expansion of $\prod_k \varepsilon_k / (s + \varepsilon_k)$ has the coefficient $\varepsilon_k \prod_{l \neq k} \varepsilon_l / (\varepsilon_l - \varepsilon_k)$ at $1/(s + \varepsilon_k)$; comparing coefficients of the linearly independent functions $e^{-\varepsilon_k \tau}$ gives the product formula. $\square$

Lemma 1.8 (bounds for v). *Let p_2 be the density of $\xi_2 + \dots + \xi_{N-1}$. Then $v' + v'' = p_2 \geq 0$ and, for all $\tau \geq 0$,*

$$\begin{aligned} 0 &\leq p_2(\tau) \leq \gamma_2 \varepsilon_2 (\varepsilon_2 - 1) e^{-\varepsilon_2 \tau}, \\ Ae^{-\tau} - \gamma_2 \varepsilon_2 e^{-\varepsilon_2 \tau} &\leq v'(\tau) \leq Ae^{-\tau}, \\ Ae^{-\tau} - \gamma_2 e^{-\varepsilon_2 \tau} &\leq 1 - v(\tau) \leq Ae^{-\tau}. \end{aligned}$$

Moreover $\ell(\tau) := Ae^{-\tau} - \gamma_2 e^{-\varepsilon_2 \tau}$ satisfies $0 \leq \ell \leq 1 - v \leq 1$.

Proof. Let p_3 be the density of $\xi_3 + \dots + \xi_{N-1}$ (the unit mass at 0 if $N = 3$). Then $p_2 = \varepsilon_2 e^{-\varepsilon_2 \cdot} * p_3$, i.e. $e^{\varepsilon_2 \tau} p_2(\tau) = \varepsilon_2 \int_0^\tau e^{\varepsilon_2 s} p_3(ds)$, which is nondecreasing with limit $\varepsilon_2 \mathbb{E} e^{\varepsilon_2(\xi_3 + \dots)} = \varepsilon_2 \prod_{l \geq 3} \frac{\varepsilon_l}{\varepsilon_l - \varepsilon_2}$. By Lemma 1.7 with $k = 2$, $\frac{\varepsilon_1}{\varepsilon_1 - \varepsilon_2} \prod_{l \geq 3} \frac{\varepsilon_l}{\varepsilon_l - \varepsilon_2} = -\gamma_2$, i.e. $\prod_{l \geq 3} \frac{\varepsilon_l}{\varepsilon_l - \varepsilon_2} = \gamma_2 (\varepsilon_2 - 1)$. This is the bound on p_2 . Next $v' = e^{-\cdot} * p_2$ (as $\varepsilon_1 = 1$), i.e. $e^\tau v'(\tau) = \int_0^\tau e^s p_2(s) ds$; differentiating gives $v' + v'' = p_2$, and the limit is $\mathbb{E} e^{\xi_2 + \dots} = \prod_{l \geq 2} \frac{\varepsilon_l}{\varepsilon_l - 1} = \gamma_1 = A$ (Lemma 1.7, $k = 1$). Hence $e^\tau v'(\tau) = A - \int_\tau^\infty e^s p_2(s) ds \in [A - \gamma_2 \varepsilon_2 e^{-(\varepsilon_2 - 1)\tau}, A]$. Integrating over (τ, ∞) gives the bounds on $1 - v$. Finally $\gamma_2 \leq A$ (because $0 \leq \cos 2x \leq \cos x$ for $x \leq \pi/6$) and $\varepsilon_2 > 1$ give $\ell \geq 0$. $\square$

1.5 The function $\varphi = u'$ for $d \geq 2$

Lemma 1.9 (bounds for φ). *Let $d \geq 2$. Then $\varphi = dv^{d-1}v' \geq 0$, $\varphi(0) = 0$, and the functions*

$$\tilde{q} := \varphi' + \varphi, \quad r := We^{-\tau} - \varphi, \quad r_1 := -r' = We^{-\tau} + \varphi' = r + \tilde{q}$$

satisfy, for all $\tau \geq 0$,

$$\begin{aligned} 0 &\leq \tilde{q} \leq C_1 e^{-2\tau} + C_2 (\varepsilon_2 - 1) e^{-\varepsilon_2 \tau}, \\ We^{-\tau} g_\ell(\tau) &\leq r \leq \bar{r}(\tau) := C_1 e^{-2\tau} + C_2 e^{-\varepsilon_2 \tau}, \\ r &\leq r_1 \leq \bar{r}_1(\tau) := 2C_1 e^{-2\tau} + C_2 \varepsilon_2 e^{-\varepsilon_2 \tau}, \end{aligned}$$

where $g_\ell = \ell$ (any $d \geq 2$) or $g_\ell = 2\ell - \ell^2$ (if $d \geq 3$). Also $r \leq We^{-\tau}$, $r(0) = W$, $r(\tau) = e^{-\tau} \int_\tau^\infty e^s \tilde{q}(s) ds$, and $1 - u \leq \min(1, We^{-\tau})$.

Proof. $\varphi' = d(d-1)v^{d-2}v'^2 + dv^{d-1}v''$, so by Lemma 1.8 $\tilde{q} = d(d-1)v^{d-2}v'^2 + dv^{d-1}p_2$, which lies between 0 and $d(d-1)A^2 e^{-2\tau} + d\gamma_2 \varepsilon_2 (\varepsilon_2 - 1) e^{-\varepsilon_2 \tau}$. Since $(e^\tau \varphi)' = e^\tau \tilde{q} \geq 0$ and $e^\tau \varphi = dv^{d-1}(e^\tau v') \rightarrow dA = W$, we get $W - e^\tau \varphi(\tau) = \int_\tau^\infty e^s \tilde{q}(s) ds$, which is the formula for r ; inserting the bound on $\tilde{q}$ gives $0 \leq r \leq \bar{r}$. $r \leq We^{-\tau}$ and $r(0) = W$ follow from $\varphi \geq 0$, $\varphi(0) = dv(0)^{d-1}v'(0) = 0$. Next $r_1 = -r' = We^{-\tau} + \varphi' = (We^{-\tau} - \varphi) + (\varphi' + \varphi) = r + \tilde{q}$, which gives $r \leq r_1 \leq \bar{r} + C_1 e^{-2\tau} + C_2 (\varepsilon_2 - 1) e^{-\varepsilon_2 \tau} = \bar{r}_1$.

Lower bound: $r = d(Ae^{-\tau} - v') + dv'(1 - v^{d-1})$. By Lemma 1.8, $0 \leq \ell \leq 1 - v$; hence $1 - v^{d-1} \geq 1 - v \geq \ell$, and if $d \geq 3$ also $v^{d-1} \leq v^2 \leq (1 - \ell)^2$, i.e. $1 - v^{d-1} \geq 2\ell - \ell^2$. In both cases $0 \leq g_\ell \leq 1$ and $1 - v^{d-1} \geq g_\ell$, so, using $Ae^{-\tau} - v' \geq 0$ and $v' \geq 0$, $r \geq d(Ae^{-\tau} - v')g_\ell + dv'g_\ell = We^{-\tau} g_\ell$. Finally $1 - u = 1 - v^d \leq d(1 - v) \leq We^{-\tau}$. $\square$

1.6 Unimodality in continuous time

Lemma 1.10 (log-concavity). *v' , v and $\varphi = dv^{d-1}v'$ are positive, C^∞ and log-concave on $(0, \infty)$ (for every $d \geq 1$).*

Proof. Step 1. Let $\Pi : [0, \infty) \rightarrow [0, \infty)$ be continuous, positive and C^1 on $(0, \infty)$, with Π'/Π nonincreasing on $(0, \infty)$. Then $F(\tau) := \int_0^\tau \Pi$ is positive on $(0, \infty)$ and F'/F is nonincreasing there. Indeed for $0 < s \leq \tau$, $\Pi'(\tau)/\Pi(\tau) \leq \Pi'(s)/\Pi(s)$, i.e. $\Pi'(\tau)\Pi(s) \leq \Pi'(s)\Pi(\tau)$; integrating over $s \in (\delta, \tau)$ and letting $\delta \downarrow 0$, $\Pi'(\tau)F(\tau) \leq \Pi(\tau)(\Pi(\tau) - \Pi(0)) \leq \Pi(\tau)^2$, which is $(F'/F)' = (F''F - F'^2)/F^2 \leq 0$ (note $F' = \Pi$, $F'' = \Pi'$ on $(0, \infty)$).

Step 2. Let p_j be the density of $\xi_j + \dots + \xi_{N-1}$ ($1 \leq j \leq N-1$). $p_{N-1}(\tau) = \varepsilon_{N-1} e^{-\varepsilon_{N-1}\tau}$ is continuous on $[0, \infty)$, positive, with constant logarithmic derivative. Suppose p_{j+1} is continuous

on $[0, \infty)$, positive and C^1 on $(0, \infty)$ with nonincreasing logarithmic derivative. Then so is $\Pi(s) := e^{\varepsilon_j s} p_{j+1}(s)$, and $p_j(\tau) = \varepsilon_j e^{-\varepsilon_j \tau} \int_0^\tau \Pi$ has, by Step 1, the logarithmic derivative $-\varepsilon_j + F'/F$, nonincreasing; p_j is continuous on $[0, \infty)$, positive and C^1 on $(0, \infty)$. By induction $v' = p_1$ is log-concave; by Step 1 with $\Pi = v'$, so is v ; and $\ln \varphi = \ln d + (d-1) \ln v + \ln v'$ is concave. Smoothness is clear from the explicit exponential sums. $\square$

Theorem 1.11 (strict unimodality, continuous time). *Let $d \geq 1$ and $N \geq 2$ with $(d, N) \neq (1, 2)$. The function $B(\tau) := g'(\tau)/\varphi(\tau)$ is nonincreasing on $(0, \infty)$. Consequently there is a unique $\tau^* \in (0, \infty)$ with $g'(\tau^*) = 0$; $g' > 0$ on $(0, \tau^*)$ and $g' < 0$ on (τ^*, ∞) . In particular the continuous-time first-passage density has the unique mode $t^* = \tau^*/\mu$, and for any $\tau > 0$*

$$g'(\tau) > 0 \iff \tau < \tau^*, \quad g'(\tau) < 0 \iff \tau > \tau^*.$$

Proof. By Proposition 1.5, $g(\tau) = \pi_a \varphi(\tau) + \int_0^\tau \varphi(s) h_{ac}(\tau - s) ds$. Differentiating and substituting $s \rightarrow \tau - s$,

$$\begin{aligned} g'(\tau) &= \pi_a \varphi'(\tau) + \varphi(\tau) h_{ac}(0) - \int_0^\tau \varphi(\tau - s) \eta(s) ds, \\ \eta &:= -h'_{ac} = \sum_j \rho_j \nu_j e^{-\nu_j s} > 0. \end{aligned}$$

Hence, for $\tau > 0$,

$$B(\tau) = \pi_a \frac{\varphi'(\tau)}{\varphi(\tau)} + h_{ac}(0) - \int_0^\infty \mathbf{1}_{\{s < \tau\}} \frac{\varphi(\tau - s)}{\varphi(\tau)} \eta(s) ds.$$

φ'/φ is nonincreasing (Lemma 1.10). For fixed $s > 0$ the function $\tau \mapsto \mathbf{1}_{\{s < \tau\}} \varphi(\tau - s)/\varphi(\tau)$ is nondecreasing: it vanishes for $\tau \leq s$, and for $s < \tau_1 < \tau_2$ concavity of $\ln \varphi$ gives $\ln \varphi(\tau_2) - \ln \varphi(\tau_2 - s) \leq \ln \varphi(\tau_1) - \ln \varphi(\tau_1 - s)$. Therefore B is nonincreasing.

Next, $\varphi(0) = 0$: for $d \geq 2$ because $\varphi = dv^{d-1}v'$ and $v(0) = 0$; for $d = 1$ (so $N \geq 3$) because $\varphi = v'$ is the density of $\xi_1 + (\xi_2 + \dots + \xi_{N-1})$, a convolution of two bounded densities on $[0, \infty)$, so $v'(\tau) \leq \tau \varepsilon_1 \sup p_2 \rightarrow 0$ as $\tau \downarrow 0$ (Lemma 1.7). Also $\varphi > 0$ on $(0, \infty)$ (Lemma 1.10). g is a finite sum of exponentials, $g(0) = \pi_a \varphi(0) = 0$, $g(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$ and $\int g = 1$; so g has a positive maximum at some $\tau^* \in (0, \infty)$, where $g'(\tau^*) = 0$. The zero set of g' in $(0, \infty)$ equals the zero set of the monotone function B (as $\varphi > 0$), hence is an interval; g' is real-analytic and not identically zero, so this interval is the single point τ^* . Monotonicity of B gives the signs. $\square$

Remark 1.12. (o) The excluded case $(d, N) = (1, 2)$ is a genuine exception: then $L = 1$, $\pi_a = \frac{1}{2}$, $\nu_0 = \frac{1}{2}$, $\rho_0 = \frac{1}{4}$, $\varphi(\tau) = v'(\tau) = e^{-\tau}$, and Proposition 1.5 gives $g(\tau) = \frac{1}{2}e^{-\tau} + \frac{1}{2}(e^{-\tau/2} - e^{-\tau}) = \frac{1}{2}e^{-\tau/2}$: μT is exponentially distributed, $g' < 0$ on $(0, \infty)$ and the density has no interior mode (its supremum is at the endpoint $\tau = 0$). The proof above uses $\varphi(0) = 0$ exactly once, to place the maximum of g in the interior. (i) The argument is a direct proof, for this situation, of Ibragimov's theorem that a log-concave density is strongly unimodal (I. A. Ibragimov, *On the composition of unimodal distributions*, Theory Probab. Appl. 1 (1956), 255–260, main theorem; see also S. Dharmadhikari and K. Joag-Dev, *Unimodality, Convexity, and Applications*, Academic Press 1988, Section 1.4), applied to U (log-concave) and T_π (unimodal with mode 0). (ii) In discrete time the same argument works whenever the sequence $u_{t+1} - u_t$ is log-concave and $t \mapsto \sum_j \rho_j (1 - \mu \nu_j)^t$ is nonincreasing; see Section 7. For $q = 1$ unimodality is false in general (parity).

2 An exact identity for g' and the window theorem

Standing assumptions for Sections 2–6: continuous time, scaled units; $d \in \{2, 3\}$; $N \geq 3$ if $d = 2$ and $N \geq 4$ if $d = 3$.

Under these assumptions $\varepsilon_k \geq \varepsilon_2 = 4 \cos^2 x \geq 3$ for $k \geq 2$, with $\varepsilon_2 > 3$ if $N \geq 4$. A multi-index k has $\mu_k < \varepsilon_2$ iff all $k_i \in \{0, 1\}$, and then $\mu_k = \#\{i : k_i = 1\}$. Hence the lowest levels and their weights are

$$m_1 = 1, \quad W_1 = W = dA; \quad m_2 = 2, \quad W_2 = \binom{d}{2}A^2 = \frac{1}{2}C_1; \quad (d = 3) \quad m_3 = 3, \quad W_3 = A^3; \\ \text{next level } \varepsilon_2.$$

We write $\theta := \nu_1 - 1 \in (0, 1)$, $\Gamma := \rho_1/\theta$, and call *near poles* the ν_j with $2 < \nu_j < \varepsilon_2$ (that is $j = 2$ if $d = 2$; $j \in \{2, 3\}$ if $d = 3$) and *far poles* the $\nu_j > \varepsilon_2$; $\mathcal{N}$ denotes the index set of the near poles. For a function f on $[0, \infty)$ and $\nu > 0$ put $(f * E_\nu)(\tau) := \int_0^\tau f(s)e^{-\nu(\tau-s)}ds$.

Proposition 2.1 (exact identity). *Let $m_0 := \hat{\varphi}(-\nu_0) = \int_0^\infty \varphi(s)e^{\nu_0 s}ds$ and $J(\tau) := \int_\tau^\infty r(s)e^{\nu_0 s}ds$. For all $\tau \geq 0$,*

$$g'(\tau) = W \sum_{j \geq 1} \frac{\rho_j}{\nu_j - 1} e^{-\nu_j \tau} - \rho_0 \nu_0 m_0 e^{-\nu_0 \tau} - \rho_0 \nu_0 e^{-\nu_0 \tau} J(\tau) - \rho_0 r(\tau) + \pi_a r_1(\tau) + \sum_{j \geq 1} \rho_j (r_1 * E_{\nu_j})(\tau).$$

All six terms have a definite sign: the first, fifth and sixth are ≥ 0 , the others ≤ 0 .

Proof. By Proposition 1.5, $g = \pi_a \varphi + \sum_j \rho_j (\varphi * E_{\nu_j})$, and $\frac{d}{d\tau}(\varphi * E_\nu) = \varphi - \nu(\varphi * E_\nu)$.

Pole $j = 0$. $(\varphi * E_{\nu_0})(\tau) = e^{-\nu_0 \tau} [m_0 - \int_\tau^\infty \varphi e^{\nu_0 s} ds]$ and, since $\varphi = W e^{-s} - r$ and $\nu_0 < 1$, $\int_\tau^\infty \varphi e^{\nu_0 s} ds = \frac{W}{1-\nu_0} e^{-(1-\nu_0)\tau} - J(\tau)$. Therefore

$$\frac{d}{d\tau}(\varphi * E_{\nu_0}) = W e^{-\tau} - r - \nu_0 m_0 e^{-\nu_0 \tau} + \frac{\nu_0 W}{1-\nu_0} e^{-\tau} - \nu_0 e^{-\nu_0 \tau} J = \frac{W}{1-\nu_0} e^{-\tau} - r - \nu_0 m_0 e^{-\nu_0 \tau} - \nu_0 e^{-\nu_0 \tau} J.$$

Poles $j \geq 1$. Integrating by parts and using $\varphi(0) = 0$, $\varphi - \nu(\varphi * E_\nu) = \varphi' * E_\nu$. With $\varphi' = -W e^{-s} + r_1$ and $(E_1 * E_\nu)(\tau) = (e^{-\tau} - e^{-\nu\tau})/(\nu - 1)$ for $\nu \neq 1$,

$$\frac{d}{d\tau}(\varphi * E_{\nu_j}) = -\frac{W}{\nu_j - 1} (e^{-\tau} - e^{-\nu_j \tau}) + (r_1 * E_{\nu_j}).$$

Summation. The total coefficient of $W e^{-\tau}$ is $\frac{\rho_0}{1-\nu_0} - \sum_{j \geq 1} \frac{\rho_j}{\nu_j - 1} = \pi_a$ by the sum rule of Lemma 1.4(c) with $m_i = 1$. Since $\pi_a \varphi' + \pi_a W e^{-\tau} = \pi_a r_1$, the identity follows. Signs: $\rho_j > 0$, $\nu_j > 1$ for $j \geq 1$, $m_0 > 0$, and $r_1 \geq r \geq 0$ (Lemma 1.9). $\square$

Define

$$a_1 := W\Gamma = \frac{W\rho_1}{\theta}, \quad a_0 := \rho_0 \nu_0 m_0, \quad \kappa(\tau) := \int_0^\tau r_1(s)e^{\nu_1 s}ds, \\ E_-(\tau) := \frac{\nu_0 J(\tau) + r(\tau)e^{\nu_0 \tau}}{\nu_0 m_0},$$

$$P_{\text{rest}}(\tau) := W \sum_{j \geq 2} \frac{\rho_j}{\nu_j - 1} e^{-\nu_j \tau} + \pi_a r_1(\tau) + \sum_{j \geq 2} \rho_j (r_1 * E_{\nu_j})(\tau).$$

Because $\rho_1(r_1 * E_{\nu_1})(\tau) = \rho_1 e^{-\nu_1 \tau} \kappa(\tau)$, Proposition 2.1 reads

$$g'(\tau) = a_1 e^{-\nu_1 \tau} \left[1 + \frac{\theta}{W} \kappa(\tau) \right] - a_0 e^{-\nu_0 \tau} [1 + E_-(\tau)] + P_{\text{rest}}(\tau). \quad (2.1)$$

Lemma 2.2 (bounds for the correction terms). *Let $s_x := \ln((d-1)A) \geq 0$, $\hat{\beta} := 1 - b_0 - b_1$, $c_\beta := 2W + 2C_1 \varepsilon_2 / (\varepsilon_2 - 2)$, and*

$$I^+(\theta) := W \frac{e^{\theta s_x} - 1}{\theta} + \frac{C_1 e^{-(1-\theta)s_x}}{1-\theta} + \frac{C_2 e^{-(\varepsilon_2-1-\theta)s_x}}{\varepsilon_2-1-\theta}, \\ \bar{\kappa} := W + (1+\theta)I^+(\theta),$$

$$\kappa_{\text{lo}}(\tau) := W - \bar{r}(\tau)e^{(1+\theta)\tau} + (1+\theta)W \int_0^\tau e^{\theta s} g_\ell(s) ds,$$

$$E_-^{\text{lo}}(\tau) := \frac{W e^{-(1-\nu_0)\tau} g_\ell(\tau)}{\nu_0 m_0},$$

$$\bar{E}_-(\tau) := \frac{1}{\nu_0 m_0} \left[\frac{2C_1 e^{-(2-\nu_0)\tau}}{2-\nu_0} + \frac{C_2 \varepsilon_2 e^{-(\varepsilon_2-\nu_0)\tau}}{\varepsilon_2 - \nu_0} \right],$$

$$\bar{P}_{\text{rest}}(\tau) := e^{-2\tau} \left[c_\beta \hat{\beta} + \sum_{i \in \mathcal{N}} \rho_i N_i(\tau) \right] + e^{-\varepsilon_2 \tau} C_2 \varepsilon_2 [\pi_a + \tau R_{\text{tot}}],$$

$$N_i(\tau) := 2C_1 \min\left(\tau, \frac{1}{\nu_i - 2}\right) + C_2 \varepsilon_2 \min\left(\tau, \frac{1}{\varepsilon_2 - \nu_i}\right).$$

Then for all $\tau > 0$: (a) $\kappa_{\text{lo}}(\tau) \leq \kappa(\tau) \leq \bar{\kappa}$; (b) $0 \leq E_-^{\text{lo}}(\tau) \leq E_-(\tau) \leq \bar{E}_-(\tau)$; (c) $0 \leq P_{\text{rest}}(\tau) \leq \bar{P}_{\text{rest}}(\tau)$.

Proof. (a) Integrating by parts ($r_1 = -r'$, $r(0) = W$): $\kappa(\tau) = W - r(\tau)e^{\nu_1 \tau} + \nu_1 \int_0^\tau r(s)e^{\nu_1 s} ds$. Lower bound: $r \leq \bar{r}$ in the second term and $r \geq W e^{-s} g_\ell(s)$ in the third (Lemma 1.9). Upper bound: κ is nondecreasing ($r_1 \geq 0$) and $r(\tau)e^{\nu_1 \tau} \rightarrow 0$ ($r = O(e^{-2\tau})$, $\nu_1 < 2$), so $\kappa(\tau) \leq W + \nu_1 \int_0^\infty r e^{\nu_1 s} ds$; use $r \leq W e^{-s}$ on $[0, s_x]$ and $r \leq \bar{r}$ on $[s_x, \infty)$ ($\theta < 1 < \varepsilon_2 - 1$ makes the integrals converge) to get $\int_0^\infty r e^{\nu_1 s} ds \leq I^+(\theta)$.

(b) $J(\tau) \leq \int_\tau^\infty \bar{r}(s)e^{\nu_0 s} ds = \frac{C_1 e^{-(2-\nu_0)\tau}}{2-\nu_0} + \frac{C_2 e^{-(\varepsilon_2-\nu_0)\tau}}{\varepsilon_2-\nu_0}$; adding $\bar{r}(\tau)e^{\nu_0 \tau}/\nu_0$ and using $\frac{1}{a-\nu_0} + \frac{1}{\nu_0} = \frac{a}{\nu_0(a-\nu_0)}$ for $a = 2, \varepsilon_2$ gives $\bar{E}_-$. The lower bound keeps only $r(\tau)e^{\nu_0 \tau} \geq W e^{-(1-\nu_0)\tau} g_\ell(\tau) \geq 0$.

(c) $P_{\text{rest}} \geq 0$ is clear. For $j \geq 2$ we have $\nu_j > 2$, so $\rho_j/(\nu_j - 1) \leq 2b_j$ and $e^{-\nu_j \tau} \leq e^{-2\tau}$: the first sum is at most $2W\beta_2 e^{-2\tau}$, $\beta_2 := \sum_{j \geq 2} b_j$. The second term is at most $\pi_a \bar{r}_1(\tau)$. For the third, $r_1 \leq \bar{r}_1 = 2C_1 E_2 + C_2 \varepsilon_2 E_{\varepsilon_2}$ and, for $a \neq \nu$,

$$(E_a * E_\nu)(\tau) = \frac{e^{-\min(a,\nu)\tau} - e^{-\max(a,\nu)\tau}}{|a - \nu|} \leq e^{-\min(a,\nu)\tau} \min\left(\tau, \frac{1}{|a - \nu|}\right).$$

For a near pole ($2 < \nu_i < \varepsilon_2$) this gives $(r_1 * E_{\nu_i}) \leq e^{-2\tau} N_i(\tau)$. For a far pole ($\nu_j > \varepsilon_2$): $(E_2 * E_{\nu_j}) \leq e^{-2\tau}/(\nu_j - 2) \leq e^{-2\tau} \frac{\varepsilon_2}{\varepsilon_2 - 2} \frac{1}{\nu_j}$ and $(E_{\varepsilon_2} * E_{\nu_j}) \leq \tau e^{-\varepsilon_2 \tau}$. Summing, with $\sum_{\text{far}} b_j \leq \beta_2$ and $\sum_{\text{far}} \rho_j \leq R_{\text{tot}}$ (Lemma 1.4(c)),

$$P_{\text{rest}} \leq e^{-2\tau} \left[2W\beta_2 + 2C_1 \pi_a + \frac{2C_1 \varepsilon_2}{\varepsilon_2 - 2} \beta_2 + \sum_{i \in \mathcal{N}} \rho_i N_i(\tau) \right] + e^{-\varepsilon_2 \tau} C_2 \varepsilon_2 [\pi_a + \tau R_{\text{tot}}].$$

Finally $\beta_2 + \pi_a = 1 - b_0 - b_1 = \hat{\beta}$ (Lemma 1.4(c)) and $2C_1 \leq c_\beta$. □

Theorem 2.3 (window theorem). *Define for $\tau > 0$*

$$\Phi_-(\tau) := a_1 e^{-\nu_1 \tau} \left[1 + \frac{\theta}{W} \kappa_{\text{lo}}(\tau) \right] - a_0 e^{-\nu_0 \tau} [1 + \bar{E}_-(\tau)],$$

$$\Phi_+(\tau) := a_1 e^{-\nu_1 \tau} \left[1 + \frac{\theta}{W} \bar{\kappa} \right] + \bar{P}_{\text{rest}}(\tau) - a_0 e^{-\nu_0 \tau} [1 + E_-^{\text{lo}}(\tau)].$$

Then $\Phi_-(\tau) \leq g'(\tau) \leq \Phi_+(\tau)$ for all $\tau > 0$. Consequently, by Theorem 1.11,

$$\Phi_-(\tau_-) > 0 \implies \tau_- < \tau^*, \quad \Phi_+(\tau_+) < 0 \implies \tau^* < \tau_+.$$

Equivalently, with $\Lambda := a_1/a_0$ and $\hat{\nu} := \nu_1 - \nu_0$: $\tau_- < \tau^*$ as soon as $\ln \Lambda - \hat{\nu} \tau_- + \ln(1 + \frac{\theta}{W} \kappa_{\text{lo}}(\tau_-)) - \ln(1 + \bar{E}_-(\tau_-)) > 0$ (when $1 + \frac{\theta}{W} \kappa_{\text{lo}}(\tau_-) > 0$), and $\tau^* < \tau_+$ as soon as $\ln \Lambda - \hat{\nu} \tau_+ + \ln(1 + \frac{\theta}{W} \bar{\kappa}) + \bar{P}_{\text{rest}}(\tau_+) e^{\nu_1 \tau_+}/a_1 - \ln(1 + E_-^{\text{lo}}(\tau_+)) < 0$.

Proof. Insert Lemma 2.2 into (2.1) (all prefactors are positive), then apply Theorem 1.11. $\square$

Remark 2.4 (the two-pole crossing). Dropping all corrections, g' vanishes at $t_c := \ln \Lambda / \hat{\nu}$. Section 3 shows $\Lambda = WX_s(1 + O(1/X_s))$, $\hat{\nu} = 1 + (W - 1)/X_s + O(X_s^{-2})$, and that every correction in Theorem 2.3 is of relative size $O(1/X_s)$; this is the rigorous form of the "two slowest poles" heuristic. The checks `scripts/40_check_thmA.py` (identity, all bounds of Lemmas 1.8, 1.9, 2.2 and Theorem 2.3 against a dense eigen-decomposition, $d = 2$: $N \leq 36$, $d = 3$: $N \leq 12$; 0 violations, exactly one sign change of g') and `scripts/41_windows_exact.py` (windows from $\Phi_{\pm}$ with float pole data against the exact modes, $d = 2$: $N \leq 1810$, $d = 3$: $N \leq 160$) are sanity checks only.

3 The pole data in terms of lattice sums

Throughout this section

$$y := \frac{1}{X_s}, \quad \sigma := \sum_{k \in \mathcal{K}} \frac{w_k}{\mu_k^2},$$

$$T_1 := \sum_{k: \mu_k \geq \varepsilon_2} \frac{w_k}{\mu_k(\mu_k - 1)}.$$

Every statement is an exact identity or a two-sided bound; no asymptotic symbol is used. Section 4 supplies $\sigma \leq Z_d$ and enclosures of X_s and T_1 .

Lemma 3.1 (the slowest pole). *Put $s := 1 - \nu_0 X_s$, so $\nu_0 = y(1 - s)$. Then*

$$s = \nu_0^2 \sigma', \quad \sigma' := \sum_k \frac{w_k}{\mu_k(\mu_k - \nu_0)} \in \left[\sigma, \frac{\sigma}{1 - \nu_0} \right];$$

$$\frac{1}{b_0} = 1 + \nu_0^2 \sigma'', \quad \sigma'' := \sum_k \frac{w_k}{(\mu_k - \nu_0)^2} \in \left[\sigma, \frac{\sigma}{(1 - \nu_0)^2} \right].$$

Consequently, since $\sigma \geq W$,

$$W\nu_0^2 \leq s \leq \frac{\sigma y^2}{1 - y}, \quad y \left(1 - \frac{\sigma y^2}{1 - y} \right) \leq \nu_0 \leq y,$$

$$0 \leq 1 - b_0 \leq \frac{\nu_0^2 \sigma}{(1 - \nu_0)^2}, \quad 0 \leq \ln \frac{1}{b_0} \leq \frac{\nu_0^2 \sigma}{(1 - \nu_0)^2}.$$

Proof. $D(\nu_0) = 0$ reads $1 = \nu_0 \sum_k w_k / (\mu_k - \nu_0) = \nu_0 \sum_k w_k \left[\frac{1}{\mu_k} + \frac{\nu_0}{\mu_k(\mu_k - \nu_0)} \right] = \nu_0 X_s + \nu_0^2 \sigma'$. As $\mu_k \geq 1 > \nu_0$, $\mu_k - \nu_0 \geq \mu_k(1 - \nu_0)$, which gives the bounds on σ' and σ'' ; the formula for $1/b_0 = \nu_0/\rho_0$ is Lemma 1.4(c). $\sigma \geq W$ because the k with $\mu_k = 1$ contribute W . Hence $s \geq W\nu_0^2 \geq 0$, so $\nu_0 \leq y$, and then $s \leq \nu_0^2 \sigma / (1 - \nu_0) \leq y^2 \sigma / (1 - y)$ ($t \mapsto t^2/(1 - t)$ is increasing). The last two bounds follow from $1 - b_0 = \nu_0^2 \sigma'' / (1 + \nu_0^2 \sigma'')$ and $\ln(1 + z) \leq z$. $\square$

Lemma 3.2 ($\mathbb{E}U$ and m_0). *Let $H_d := \sum_{j=1}^d 1/j$ and $h_d(\nu) := \sum_{j=1}^d (-1)^{j+1} \binom{d}{j} \frac{1}{j-\nu}$ for $0 \leq \nu < 1$ (so $h_2(\nu) = \frac{2}{1-\nu} - \frac{1}{2-\nu}$, $h_3(\nu) = \frac{3}{1-\nu} - \frac{3}{2-\nu} + \frac{1}{3-\nu}$, $h_d(0) = H_d$). Then*

$$\mathbb{E}U_{\text{lo}} := H_d - \frac{d\gamma_2 A^{-\varepsilon_2}}{\varepsilon_2} + (\ln A)(1 - (\gamma_2 A^{-\varepsilon_2})^d) \leq \mathbb{E}U \leq \ln A + H_d =: \mathbb{E}U_{\text{hi}},$$

$$\nu_0 \mathbb{E}U \leq \ln m_0 \leq \nu_0 (\ln A + h_d(\nu_0)).$$

Proof. $\mathbb{E}U = \int_0^\infty (1 - v^d) ds$ and $m_0 = \mathbb{E}e^{\nu_0 U} = 1 + \nu_0 \int_0^\infty (1 - v(s)^d) e^{\nu_0 s} ds$ (integrate by parts; $1 - u \leq W e^{-s}$, $\nu_0 < 1$). By Lemma 1.8, $v \geq (1 - A e^{-s})_+$, and $A \geq \frac{3}{2}$. Substituting $z = A e^{-s}$ on $s \geq \ln A$,

$$\int_0^\infty [1 - (1 - A e^{-s})_+^d] e^{\nu s} ds = \frac{A^\nu - 1}{\nu} + A^\nu \int_0^1 \frac{1 - (1 - z)^d}{z^{1+\nu}} dz = \frac{A^\nu - 1}{\nu} + A^\nu h_d(\nu)$$

$$(0 < \nu < 1),$$

and for $\nu = 0$ the value is $\ln A + H_d$ (note $\int_0^1 (1 - (1 - z)^d) z^{-1} dz = \int_0^1 \frac{1-w^d}{1-w} dw = H_d$). This gives $\mathbb{E}U \leq \ln A + H_d$ and $m_0 \leq A^{\nu_0}(1 + \nu_0 h_d(\nu_0))$, hence $\ln m_0 \leq \nu_0(\ln A + h_d(\nu_0))$. Jensen's inequality gives $m_0 \geq e^{\nu_0 \mathbb{E}U}$.

Lower bound for $\mathbb{E}U$: by Lemma 1.8, $v \leq 1 - \ell$ with $\ell = Ae^{-s} - \gamma_2 e^{-\varepsilon_2 s} \in [0, 1]$. For $s \geq s_A := \ln A$ both $1 - \ell$ and $1 - Ae^{-s}$ lie in $[0, 1]$ and differ by $\gamma_2 e^{-\varepsilon_2 s}$, so $(1 - \ell)^d \leq (1 - Ae^{-s})^d + d\gamma_2 e^{-\varepsilon_2 s}$ and $\int_{s_A}^\infty (1 - v^d) \geq H_d - d\gamma_2 A^{-\varepsilon_2}/\varepsilon_2$. For $s < s_A$, $v(s) \leq v(s_A) \leq 1 - \ell(s_A) = \gamma_2 A^{-\varepsilon_2} \leq 1$, so $\int_0^{s_A} (1 - v^d) \geq (\ln A)(1 - (\gamma_2 A^{-\varepsilon_2})^d)$. $\square$

Lemma 3.3 (the second pole and the leading coefficient). *Put*

$$\begin{aligned} S_\nu &:= \sum_{k: \mu_k \geq 2} \frac{w_k}{\mu_k(\mu_k - \nu_1)}, \\ S_{2\nu} &:= \sum_{k: \mu_k \geq 2} \frac{w_k}{(\mu_k - \nu_1)^2}, \quad e := \frac{\theta}{1 + \theta} + (1 + \theta)S_\nu, \\ \zeta &:= \frac{1}{W(1 + \theta)^2} + \frac{S_{2\nu}}{W}. \end{aligned}$$

Then (a) $\frac{W}{\theta} = X_s - W - 1 + e$, equivalently $\frac{\theta}{1 + \theta} = \frac{Wy}{1 + (e - 1)y}$;

(b) $a_1 = W\Gamma = \frac{\theta}{1 + \theta} \cdot \frac{1}{1 + \theta^2 \zeta}$ and $b_1 = \frac{\theta \Gamma}{1 + \theta}$;

(c) with $[\cdot]_3$ meaning "only if $d = 3$ ",

$$\frac{W_2}{2(1 - \theta)} + \left[\frac{W_3}{3(2 - \theta)} \right]_3 + T_1 \leq S_\nu \leq \frac{W_2}{2(1 - \theta)} + \left[\frac{W_3}{3(2 - \theta)} \right]_3 + T_1 \frac{\varepsilon_2 - 1}{\varepsilon_2 - 1 - \theta},$$

$$W_2 \leq S_{2\nu} \leq \frac{W_2}{(1 - \theta)^2} + \left[\frac{W_3}{(2 - \theta)^2} \right]_3 + T_1 \frac{\varepsilon_2}{\varepsilon_2 - 1} \left(\frac{\varepsilon_2 - 1}{\varepsilon_2 - 1 - \theta} \right)^2;$$

(d) with s as in Lemma 3.1,

$$\ln \Lambda = \ln \frac{a_1}{a_0} = \ln(WX_s) - \ln(1 + (e - 1)y) - \ln(1 + \theta^2 \zeta) + \ln \frac{1}{b_0} - 2 \ln(1 - s) - \ln m_0.$$

Proof. (a) $D(\nu_1) = 0$ gives $\frac{1}{\nu_1} = -\frac{W}{\theta} + \sum_{\mu_k \geq 2} \frac{w_k}{\mu_k - \nu_1}$ and $\sum_{\mu_k \geq 2} \frac{w_k}{\mu_k - \nu_1} = \sum_{\mu_k \geq 2} \frac{w_k}{\mu_k} + \nu_1 S_\nu = X_s - W + \nu_1 S_\nu$. Hence $\frac{W}{\theta} = X_s - W - 1 + (1 - \frac{1}{\nu_1}) + \nu_1 S_\nu$, and $1 - \frac{1}{\nu_1} = \frac{\theta}{1 + \theta}$. Adding W : $W \frac{1 + \theta}{\theta} = X_s - 1 + e$.

(b) By Lemma 1.4(c), $\frac{1}{\rho_1} = \frac{1}{\nu_1} + \nu_1 \left[\frac{W}{\theta^2} + S_{2\nu} \right]$. Multiply by θ : $\frac{1}{\Gamma} = \frac{\theta}{1 + \theta} + \frac{(1 + \theta)W}{\theta} + \theta(1 + \theta)S_{2\nu} = \frac{(1 + \theta)W}{\theta} [1 + \theta^2 \zeta]$. $b_1 = \rho_1 / \nu_1$.

(c) The levels ≥ 2 are 2 (weight W_2), 3 (weight W_3 , $d = 3$), and the $\mu_k \geq \varepsilon_2$. For the latter, $\frac{1}{\mu_k - 1} \leq \frac{1}{\mu_k - \nu_1} = \frac{1}{\mu_k - 1} \cdot \frac{\mu_k - 1}{\mu_k - 1 - \theta} \leq \frac{1}{\mu_k - 1} \cdot \frac{\varepsilon_2 - 1}{\varepsilon_2 - 1 - \theta}$, which gives the bounds for S_ν ; and $\frac{w_k}{(\mu_k - \nu_1)^2} = \frac{w_k}{\mu_k(\mu_k - 1)} \cdot \frac{\mu_k}{\mu_k - 1} \left(\frac{\mu_k - 1}{\mu_k - 1 - \theta} \right)^2$ with both factors decreasing in μ_k .

(d) $a_0 = \rho_0 \nu_0 m_0 = b_0 \nu_0^2 m_0 = b_0 y^2 (1 - s)^2 m_0$; combine with (a), (b). $\square$

Lemma 3.4 (near poles). *For $i \in \mathcal{N}$ let $m_i \in \{2, 3\}$ be the level just below ν_i and $\theta_i := \nu_i - m_i > 0$. Then $\rho_i \leq \theta_i^2 / (m_i W_i)$ and*

$$\begin{aligned} \frac{W_i}{\theta_i} &= X_s - \sum_{l \leq i} \frac{W_l}{m_l} + \nu_i \Sigma_i - \sum_{l < i} \frac{W_l}{\nu_i - m_l} - \frac{1}{\nu_i}, \\ \Sigma_i &:= \sum_{l > i} \frac{W_l}{m_l(m_l - \nu_i)}, \end{aligned}$$

where

$$\left[\frac{W_3}{3(1-\theta_2)} \right]_3 + T_1 \leq \Sigma_2 \leq \left[\frac{W_3}{3(1-\theta_2)} \right]_3 + T_1 \frac{\varepsilon_2 - 1}{\varepsilon_2 - 2 - \theta_2},$$

$$T_1 \leq \Sigma_3 \leq T_1 \frac{\varepsilon_2 - 1}{\varepsilon_2 - 3 - \theta_3} \quad (d = 3).$$

Proof. $D(\nu_i) = 0$: $\frac{1}{\nu_i} = -\sum_{l < i} \frac{W_l}{\nu_i - m_l} - \frac{W_i}{\theta_i} + \sum_{l > i} \frac{W_l}{m_l - \nu_i}$ and $\sum_{l > i} \frac{W_l}{m_l - \nu_i} = \sum_{l > i} \frac{W_l}{m_l} + \nu_i \Sigma_i$, $\sum_{l > i} W_l/m_l = X_s - \sum_{l \leq i} W_l/m_l$. The bounds for Σ_i are obtained as in Lemma 3.3(c) ($\mu_k \geq \varepsilon_2$: $\frac{1}{\mu_k - 1} \leq \frac{1}{\mu_k - \nu_i} \leq \frac{1}{\mu_k - 1} \frac{\varepsilon_2 - 1}{\varepsilon_2 - \nu_i}$; for $i = 2$, $d = 3$ the level 3 contributes $W_3/(3(3 - \nu_2))$). Finally $1/\rho_i = \sum_l W_l m_l / (m_l - \nu_i)^2 \geq W_i m_i / \theta_i^2$. $\square$

Lemma 3.5. $\hat{\beta} = 1 - b_0 - b_1 = \pi_a + \sum_{j \geq 2} b_j$, $\pi_a = N^{-d}$ and $R_{\text{tot}} = \frac{d}{4N^d \sin^2 x} \leq \frac{d}{4} N^{2-d}$.

Proof. Lemma 1.4(c), and $\sin x \geq \frac{2}{\pi}x = \frac{1}{N}$. $\square$

Remark 3.6 (first-order picture). As $X_s \rightarrow \infty$ (i.e. $N \rightarrow \infty$) the lemmas give $\nu_0 = y(1 + O(y^2))$, $b_0 = 1 + O(y^2)$, $\theta = Wy(1 + O(y))$, $a_1 = Wy(1 + O(y))$, $a_0 = y^2(1 + O(y))$, so $\Lambda = WX_s(1 + O(y))$ and $t_c = \ln \Lambda / (\nu_1 - \nu_0) = \ln(WX_s)(1 - (W - 1)y) + O(y)$, while every correction term in Theorem 2.3 is $O(y)$ relative to the two main terms. This is made quantitative, with explicit constants, in Section 5.

4 Lattice sums

4.1 Comparison with $|k|^{-4}$

Lemma 4.1. (a) For $0 \leq a \leq \pi/2$: $a^2 \cos a \leq \sin^2 a$. (b) For every $k \in \mathcal{K}$, with $nz(k)$ the number of nonzero coordinates,

$$\frac{w_k}{\mu_k^2} \leq \frac{2^{nz(k)}}{|k|^4}, \quad \text{hence}$$

$$\sum_{k \in S} \frac{w_k}{\mu_k^2} \leq \sum_{k' \in \mathbb{Z}^d: (|k'_1|, \dots, |k'_d|) \in S} \frac{1}{|k'|^4} \quad \text{for every } S \subset \mathcal{K},$$

$$\sigma \leq Z_d := \sum_{k' \in \mathbb{Z}^d \setminus 0} \frac{1}{|k'|^4}.$$

(c) For fixed integer $k \geq 1$ the functions $x \mapsto \varepsilon_k = \sin^2(kx)/\sin^2 x$ and $x \mapsto \gamma_k = 2 \cos^2(kx)$ are nonincreasing on $(0, \pi/(2k)]$, $\varepsilon_k \leq k^2$, and $\varepsilon_k \geq \cot^2 \frac{\pi}{2(k+1)} \geq \frac{4(k+1)^2}{\pi^2} - 1$ whenever $N \geq k + 1$.

Proof. (a) For $0 \leq a \leq \pi/2$, $\sin a \geq a - a^3/6 \geq 0$ and $\cos a \leq 1 - a^2/2 + a^4/24$, so $\sin^2 a - a^2 \cos a \geq a^2 \left[(1 - \frac{a^2}{6})^2 - 1 + \frac{a^2}{2} - \frac{a^4}{24} \right] = \frac{a^4}{6} (1 - \frac{a^2}{12}) \geq 0$.

(b) Put $a_i = k_i x \in [0, \pi/2)$. Then $w_k/2^{nz(k)} = \prod_i \cos^2 a_i$ and $\mu_k = \sum_i \sin^2 a_i / \sin^2 x$. By (a), $\prod_j \cos a_j \cdot \sum_i a_i^2 \leq \sum_i a_i^2 \cos a_i \leq \sum_i \sin^2 a_i$; squaring and using $\sin x \leq x$: $\prod_i \cos^2 a_i \cdot \sin^4 x / (\sum_i \sin^2 a_i)^2 \leq x^4 / (\sum_i a_i^2)^2 = |k|^{-4}$. The $2^{nz(k)}$ points k' with $(|k'_i|) = k$ have $|k'| = |k|$.

(c) $\frac{d}{dx} \frac{\sin kx}{\sin x} = \frac{\cos kx \cos x}{\sin x} - \frac{k \tan x - \tan kx}{\sin x} \leq 0$ for $kx < \pi/2$ because $\tan$ is convex on $[0, \pi/2)$ with $\tan 0 = 0$ (at $kx = \pi/2$ the derivative is $-\cos x / \sin^2 x < 0$). $|\sin kx| \leq k \sin x$ gives $\varepsilon_k \leq k^2$. If $N \geq k + 1$ then $x \leq \frac{\pi}{2(k+1)}$ and by monotonicity $\varepsilon_k \geq \sin^2 \frac{k\pi}{2(k+1)} / \sin^2 \frac{\pi}{2(k+1)} = \cot^2 \frac{\pi}{2(k+1)}$; finally $\cot^2 a = \sin^{-2} a - 1 \geq a^{-2} - 1$. $\square$

Lemma 4.2 (tails of Z_d ; certified values). For integers $K \geq 1$ let $\text{tail}_d(K) := \sum_{k \in \mathbb{Z}^d, |k|_\infty > K} |k|^{-4}$. Then

$$\text{tail}_2(K) \leq \frac{c_{\square}^{(2)}}{K^2} + \frac{4}{3K^3},$$

$$\begin{aligned}\text{tail}_3(K) &\leq \frac{c_{\square}^{(3)}}{K} + \frac{3c_{\square}^{(2)}}{K^2} + \frac{2}{K^3}, \\ c_{\square}^{(2)} &= \frac{\pi}{2} + 1, \quad c_{\square}^{(3)} = 24 \int_{[0,1]^2} \frac{da db}{(1+a^2+b^2)^2}.\end{aligned}$$

Certified (Arb, scripts/50_cert_constants.py): $c_{\square}^{(3)} = 10.44503701\dots$,

$$6.026748 \leq Z_2 \leq 6.0268126, \quad 16.35967 \leq Z_3 \leq 16.53592.$$

Proof. $f(y) = |y|^{-4}$ is decreasing in each $|y_i|$. For $k \in \{1, 2, \dots\}^m$, $f(k) \leq \int_{\prod_i [k_i-1, k_i]} f$; the cells with $\max_i k_i > K$ tile $\{y \in [0, \infty)^m : \max y_i > K\}$, so $\sum_{k \in \mathbb{N}_{\geq 1}^m, \max k_i > K} f(k) \leq 2^{-m} \int_{|y|_{\infty} > K} |y|^{-4} dy = 2^{-m} c_{\square}^{(m)} K^{m-4}$ with $c_{\square}^{(m)} := \int_{\mathbb{R}^m, |y|_{\infty} > 1} |y|^{-4} dy$ ($c_{\square}^{(1)} = 2/3$). Sorting $k \in \mathbb{Z}^3$, $|k|_{\infty} > K$, by the set of nonzero coordinates gives $8 \cdot \frac{c_{\square}^{(3)}}{8K} + 12 \cdot \frac{c_{\square}^{(2)}}{4K^2} + 6 \cdot \frac{1}{3K^3}$; in $d = 2$: $4 \cdot \frac{c_{\square}^{(2)}}{4K^2} + 4 \cdot \frac{1}{3K^3}$. In polar coordinates $c_{\square}^{(2)} = \int_0^{2\pi} \frac{1}{2} \max(|\cos \vartheta|, |\sin \vartheta|)^2 d\vartheta = 4 \int_0^{\pi/4} \cos^2 \vartheta d\vartheta = \frac{\pi}{2} + 1$. For $c_{\square}^{(3)}$: the region is the union of six congruent pieces $\{|y_3| > 1, |y_1|, |y_2| < |y_3|\}$ etc.; substituting $y_1 = y_3 a$, $y_2 = y_3 b$ gives $c_{\square}^{(3)} = 6 \int_1^{\infty} z^{-2} dz \int_{[-1,1]^2} (1+a^2+b^2)^{-2} = 24 \int_{[0,1]^2} (1+a^2+b^2)^{-2}$; the inner integral in b is elementary and the remaining one-dimensional integral is enclosed by `acb.integral`. The enclosures of Z_d are (exact partial sum over $|k|_{\infty} \leq K$, evaluated in ball arithmetic) $+ [0, \text{tail}_d(K)]$ with $K = 200$ ($d = 2$) and $K = 60$ ($d = 3$). $\square$

4.2 The kernel κ and the one-dimensional return function ω

Recall $\omega(s) = \sum_{k=0}^{N-1} \gamma_k e^{-\varepsilon_k s}$, so that $X_s = \int_0^{\infty} (\omega(s)^d - 1) ds$ (integrate $\omega^d - 1 = \sum_{k \in \mathcal{K}} w_k e^{-\mu_k s}$). Put

$$\begin{aligned}\lambda &:= \frac{1}{2 \sin^2 x}, \\ \kappa(z) &:= \frac{1}{\pi} \int_0^{\pi} (1 + \cos \vartheta) e^{-z(1 - \cos \vartheta)} d\vartheta = \frac{4}{\pi} \int_0^1 e^{-2zt^2} \sqrt{1-t^2} dt \\ &\quad (z \geq 0).\end{aligned}$$

(The second form follows from $\vartheta = 2\phi$, $t = \sin \phi$. By the integral representation of the modified Bessel functions, DLMF 10.32.3, $\kappa(z) = e^{-z}(I_0(z) + I_1(z))$; this is used only to evaluate κ numerically in Arb.)

Lemma 4.3 (bounds for κ). *κ is decreasing, $\kappa(0) = 1$, and for $z > 0$*

$$\sqrt{\frac{2}{\pi z}} \left(1 - \frac{1}{4z}\right) - \frac{e^{-2z}}{\pi z} \leq \kappa(z) \leq \min\left(1, \sqrt{\frac{2}{\pi z}}\right).$$

Consequently, for $z \geq 1$: $\kappa(z) \geq \sqrt{2/(\pi z)} (1 - \frac{1}{3z})$ and $\kappa(z)^3 \geq (\frac{2}{\pi z})^{3/2} (1 - \frac{1}{z})$; and with $K_2(Z) := \int_0^Z \kappa^2$, for $Z \geq Z_1 \geq 1$,

$$\begin{aligned}\frac{2}{\pi} \ln Z + c_{\kappa}^{\text{lo}}(Z_1) &\leq K_2(Z) \leq \frac{2}{\pi} \ln Z + c_{\kappa}^{\text{hi}}(Z_1), \\ c_{\kappa}^{\text{hi}} &:= K_2(Z_1) - \frac{2}{\pi} \ln Z_1, \quad c_{\kappa}^{\text{lo}} := c_{\kappa}^{\text{hi}} - \frac{1}{\pi Z_1} - \frac{\sqrt{2/\pi}}{\pi} \frac{e^{-2Z_1}}{Z_1^{3/2}}.\end{aligned}$$

Certified (Arb): with $Z_1 = 10$, $c_{\kappa}^{\text{lo}} \geq 0.48012$, $c_{\kappa}^{\text{hi}} \leq 0.51196$; and $J := \int_0^{\infty} \kappa^3 \in [1.4401510, 1.4401549]$, so that

$$c_3 := \frac{\pi^2}{2} J \in [7.106860, 7.106880], \quad \ln(6c_3) \in [3.752820, 3.752823].$$

Proof. Upper bound: $\sqrt{1-t^2} \leq 1$ and $\int_0^\infty e^{-2zt^2} dt = \frac{1}{2} \sqrt{\pi/(2z)}$. Lower bound: $\sqrt{1-t^2} \geq 1-t^2$ on $[0, 1]$, $\int_1^\infty e^{-2zt^2} dt \leq \int_1^\infty te^{-2zt^2} dt = \frac{e^{-2z}}{4z}$ and $\int_0^\infty t^2 e^{-2zt^2} dt = \frac{\sqrt{\pi}}{4(2z)^{3/2}}$, so $\kappa \geq \frac{4}{\pi} [\frac{1}{2} \sqrt{\frac{\pi}{2z}} - \frac{e^{-2z}}{4z} - \frac{\sqrt{\pi}}{4(2z)^{3/2}}]$, which is the claim. For $z \geq 1$, $\frac{e^{-2z}}{\pi z} \leq \sqrt{\frac{2}{\pi z}} \frac{1}{12z}$ (equivalent to $e^{-2z} \sqrt{z} \leq \sqrt{2\pi}/12$, true at $z = 1$ and the left side decreases), whence $\kappa \geq \sqrt{2/(\pi z)}(1 - \frac{1}{3z})$ and, by Bernoulli, the bound for κ^3 . For K_2 : $\kappa^2 \leq \frac{2}{\pi z}$ gives the upper bound; and for $z \geq 1$, squaring the (positive) lower bound, $\kappa^2 \geq \frac{2}{\pi z}(1 - \frac{1}{2z}) - 2\sqrt{\frac{2}{\pi}} \frac{e^{-2z}}{\pi z^{3/2}}$, whose integral over $[Z_1, Z]$ is at least $\frac{2}{\pi} \ln \frac{Z}{Z_1} - \frac{1}{\pi Z_1} - \frac{\sqrt{2/\pi}}{\pi} e^{-2Z_1} Z_1^{-3/2}$. $K_2(10)$ and $\int_0^{2000} \kappa^3$ are enclosed by `acb.integral`; the tail $\int_{2000}^\infty \kappa^3$ lies between $(\frac{2}{\pi})^{3/2}(2Z^{-1/2} - \frac{2}{3}Z^{-3/2})$ and $(\frac{2}{\pi})^{3/2}2Z^{-1/2}$ at $Z = 2000$. $\square$

Lemma 4.4 (bounds for ω). *Let $c_0 := \frac{1}{2}(\cosh 2 - 1)$, $G_A := (1 - e^{-4/c_0})^{-1}$, $\beta_N := (2N - 1)^2 \sin^2 x / c_0$,*

$$\begin{aligned} \iota_A(s) &:= G_A \pi \sqrt{\pi/s} e^{-\beta_N/s}, & \iota_B &:= \frac{4Ne^{-(2N-1)}}{1 - e^{-2N}}, \\ \bar{\omega}(s) &:= 2e^{-s} + 2e^{-3s} + \frac{2e^{-(64/\pi^2 - 1)s}}{1 - e^{-32s/\pi^2}}. \end{aligned}$$

Then for all $N \geq 3$: (a) $\omega(s) \geq \max(1, N\kappa(\lambda s))$ for $s > 0$; (b) $\omega(s) \leq N\kappa(\lambda s) + \iota_A(s) + \iota_B$ for $0 < s \leq 1$; (c) $\omega(s) \leq 1 + \bar{\omega}(s)$ for $s > 0$.

Proof. Since $\varepsilon_k = \lambda(1 - \cos \frac{\pi k}{N})$ and $\gamma_k = 1 + \cos \frac{\pi k}{N}$ ($k \geq 1$), we have $\omega(s) = \frac{1}{2} \sum_{j \in \mathbb{Z}/2N} F(\pi j/N)$ with $F(\vartheta) := (1 + \cos \vartheta)e^{-z(1 - \cos \vartheta)}$, $z = \lambda s$ (the terms j and $-j$ coincide, $j = N$ contributes 0, $j = 0$ contributes 2). For $n \in \mathbb{Z}$ let $p_z(n) := \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{in\vartheta} e^{-z(1 - \cos \vartheta)} d\vartheta$. Expanding $e^{z \cos \vartheta} = \sum_m \frac{z^m}{m!} (\frac{e^{i\vartheta} + e^{-i\vartheta}}{2})^m$ shows $p_z(n) = p_z(-n) \geq 0$. The Fourier coefficients of F are $\hat{F}(n) = p_z(n) + \frac{1}{2}p_z(n-1) + \frac{1}{2}p_z(n+1)$, the Fourier series converges absolutely, and $\sum_{j \in \mathbb{Z}/2N} e^{in\pi j/N} = 2N \cdot \mathbf{1}_{2N|n}$; hence

$$\omega(s) = N \sum_{m \in \mathbb{Z}} \hat{F}(2Nm) = N\kappa(z) + N \sum_{m \geq 1} [2p_z(2Nm) + p_z(2Nm-1) + p_z(2Nm+1)],$$

because $\hat{F}(0) = \frac{1}{2\pi} \int F = \kappa(z)$. (a) follows from $p_z \geq 0$, and from $\omega \geq \gamma_0 = 1$.

(b) Let $n \geq 1$, $\eta \geq 0$. The integrand of $p_z(n)$ is entire and 2π -periodic, so the contour may be shifted to $\vartheta + i\eta$; with $\operatorname{Re} \cos(\vartheta + i\eta) = \cos \vartheta \cosh \eta$,

$$\begin{aligned} p_z(n) &\leq e^{-n\eta + z(\cosh \eta - 1)} p_{z \cosh \eta}(0), \\ p_{z'}(0) &= \frac{1}{\pi} \int_0^\pi e^{-z'(1 - \cos \vartheta)} d\vartheta \leq \min\left(1, \frac{1}{2} \sqrt{\frac{\pi}{2z'}}\right), \end{aligned}$$

the last bound by $1 - \cos \vartheta \geq 2\vartheta^2/\pi^2$ on $[0, \pi]$. Since $(\cosh \eta - 1)/\eta^2$ is increasing, $\cosh \eta - 1 \leq \frac{c_0}{2}\eta^2$ for $0 \leq \eta \leq 2$. If $z \geq n/(2c_0)$ take $\eta = n/(c_0 z) \leq 2$: $p_z(n) \leq \frac{1}{2} \sqrt{\frac{\pi}{2z}} e^{-n^2/(2c_0 z)}$. If $z < n/(2c_0)$ take $\eta = 2$: $p_z(n) \leq e^{-2n+2c_0 z} \leq e^{-n}$. So in all cases $p_z(n) \leq Q(n) := \frac{1}{2} \sqrt{\frac{\pi}{2z}} e^{-n^2/(2c_0 z)} + e^{-n}$, which is decreasing in n , and the bracket above is at most $4Q(2Nm-1)$. Now $(2Nm-1)^2 - (2N-1)^2 \geq 4N^2(m-1)$ and $z \leq \lambda \leq N^2/2$ (as $\sin x \geq 1/N$), so $\sum_{m \geq 1} e^{-(2Nm-1)^2/(2c_0 z)} \leq G_A e^{-(2N-1)^2/(2c_0 z)}$; with $z = \lambda s$: $\frac{1}{2} \sqrt{\frac{\pi}{2z}} = \frac{\sin x}{2} \sqrt{\frac{\pi}{s}}$, $\frac{(2N-1)^2}{2c_0 z} = \frac{\beta_N}{s}$ and $N \sin x \leq \frac{\pi}{2}$. This gives $N \cdot 4 \sum_m Q(2Nm-1) \leq \iota_A(s) + \iota_B$.

(c) $\gamma_k \leq 2$, $\varepsilon_1 = 1$, $\varepsilon_2 \geq 3$ and, by Lemma 4.1(c), $\varepsilon_k \geq \frac{4(k+1)^2}{\pi^2} - 1$ for $3 \leq k \leq N-1$. Hence $\omega - 1 \leq 2e^{-s} + 2e^{-3s} + 2e^s \sum_{j \geq 4} e^{-4j^2 s/\pi^2}$ and $j^2 - 16 \geq 8(j-4)$ for $j \geq 4$. $\square$

4.3 Two-sided bounds for X_s

Let $T_d(1) := \int_1^\infty [(1 + \bar{\omega})^d - 1] ds$; certified: $T_2(1) \leq 1.852100$, $T_3(1) \leq 3.415423$. Let ϵ_2, ϵ_3 be the numbers

$$\begin{aligned}\epsilon_2 &:= \frac{2G_A\pi^2 e^{-\beta}}{\beta} + 4\sqrt{\pi}\iota_B + (\iota_A(1) + \iota_B)^2, \\ \epsilon_3 &:= \frac{3G_A\pi^{5/2} e^{-\beta}}{\beta} + 3\pi\iota_B \left(1 + \ln \frac{N^2}{\pi}\right) + 6\sqrt{\pi}(\iota_A(1) + \iota_B)^2 + (\iota_A(1) + \iota_B)^3\end{aligned}$$

evaluated at $N = 10$ ($\beta = \beta_{10}$); certified: $\epsilon_2 \leq 0.00555$, $\epsilon_3 \leq 0.01552$. Both expressions are decreasing in N for $N \geq 10$.

Proposition 4.5 ($d = 2$). *For all $N \geq 10$, $X_{\text{lo}}^{(2)}(N) \leq X_s \leq \bar{X}^{(2)}(N)$, where*

$$\begin{aligned}X_{\text{lo}}^{(2)}(N) &:= 2N^2 \sin^2 x \left[\frac{2}{\pi} \ln \frac{2N^2}{\pi} + c_\kappa^{\text{lo}} - \frac{2}{\pi} \right], \\ \bar{X}^{(2)}(N) &:= 2N^2 \sin^2 x \left[\frac{2}{\pi} \ln \lambda + c_\kappa^{\text{hi}} \right] - 1 + T_2(1) + \epsilon_2.\end{aligned}$$

Both bounds are increasing in N , and

$$2\pi \ln N - 2.32 \leq X_s \leq 2\pi \ln N - 1.60 \quad (N \geq 10).$$

Proof. Lower bound. By Lemma 4.4(a), $\omega^2 - 1 \geq (N^2\kappa(\lambda s)^2 - 1)_+$, so for any $Z_0 > 0$, $X_s \geq \lambda^{-1} \int_0^{Z_0} (N^2\kappa^2 - 1) dz = 2\sin^2 x [N^2 K_2(Z_0) - Z_0]$. Take $Z_0 = 2N^2/\pi \geq 10$ and use Lemma 4.3.

Upper bound. Split at $s = 1$. On $(1, \infty)$ use Lemma 4.4(c): $\int_1^\infty (\omega^2 - 1) \leq T_2(1)$. On $(0, 1]$ use Lemma 4.4(b) with $\iota := \iota_A + \iota_B$: $\omega^2 - 1 \leq N^2\kappa^2 - 1 + 2\iota N\kappa + \iota^2$. First, $\int_0^1 N^2\kappa(\lambda s)^2 ds = 2N^2 \sin^2 x K_2(\lambda)$ and $\lambda \geq 10$ for $N \geq 10$. Next $N\kappa(\lambda s) \leq N\sqrt{2/(\pi\lambda s)} = 2N \sin x / \sqrt{\pi s} \leq \sqrt{\pi/s}$, so $\int_0^1 2\iota N\kappa \leq 2G_A\pi^2 \int_0^1 s^{-1} e^{-\beta_N/s} ds + 4\sqrt{\pi}\iota_B \leq 2G_A\pi^2 e^{-\beta_N}/\beta_N + 4\sqrt{\pi}\iota_B$ (substitute $u = 1/s$), and $\int_0^1 \iota^2 \leq (\iota_A(1) + \iota_B)^2$ since ι_A is increasing on $(0, 1]$ ($\beta_N \geq \frac{1}{2}$). As β_N increases and ι_B decreases with N , these corrections are at most ϵ_2 .

Monotonicity. $N \sin x = \frac{\pi}{2} \frac{\sin x}{x}$ and λ are increasing in N and the brackets are positive.

Simple form. $2N^2 \sin^2 x = \frac{\pi^2}{2} (\frac{\sin x}{x})^2 \in [\frac{\pi^2}{2} (1 - \frac{x^2}{3}), \frac{\pi^2}{2}]$. Lower: $X_{\text{lo}}^{(2)} = (\frac{\sin x}{x})^2 [2\pi \ln N - c_L]$ with $c_L = -\frac{\pi^2}{2} (\frac{2}{\pi} \ln \frac{2}{\pi} + c_\kappa^{\text{lo}} - \frac{2}{\pi}) \leq 2.1914$, and $\frac{x^2}{3} 2\pi \ln N \leq 0.12$ for $N \geq 10$. Upper: $\ln \lambda \leq \ln \frac{2N^2}{\pi^2} - 2 \ln(1 - \frac{x^2}{6}) \leq \ln \frac{2N^2}{\pi^2} + 0.0083$, so $\bar{X}^{(2)} \leq 2\pi \ln N + \pi \ln \frac{2}{\pi^2} + 0.0261 + \frac{\pi^2}{2} c_\kappa^{\text{hi}} - 1 + T_2(1) + \epsilon_2 \leq 2\pi \ln N - 1.60$. $\square$

Proposition 4.6 ($d = 3$). *For all $N \geq 10$, $X_{\text{lo}}^{(3)}(N) \leq X_s \leq \bar{X}^{(3)}(N)$, where*

$$\begin{aligned}X_{\text{lo}}^{(3)}(N) &:= 2\sin^2 x \left[N^3 J - \frac{6N^2}{\pi} \right], \\ \bar{X}^{(3)}(N) &:= 2\sin^2 x N^3 J - 2(N \sin x)^3 \left(\frac{4}{\pi} \right)^{3/2} \left(1 - \frac{1}{3\lambda} \right) - 1 + T_3(1) + \epsilon_3.\end{aligned}$$

Both bounds are increasing in N , and

$$c_3 N - 3\pi - \frac{5.85}{N} \leq X_s \leq c_3 N - 8.38 \quad (N \geq 10).$$

Proof. Lower bound. By Lemma 4.4(a), $X_s \geq \lambda^{-1} \int_0^\infty (N^3\kappa^3 - 1)_+ dz = \lambda^{-1} [N^3 J - \int_0^\infty \min(N^3\kappa^3, 1) dz]$, and by Lemma 4.3 $\int_0^\infty \min(N^3\kappa^3, 1) \leq \int_0^\infty \min(N^3(\frac{2}{\pi z})^{3/2}, 1) dz = 3z_0$ with $z_0 = 2N^2/\pi$.

Upper bound. On $(1, \infty)$: $\int_1^\infty (\omega^3 - 1) \leq T_3(1)$. On $(0, 1]$: $\omega^3 - 1 \leq N^3\kappa^3 - 1 + 3\iota N^2\kappa^2 + 3\iota^2 N\kappa + \iota^3$. First $\int_0^1 N^3\kappa(\lambda s)^3 ds = \frac{N^3}{\lambda} [J - \int_\lambda^\infty \kappa^3]$ and, by Lemma 4.3 ($\lambda \geq 1$), $\int_\lambda^\infty \kappa^3 \geq (\frac{2}{\pi})^{3/2} (2\lambda^{-1/2} -$

$\frac{2}{3}\lambda^{-3/2}$); note $\frac{N^3}{\lambda}(\frac{2}{\pi})^{3/2}2\lambda^{-1/2} = 2(N \sin x)^3(\frac{4}{\pi})^{3/2}$. Next $N^2\kappa(\lambda s)^2 \leq \min(N^2, \pi/s)$, so $\int_0^1 3\iota N^2\kappa^2 \leq 3G_A\pi^{5/2} \int_0^1 s^{-3/2}e^{-\beta_N/s}ds + 3\iota_B\pi(1 + \ln \frac{N^2}{\pi}) \leq 3G_A\pi^{5/2}\frac{e^{-\beta_N}}{\beta_N} + 3\pi\iota_B(1 + \ln \frac{N^2}{\pi})$; and $\int_0^1 3\iota^2 N\kappa \leq 3(\iota_A(1) + \iota_B)^2 \cdot 2\sqrt{\pi}$, $\int_0^1 \iota^3 \leq (\iota_A(1) + \iota_B)^3$. All these are decreasing in N , hence at most ϵ_3 .

Monotonicity. $X_{\text{lo}}^{(3)} = 2(N \sin x)^2[NJ - \frac{6}{\pi}]$ is increasing. $\bar{X}^{(3)} = (N \sin x)^2[2NJ - 2(N \sin x)(\frac{4}{\pi})^{3/2}(1 - \frac{2}{3}\sin^2 x)] + \text{const}$; the derivative in N of the subtracted term is at most $2(\frac{4}{\pi})^{3/2}[\frac{x^3}{3} + \frac{\pi}{2} \cdot \frac{4}{3}\frac{x^2}{N}] \leq 0.02 < 2J$ for $N \geq 10$, so the bracket is increasing (and positive).

Simple form. $2\sin^2 x N^3 J = c_3 N(\frac{\sin x}{x})^2 \in [c_3 N - \frac{c_3\pi^2}{12N}, c_3 N]$ and $2\sin^2 x \cdot 6N^2/\pi \leq 3\pi$. For the upper bound, $N \sin x \geq \frac{\pi}{2}(1 - \frac{x^2}{6})$ and $\frac{1}{3\lambda} = \frac{2}{3}\sin^2 x$ give, for $N \geq 10$, $2(N \sin x)^3(\frac{4}{\pi})^{3/2}(1 - \frac{1}{3\lambda}) \geq 10.82$, so $\bar{X}^{(3)} \leq c_3 N - 10.82 - 1 + 3.4155 + 0.0156 \leq c_3 N - 8.38$. $\square$

Corollary 4.7. *For $N \geq 10$: $X = X_s + \mathbb{E}U$ satisfies $2\pi \ln N - 0.24 \leq X \leq 2\pi \ln N + 0.60$ if $d = 2$, and $c_3 N - 7.63 \leq X \leq c_3 N - 5.85$ if $d = 3$. Moreover $\sin^2 x \cdot X_s \leq \frac{\pi^2}{4N^2}(2\pi \ln N + 1.3)$ ($d = 2$), $\leq \frac{\pi^2 c_3}{4N}$ ($d = 3$), and both right-hand sides are decreasing in N .*

Proof. For $N \geq 10$ we have $A = 2\cos^2 x \in [1.9507, 2]$ and $\varepsilon_2 = 2A \geq 3.9014$. The function $f(x) := \gamma_2 A^{-\varepsilon_2} = 2\cos^2(2x) \exp(-4\cos^2 x \ln(2\cos^2 x))$ satisfies $(\ln f)'(x) = 4\sin(2x)[1 + \ln(2\cos^2 x) - \frac{1}{\cos 2x}] > 0$ on $(0, \frac{\pi}{20}]$, because there $1 + \ln(2\cos^2 x) \geq 1.668$ and $1/\cos 2x \leq 1.052$; so f is decreasing in N and $f \leq f(\frac{\pi}{20}) = 0.13328 \dots \leq 0.1333$ for $N \geq 10$. Since $\mathbb{E}U_{\text{lo}}$ of Lemma 3.2 is decreasing in f and increasing in ε_2 and $\ln A$, it is at least $\frac{3}{2} - \frac{2 \cdot 0.1333}{3.9014} + \ln(1.9507)(1 - 0.1333^2) \geq 2.0879$ for $d = 2$ and $\frac{11}{6} - \frac{3 \cdot 0.1333}{3.9014} + \ln(1.9507)(1 - 0.1333^3) \geq 2.3974$ for $d = 3$; and $\mathbb{E}U_{\text{hi}} \leq \ln 2 + H_d$. Hence $2.08 \leq \mathbb{E}U \leq 2.20$ for $d = 2$ and $2.39 \leq \mathbb{E}U \leq 2.53$ for $d = 3$. Add Propositions 4.5 and 4.6 (with $3\pi + 0.585 \leq 10.02$). The last claim follows from $\sin x \leq x$, $X_s \leq \bar{X}^{(d)}(N)$ and $\frac{d}{dN} \frac{a \ln N + b}{N^2} < 0$ for $\ln N > \frac{1}{2} - \frac{b}{a}$. $\square$

4.4 The lattice constant T_1

Lemma 4.8 (enclosures of T_1). *Let $N_a \leq N \leq N_b \leq \infty$ be integers, $x_b := \frac{\pi}{2N_a}$, $x_a := \frac{\pi}{2N_b}$ ($x_a = 0$ if $N_b = \infty$), and $1 \leq K \leq N_a - 1$. Write $\varepsilon_j(\xi) = \sin^2(j\xi)/\sin^2 \xi$ (with $\varepsilon_j(0) = j^2$), $\gamma_j(\xi) = 2\cos^2(j\xi)$ for $j \geq 1$, $\gamma_0 = 1$, $\varepsilon_0 = 0$, and let $\mathcal{K}_K := \{k \in \{0, \dots, K\}^d : \max_i k_i \geq 2\}$. Then*

$$\sum_{k \in \mathcal{K}_K} \frac{\prod_i \gamma_{k_i}(x_b)}{\mu_k^+(\mu_k^+ - 1)} \leq T_1(N) \leq \sum_{k \in \mathcal{K}_K} \frac{\prod_i \gamma_{k_i}(x_a)}{\mu_k^-(\mu_k^- - 1)} + \frac{M}{M-1} \text{tail}_d(K),$$

$$\mu_k^+ := \sum_i \varepsilon_{k_i}(x_a), \quad \mu_k^- := \sum_i \varepsilon_{k_i}(x_b), \quad M := \varepsilon_{K+1}(x_b).$$

If $K = N - 1$ (so $N_a = N_b = N$) the tail term is absent and both sides equal $T_1(N)$.

Proof. The $k \in \mathcal{K}$ with $\mu_k \geq \varepsilon_2$ are exactly those with $\max_i k_i \geq 2$ (Section 2). For $k \in \mathcal{K}_K$ ($K \leq N - 1$), Lemma 4.1(c) gives $\mu_k^- \leq \mu_k \leq \mu_k^+$ and $\prod \gamma_{k_i}(x_b) \leq w_k \leq \prod \gamma_{k_i}(x_a)$; $t \mapsto 1/(t(t-1))$ is decreasing on $t > 1$ and $\mu_k^- \geq \varepsilon_2(x_b) \geq 3$. For the remaining k (some $k_i \geq K + 1$), $\mu_k \geq \varepsilon_{K+1}(x) \geq M$ so $\frac{w_k}{\mu_k(\mu_k-1)} \leq \frac{M}{M-1} \frac{w_k}{\mu_k^2}$, and Lemma 4.1(b) bounds the sum of w_k/μ_k^2 over these k by $\text{tail}_d(K)$. All remaining terms are nonnegative, which gives the lower bound. $\square$

5 The closed form: an explicit, computer-assisted theorem

Recall $X = \mu \mathbb{E}T = X_s + \mathbb{E}U$ and the closed form of the computations of Sections 5–6 of the article, in scaled time,

$$\tau_{\text{cf}} := \frac{X \ln(2dX)}{X + 2d - 1}$$

$$\left(\text{i.e. } t_{\text{cf}} = \frac{\tau_{\text{cf}}}{\mu} = \text{MFPT} \cdot \frac{\ln(2dX)}{X + 2d - 1} \right).$$

Theorem 5.1 (accuracy of the closed form; continuous time). *Let $d \in \{2, 3\}$ and let $\tau^* = \mu t^*$ be the scaled mode of the corner-to-corner first-passage density. For every $N \geq N_0$,*

$$-K_d^-(N_0) < X(\tau^* - \tau_{\text{cf}}) < K_d^+(N_0),$$

with the following certified constants:

$d = 2: N_0$	12	15	20	30	50	100	200	1000	10^6
K_2^-	7.653	5.669	4.242	3.023	2.058	1.320	0.767	-0.116	-1.382
K_2^+	10.280	8.774	7.486	6.348	5.449	4.839	4.641	4.641	4.641

$d = 3: N_0$	10	15	20	30	50	100	200	1000	10^6
K_3^-	3.120	1.995	1.564	1.214	1.160	1.068	0.995	0.862	0.828
K_3^+	7.120	4.659	3.663	2.786	2.149	1.740	1.598	1.465	1.427

(A negative K^- means that $\tau^* > \tau_{\text{cf}}$. $K_d^\pm(N_0)$ is the maximum of the block constants over all blocks of Section 5.3 that meet $[N_0, \infty)$.) In particular, with $K_d := \max(K_d^-, K_d^+)$,

$$|t^* - t_{\text{cf}}| \leq \frac{K_d(N_0)}{X} \cdot \frac{1}{\mu},$$

$$\frac{|t^* - t_{\text{cf}}|}{t_{\text{cf}}} \leq \frac{K_d(N_0) \left(1 + \frac{2d-1}{X}\right)}{X \ln(2dX)} \quad (N \geq N_0),$$

so the relative error of the closed form is $O(1/(\ln N \ln \ln N))$ in $d = 2$ and $O(1/(N \ln N))$ in $d = 3$, with the explicit constants above and the explicit X of Corollary 4.7.

The proof occupies the rest of this section. It is computer-assisted: the analytic reduction (Lemma 5.2) and the enclosures of Sections 3–4 turn the sign conditions of Theorem 2.3 into finitely many inequalities between explicit elementary expressions over parameter boxes, which are verified in ball arithmetic (Arb, via python-flint 0.9.0) by `scripts/52_cert_theoremB.py` using `scripts/certlib.py`.

5.1 The scaled sign function

Fix $c \in \mathbb{R}$ and put

$$y' = \frac{1}{X} = \frac{y}{1 + y \mathbb{E}U}, \quad t := \tau_{\text{cf}} + c y', \quad \xi := \frac{\sin^2 x}{y},$$

$$\delta := t - \ln \frac{W}{y}, \quad a := W + 1 - e.$$

Lemma 5.2 (decomposition). *With the notation of Lemmas 3.1–3.3,*

$$\delta = -\ln \cos^2 x + \ln(1 + y \mathbb{E}U) - (2d - 1) y' \tau_{\text{cf}} + c y', \quad e^{-t} = \frac{y}{W} e^{-\delta},$$

$$e^{-(\varepsilon_2 - 2)t} = \frac{y^2}{W^2} e^{-2\delta} e^{4\xi y t}, \quad (5.1)$$

and

$$\frac{\ln \Lambda - \hat{\nu} t}{y} = T_{\text{I}} + T_{\text{II}} + T_{\text{III}} + T_{\text{IV}} + T_{\text{V}}, \quad (5.2)$$

$$T_{\text{I}} := -\frac{\ln(1 + (e - 1)y)}{y} - \frac{\ln(1 + \theta^2 \zeta)}{y} + \frac{\ln(1/b_0)}{y} - \frac{2 \ln(1 - s)}{y},$$

$$T_{\text{II}} := -\frac{\ln m_0 + \ln(1 + y \mathbb{E}U)}{y}, \quad T_{\text{III}} := \frac{\ln \cos^2 x}{y},$$

$$T_{\text{IV}} := y\tau_{\text{cf}} \cdot (2d\xi + q_4), \quad q_4 := -\frac{(2d-1)\mathbb{E}U}{1+y\mathbb{E}U} - \frac{Wa}{1-ay} - \frac{s}{y},$$

$$T_{\text{V}} := -c \frac{1 + \theta - \nu_0}{1 + y \mathbb{E}U}.$$

Proof. $\tau_{\text{cf}}(1 + (2d-1)y') = \ln(2dX) = \ln \frac{2d}{W} + \ln \frac{W}{y} + \ln(1 + y \mathbb{E}U)$ and $2d/W = 1/\cos^2 x$ give the formula for δ ; the other two relations in (5.1) follow from $t = \ln(W/y) + \delta$ and $\varepsilon_2 - 2 = 2 - 4\sin^2 x$, $\sin^2 x \cdot t = \xi y t$. By Lemma 3.3(d), $\ln \Lambda = \ln(W/y) + y T_{\text{I}} - \ln m_0$, and $\hat{\nu}t = t + (\theta - \nu_0)t$, so

$$\ln \Lambda - \hat{\nu}t = yT_{\text{I}} - \ln m_0 - \delta - (\theta - \nu_0)(\tau_{\text{cf}} + cy').$$

Inserting δ : the terms $-\ln m_0 - \ln(1 + y \mathbb{E}U)$ give yT_{II} , $\ln \cos^2 x = yT_{\text{III}}$, the terms proportional to c give $-cy'(1 + \theta - \nu_0) = yT_{\text{V}}$, and the terms proportional to τ_{cf} give $\tau_{\text{cf}}[(2d-1)y' - \theta + \nu_0]$. By Lemma 3.3(a), $\theta = Wy/(1-ay)$, by Lemma 3.1 $\nu_0 = y(1-s)$, and $y' = y/(1+y\mathbb{E}U)$; hence $(2d-1)y' - \theta + \nu_0 = y[(2d-1) - \frac{(2d-1)y\mathbb{E}U}{1+y\mathbb{E}U} - W - \frac{Way}{1-ay} + 1-s] = y[(2d-W) + yq_4]$, and $2d-W = 2d\sin^2 x = 2d\xi y$. $\square$

By Theorem 2.3, $\tau^* < t$ if $S^+(t) < 0$ and $\tau^* > t$ if $S^-(t) > 0$, where

$$S^+(t) := \ln \Lambda - \hat{\nu}t + \ln\left(1 + \frac{\theta}{W}\bar{\kappa} + \frac{\bar{P}_{\text{rest}}(t)e^{\nu_1 t}}{a_1}\right) - \ln(1 + E_-^{\text{lo}}(t)),$$

$$S^-(t) := \ln \Lambda - \hat{\nu}t + \ln\left(1 + \frac{\theta}{W}\kappa_{\text{lo}}(t)\right) - \ln(1 + \bar{E}_-(t)).$$

Using (5.1), all correction terms are expressed through quantities that stay bounded as $y \rightarrow 0$: with $\theta t = \frac{\theta}{y} y t$, $\nu_0 t = (1-s)y t$, $a_1 = y\alpha_1$ where $\alpha_1 := W/[(1+(e-1)y)(1+\theta^2\zeta)]$ (Lemma 3.3),

$$\frac{\bar{P}_{\text{rest}}(t)e^{\nu_1 t}}{a_1 y} = \frac{e^{-\delta+\theta t}}{W\alpha_1} \left[\frac{c_\beta \hat{\beta} + \sum_{i \in \mathcal{N}} \rho_i N_i(t)}{y} + \frac{e^{-2\delta+4\xi y t}}{W^2} C_2 \varepsilon_2 (y\pi_a + y t R_{\text{tot}}) \right], \quad (5.3)$$

$$\frac{E_-^{\text{lo}}(t)}{y} = \frac{e^{-\delta+\nu_0 t}}{(1-s)m_0} \cdot \frac{g_\ell(t)}{y},$$

$$\frac{\ell(t)}{y} = \frac{e^{-\delta}}{W} [A - \gamma_2 e^{-(\varepsilon_2-1)t}],$$

$$\frac{\bar{E}_-(t)}{y} = \frac{e^{-2\delta+\nu_0 t}}{W^2(1-s)m_0} \left[\frac{2C_1}{2-\nu_0} + \frac{C_2 \varepsilon_2 e^{-(\varepsilon_2-2)t}}{\varepsilon_2 - \nu_0} \right], \quad (5.4)$$

and $\kappa_{\text{lo}}(t)$ involves t only through $e^{-(1-\theta)t} = \frac{y}{W} e^{-\delta+\theta t}$ and smaller exponentials. (These are identities; e.g. $e^{-2t} e^{\nu_1 t} = \frac{y}{W} e^{-\delta} e^{\theta t}$.)

5.2 Enclosures used by the certificate

A *block* is a set of integers $\{N_a, \dots, N_b\}$ ($N_a \geq 10$; $N_b = \infty$ allowed). For N in a block:

(E1) $x \in [\frac{\pi}{2N_b}, \frac{\pi}{2N_a}]$; hence enclosures of $A, W, \varepsilon_2, \gamma_2, C_1, C_2, W_2, W_3$ (monotone in x).

(E2) $y \in [1/\bar{X}^{(d)}(N_b), 1/X_{\text{lo}}^{(d)}(N_a)]$ by Propositions 4.5/4.6 and their monotonicity ($y \geq 0$ if $N_b = \infty$).

(E3) $\xi = \sin^2 x \cdot X_s \leq \sin^2 \frac{\pi}{2N_a} \bar{X}^{(d)}(N_b)$ if $N_b < \infty$, and $\xi \leq \frac{\pi^2}{4N_a^2} (2\pi \ln N_a + 1.3)$ resp. $\frac{\pi^2 c_3}{4N_a}$ if $N_b = \infty$ (Corollary 4.7); $\ln \cos^2 x \in [-\xi y / \cos^2 x, 0]$.

(E4) $\sigma \leq Z_d$ (Lemmas 4.1, 4.2), and s, ν_0, b_0 by Lemma 3.1: $1-s \in [1 - \frac{Z_d y^2}{1-y}, 1 - W\nu_0^2]$.

(E5) $\mathbb{E}U \in [\mathbb{E}U_{\text{lo}}, \mathbb{E}U_{\text{hi}}]$ and $\ln m_0/\nu_0 \in [\mathbb{E}U, \ln A + h_d(\nu_0)]$ (Lemma 3.2).

(E6) $T_1 \in [T_1^{\text{lo}}, T_1^{\text{hi}}]$ by Lemma 4.8, with $K = N - 1$ (exact sums) for blocks $\{N\}$, $N \leq 64$, and $K = \min(K_{\text{max}}, N_a - 1)$, $K_{\text{max}} = 24$ ($d = 2$), 48 ($d = 3$) otherwise.

(E7) e and θ : by Lemma 3.3(a),(c), $\theta = Wy/(1 - (W + 1 - e)y)$ is a decreasing function of e , and $e = \frac{\theta}{1+\theta} + (1+\theta)S_\nu$ is bounded below/above by increasing functions $e_{\text{lo}}(\theta)$, $e_{\text{hi}}(\theta)$ of θ . Starting from $\theta \geq 0$, the monotone iteration $e \geq e_{\text{lo}}(\theta_{\text{lo}}) \Rightarrow \theta \leq \bar{\theta}(e_{\text{lo}}) \Rightarrow e \leq e_{\text{hi}}(\bar{\theta}) \Rightarrow \theta \geq \theta_{\text{lo}}(e_{\text{hi}}) \Rightarrow \dots$ produces valid enclosures at every step (six steps are used). Then ζ , α_1 , $b_1 = \theta\alpha_1y/(W(1+\theta))$ by Lemma 3.3(b),(c).

(E8) Near poles: Lemma 3.4 gives $W_i/\theta_i = X_s - (\dots)$ with the right side enclosed by monotone functions of θ_i ; the same monotone iteration (four steps) gives $\theta_i/y \in [\vartheta_i, \bar{\vartheta}_i]$, and $\rho_i/y \leq \bar{\vartheta}_i^2 y/(m_i W_i)$. In $N_i(t)$ we use $\min(t, \frac{1}{\nu_2-2}) \leq \min(t, \frac{1}{\theta_{2,\text{lo}}})$, $\frac{1}{\varepsilon_2-\nu_2} \leq \frac{1}{\varepsilon_2-2-\theta_2}$, and for $d = 3$: $\frac{1}{\nu_3-2} \leq \frac{1}{1+\theta_{3,\text{lo}}}$, $\frac{1}{\varepsilon_2-\nu_3} \leq \frac{1}{\varepsilon_2-3-\theta_3}$.

(E9) $\hat{\beta}/y \leq \frac{(1-s)\nu_0 Z_d}{(1-\nu_0)^2} - \frac{b_1}{y}$ (Lemmas 3.1, 3.5), $\pi_a \leq N_a^{-d}$, $R_{\text{tot}} \leq \frac{d}{4} N_a^{2-d}$.

(E10) Elementary monotonicity: $y \mapsto y \ln(1/y)$ is increasing on $(0, 1/e)$; $(\alpha, y) \mapsto \ln(1 + \alpha y)/y$ is increasing in α and decreasing in $y > 0$ (limit α at $y = 0$); $-\ln(1 - s) \in [s, s/(1 - s)]$; $\ln(1 + z) \in [z - z^2/2, z]$ for $z \geq 0$; $(e^u - 1)/u \in [1, (1 + e^u)/2]$ for $u \geq 0$. In particular $y\tau_{\text{cf}} = [y \ln(2d) + y \ln(1 + y\mathbb{E}U) + y \ln \frac{1}{y}]/(1 + (2d - 1)y')$ and $yt = y\tau_{\text{cf}} + cy y'$ are enclosed without singularity at $y = 0$.

5.3 The certificate

Blocks. $d = 2$: the singletons $\{N\}$, $10 \leq N \leq 64$; then $\{N_a, \dots, \lfloor 1.08N_a \rfloor\}$ up to 10^7 ; then $\{N_a, \dots, \lfloor N_a^{1.3} \rfloor\}$ up to 10^{60} ; then one unbounded block (220 blocks). $d = 3$: singletons for $10 \leq N \leq 200$; ratio 1.06 up to 10^7 ; ratio 1.5 up to 10^9 ; one unbounded block (389 blocks). The blocks partition $\{N \geq 10\}$.

Verification. For each block the script computes the balls (E1)–(E9), searches (scan with step 0.5, then bisection, rounded up to 10^{-3}) for the smallest numbers c_+ and c_- for which it can **verify in ball arithmetic** that

$$\sup \frac{S^+(\tau_{\text{cf}} + c_+ y')}{y} < 0 \quad \text{and} \quad \inf \frac{S^-(\tau_{\text{cf}} - c_- y')}{y} > 0$$

over the whole parameter box, using (5.2)–(5.4) with $T_I, \dots, T_V$ and the correction terms evaluated as balls (for S^+ an upper bound of the correction is used, for S^- a lower bound; for every c the script also checks $t > 0$, and constants c printed with three decimals are entered as balls containing the decimal value). The certificate files `data/52_cert_theoremB_d2.jsonl`, `data/52_cert_theoremB_d3.jsonl` list, for every block, $[y_{\text{lo}}, y_{\text{hi}}]$, the enclosures of e, θ, T_1, ξ , the constants $c_\pm$ and the two Boolean results (all **true**).

Proof of Theorem 5.1. Let $N \geq N_0 \geq 10$ and let B be the block containing N . The true parameter tuple of N lies in the box of B by (E1)–(E9). Hence $S^+(\tau_{\text{cf}} + c_+(B)/X) < 0$ and $S^-(\tau_{\text{cf}} - c_-(B)/X) > 0$, and Theorem 2.3 gives $\tau_{\text{cf}} - c_-(B)/X < \tau^* < \tau_{\text{cf}} + c_+(B)/X$. The table lists $K_d^\pm(N_0) = \max\{c_\pm(B) : B \cap [N_0, \infty) \neq \emptyset\}$, the maximum over all blocks that meet $[N_0, \infty)$ (including the block that contains N_0 when N_0 is not the left end of a block); so every $N \geq N_0$ lies in a block whose constants are at most $K_d^\pm(N_0)$ (`scripts/80_tables.py`, which also records the block containing each N_0). The relative bound follows from $t_{\text{cf}} = \tau_{\text{cf}}/\mu$ and $\tau_{\text{cf}} = \ln(2dX)/(1 + (2d - 1)/X)$. $\square$

Remark 5.3 (what is and is not sharp). (i) Sanity checks: `scripts/54_bruteforce_theorem51.py` (for $d = 2$, $N \in \{12, 16, 24, 36, 48\}$ and $d = 3$, $N \in \{10, 12, 14\}$ the exact g' , from a dense eigen-decomposition, is positive at $\tau_{\text{cf}} - c_-/X$ and negative at $\tau_{\text{cf}} + c_+/X$; likewise for the discrete forward differences with $q = \frac{1}{2}, \frac{1}{4}$), `scripts/51_validate_enclosures.py` (every enclosure contains the float value computed from the full lattice sums; the certified interval contains $X(\tau^* - \tau_{\text{cf}})$ computed from the exact modes of Sections 5–6 of the article, $d = 2$: $N \leq 1810$, $d = 3$: $N \leq 160$) and

scripts/53_check_decomposition.py (identity (5.2) to 10^{-12} ; exact pieces inside the balls). (ii) The observed values of $X(\tau^* - \tau_{\text{cf}})$ are 0.40 ($N = 20$) to 1.99 ($N = 1810$) in $d = 2$ and 0.30 ($N = 10$) to -0.12 ($N = 160$) in $d = 3$; the certified windows are a few units wide at moderate N because the pole data are replaced by enclosures. (iii) **Numerical observation (not proved):** $X(\tau^* - \tau_{\text{cf}})$ converges, with limits $K_2^\infty \approx 3.74$ and $K_3^\infty \approx -0.22$ (the latter extrapolated; the value at $N = 4096$ is -0.214); formally $K_d^\infty = \lim_N [1 - S_1 - 2\mathbb{E}U + W + \int_0^\infty re^s ds - \frac{d-1}{d}]$ with $S_1 = \sum_{\mu_k \geq 2} w_k/(\mu_k(\mu_k - 1))$. The certified asymptotic windows (2.52, 4.65) for $d = 2$ ($N \geq 10^{40}$; block maximum $c_- = -2.529$, $c_+ = 4.641$) and $(-0.83, 1.43)$ for $d = 3$ ($N \geq 10^6$) are consistent with this. The approach to the limit is slow in $d = 2$ (error of order $\ln X/X$).

6 Asymptotic theorems and two-sided bounds (continuous time)

In this section t^* is the mode of the continuous-time first-passage density in real time (jump-attempt rate q ; $q \in (0, 1]$ only rescales time here), $\tau^* = \mu t^*$, and

$$\frac{1}{\mu} = \frac{d}{2q \sin^2 x} = \frac{2dN^2}{\pi^2 q} \left(\frac{x}{\sin x} \right)^2, \\ 1 \leq \left(\frac{x}{\sin x} \right)^2 \leq 1 + \frac{0.83}{N^2} \quad (N \geq 20). \quad (6.1)$$

(The last bound: $\sin x \geq x(1 - x^2/6)$, and $(1 - u)^{-2} \leq 1 + 2.01u$ for $0 \leq u \leq 1.1 \cdot 10^{-3}$ (indeed $(1 - u)^{-2} - 1 - 2u = u^2 \frac{3-2u}{(1-u)^2} \leq 3.01u^2$), applied with $u = x^2/6 = \pi^2/(24N^2) \leq 1.03 \cdot 10^{-3}$; then $2.01 \cdot \frac{\pi^2}{24} = 0.8266 \dots \leq 0.83$.)

Theorem 6.1 ($d = 2$). *For every $N \geq 20$,*

$$\ln \ln N + \ln(8\pi) - \frac{4.6 + 3 \ln(8\pi \ln N + 2.4)}{2\pi \ln N - 0.24} \leq \tau^* < \ln \ln N + \ln(8\pi),$$

and consequently

$$t^* = \frac{4N^2}{\pi^2 q} \left[\ln \ln N + \ln(8\pi) + R_N \right] (1 + \rho_N), \\ -\frac{4.6 + 3 \ln(8\pi \ln N + 2.4)}{2\pi \ln N - 0.24} \leq R_N < 0, \quad 0 \leq \rho_N \leq \frac{0.83}{N^2}.$$

In particular $t^ = \frac{4}{\pi^2 q} N^2 \ln \ln N (1 + o(1))$ with the explicit rate*

$$\frac{t^*}{\frac{4}{\pi^2 q} N^2 \ln \ln N} - 1 = \frac{\ln(8\pi)}{\ln \ln N} + O\left(\frac{1}{\ln N}\right) \in \left[\frac{\ln(8\pi) + R_N^{\text{lo}}}{\ln \ln N}, \frac{\ln 8\pi}{\ln \ln N} \left(1 + \frac{0.83}{N^2}\right) + \frac{0.83}{N^2} \right],$$

where R_N^{lo} is the lower bound for R_N above. (The leading correction $\ln(8\pi)/\ln \ln N = 3.224/\ln \ln N$ is positive and decays extremely slowly; the two-term form is the practically relevant statement.)

Proof. By Theorem 5.1 ($N_0 = 20$), $\tau^* = \tau_{\text{cf}} + \vartheta/X$ with $-4.242 < \vartheta < 7.486$, and $\tau_{\text{cf}} = \ln(4X) - \frac{3 \ln(4X)}{X+3}$. By Corollary 4.7, $L - 0.24 \leq X \leq L + 0.60$ with $L := 2\pi \ln N \geq 18.82$. Hence $\ln(4X) - \ln(8\pi \ln N) = \ln \frac{X}{L} \in \left[-\frac{0.24}{L-0.24}, \frac{0.60}{L}\right]$.

Upper bound. $\tau^* - \ln(8\pi \ln N) \leq \frac{0.60}{L} + \frac{7.486}{X} - \frac{3 \ln(4X)}{X+3}$. Since $X \geq 18.58$: $\ln(4X) \geq 4.30$, $\frac{X}{X+3} \geq 0.86$, so $\frac{3 \ln(4X)}{X+3} \geq \frac{11.09}{X}$, while $\frac{0.60}{L} \leq \frac{0.62}{X}$ (as $X \leq L + 0.6 \leq 1.032L$). Thus the right side is $\leq (0.62 + 7.486 - 11.09)/X < 0$.

Lower bound. $\tau^* - \ln(8\pi \ln N) \geq -\frac{0.24}{L-0.24} - \frac{3 \ln(4X)}{X+3} - \frac{4.242}{X} \geq -\frac{0.24+4.242+3 \ln(4L+2.4)}{L-0.24}$, using $X \geq L - 0.24$, $X + 3 \geq L - 0.24$ and $\ln(4X) \leq \ln(4L + 2.4)$.

The statements for t^* follow from (6.1) with $d = 2$. $\square$

Theorem 6.2 ($d = 3$). For every $N \geq 20$, with $c_3 = \frac{\pi^2}{2} \int_0^\infty (e^{-z}(I_0(z) + I_1(z)))^3 dz \in [7.106860, 7.106880]$,

$$\ln N + \ln(6c_3) - \frac{9.2 + 5 \ln(6c_3 N)}{c_3 N - 7.63} \leq \tau^* < \ln N + \ln(6c_3),$$

$$\ln(6c_3) \in [3.752820, 3.752823],$$

and consequently

$$t^* = \frac{6N^2}{\pi^2 q} \left[\ln N + \ln(6c_3) + R_N \right] (1 + \rho_N),$$

$$-\frac{9.2 + 5 \ln(6c_3 N)}{c_3 N - 7.63} \leq R_N < 0, \quad 0 \leq \rho_N \leq \frac{0.83}{N^2}.$$

In particular $t^* = \frac{6}{\pi^2 q} N^2 \ln N (1 + o(1))$ with $\frac{t^*}{\frac{6}{\pi^2 q} N^2 \ln N} - 1 = \frac{\ln(6c_3)}{\ln N} + O(\frac{1}{N})$; this is the law of the computations of Sections 5–6 of the article, “ $(6/(\pi^2 q)) N^2 (\ln N + 3.753)$ ” with a proven remainder $O(N \ln N)$ in t^* .

Proof. By Theorem 5.1 ($N_0 = 20$), $\tau^* = \tau_{\text{cf}} + \vartheta/X$, $-1.564 < \vartheta < 3.663$, $\tau_{\text{cf}} = \ln(6X) - \frac{5 \ln(6X)}{X+5}$; by Corollary 4.7, $c_3 N - 7.63 \leq X \leq c_3 N - 5.85$, so $\ln(6X) - \ln(6c_3 N) \in [-\frac{7.63}{c_3 N - 7.63}, -\frac{5.85}{c_3 N}]$. Upper bound: $\tau^* - \ln(6c_3 N) \leq -\frac{5 \ln(6X)}{X+5} + \frac{3.663}{X} < 0$ because $X \geq 134$ gives $5 \ln(6X) \frac{X}{X+5} \geq 32$. Lower bound: $\tau^* - \ln(6c_3 N) \geq -\frac{7.63}{c_3 N - 7.63} - \frac{5 \ln(6c_3 N)}{c_3 N - 7.63} - \frac{1.564}{c_3 N - 7.63}$. Then use (6.1) with $d = 3$. $\square$

Theorem 6.3 (two-sided bounds with explicit constants). For all $N \geq 20$:

$$d = 2 :$$

$$\frac{4}{\pi^2 q} N^2 (\ln \ln N + 2.27) \leq t^* \leq \frac{4}{\pi^2 q} N^2 (\ln \ln N + 3.2242) \left(1 + \frac{0.83}{N^2}\right),$$

$$\frac{0.405}{q} N^2 \ln \ln N \leq t^* \leq \frac{1.60}{q} N^2 \ln \ln N;$$

$$d = 3 :$$

$$\frac{6}{\pi^2 q} N^2 (\ln N + 3.43) \leq t^* \leq \frac{6}{\pi^2 q} N^2 (\ln N + 3.7529) \left(1 + \frac{0.83}{N^2}\right),$$

$$\frac{0.607}{q} N^2 \ln N \leq t^* \leq \frac{1.38}{q} N^2 \ln N.$$

Proof. The error terms $E_2(N) := \frac{4.6 + 3 \ln(8\pi \ln N + 2.4)}{2\pi \ln N - 0.24}$ and $E_3(N) := \frac{9.2 + 5 \ln(6c_3 N)}{c_3 N - 7.63}$ are decreasing in $N \geq 20$, with $E_2(20) \leq 0.951$ and $E_3(20) \leq 0.320$; $\ln(8\pi) \in [3.2241, 3.2242]$. This gives the refined bounds from Theorems 6.1 and 6.2. For the crude ones: $\tau^* \geq \ln \ln N$ resp. $\ln N$ gives the lower constants $4/\pi^2 \geq 0.405$, $6/\pi^2 \geq 0.607$; and $(1 + \frac{3.2242}{\ln \ln 20})(1 + \frac{0.83}{400}) \leq 3.947$, $\frac{4}{\pi^2} \cdot 3.947 \leq 1.60$; $(1 + \frac{3.7529}{\ln 20})(1 + \frac{0.83}{400}) \leq 2.258$, $\frac{6}{\pi^2} \cdot 2.258 \leq 1.38$ (the ratios $(\ln \ln N + c)/\ln \ln N$ and $(\ln N + c)/\ln N$ are decreasing). $\square$

Corollary 6.4 (mode versus mean). For $N \geq 20$, $\frac{t^*}{\mathbb{E}T} = \frac{\tau^*}{X} = \frac{\ln(2dX)}{X + 2d - 1} + \frac{\vartheta_N}{X^2}$ with ϑ_N as in Theorem 5.1; in particular $t^*/\mathbb{E}T \rightarrow 0$, like $\frac{\ln \ln N}{2\pi \ln N}$ in $d = 2$ and like $\frac{\ln N}{c_3 N}$ in $d = 3$.

Proof. Theorem 5.1 and $t^*/\mathbb{E}T = \tau^*/X$. $\square$

Remark 6.5. `scripts/60_check_asymptotics.py` compares Theorems 6.1–6.3, Corollary 4.7 and Theorem 5.1 with the exact continuous-time modes and means of the computations of Sections 5–6 of the article (`data/msc_modes/`) ($d = 2$: 38 values of N from 20 to $1.7 \cdot 10^7$; $d = 3$: 21 values from 20 to 4096): all inequalities hold. Observed: $X - 2\pi \ln N \rightarrow 0.3816$ ($d = 2$), $X - c_3 N \approx -6.4$ ($d = 3$), $R_N \approx -0.55$ at $N = 20$ and -0.14 at $N = 1.7 \cdot 10^7$ ($d = 2$); $R_N \approx -0.29$ at $N = 20$ and -0.002 at $N = 4096$ ($d = 3$).

7 Discrete time

In discrete time $\mathbb{P}(T = t) = \mu g(t-1)$ with $g(t) := \sum_j a_j(1 - \mu\nu_j)^t$, $a_j = \rho_j \hat{\varphi}(-\nu_j)$, $t \in \mathbb{N}_0$ (Proposition 1.5), and $\mu \mathbb{E}T = X$ as in continuous time. We use the following calculus on sequences $f : \mathbb{N}_0 \rightarrow \mathbb{R}$ (for rates $k \geq 0$ with $\mu k \leq 1$):

$$E_k(t) := (1 - \mu k)^t, \quad (Df)(t) := \frac{f(t+1) - f(t)}{\mu},$$

$$(f \star E_\nu)(t) := \mu \sum_{s=0}^{t-1} f(s) E_\nu(t-1-s).$$

Then $DE_k = -kE_k$; $D(f \star E_\nu) = f - \nu(f \star E_\nu)$; $(Df) \star E_\nu = f - \nu(f \star E_\nu) - f(0)E_\nu$; $E_a \star E_\nu = (E_a - E_\nu)/(\nu - a)$ for $a \neq \nu$; and for $a > \nu \geq 0$, $\mu a < 1$: $\mu \sum_{s \geq t} E_a(s)/E_\nu(s+1) = \frac{E_a(t)/E_\nu(t)}{a-\nu}$ (geometric sums). With $u_t := N^d P^t(x_0, a) = 1 + \sum_k c_k E_{\mu_k}(t)$ define

$$\varphi := Du = - \sum_k c_k \mu_k E_{\mu_k}, \quad r := WE_1 - \varphi = \sum_{\mu_k \geq 2} c_k \mu_k E_{\mu_k},$$

$$r_1 := -Dr = \sum_{\mu_k \geq 2} c_k \mu_k^2 E_{\mu_k}, \quad \tilde{J} := \sum_{\mu_k \geq 2} \frac{c_k \mu_k}{\mu_k - \nu_0} E_{\mu_k}.$$

7.1 The identity (all $q \in (0, 1]$)

Proposition 7.1 (discrete identity). *Let $d \in \{2, 3\}$, $N \geq 3$ ($N \geq 4$ if $d = 3$), $q \in (0, 1]$. Then $g = \pi_a \varphi + \sum_j \rho_j (\varphi \star E_{\nu_j})$ and, for all $t \in \mathbb{N}_0$,*

$$(Dg)(t) = W \sum_{j \geq 1} \frac{\rho_j}{\nu_j - 1} E_{\nu_j}(t) - \rho_0 \nu_0 m_0 E_{\nu_0}(t) - \rho_0 \nu_0 \tilde{J}(t) - \rho_0 r(t) + \pi_a r_1(t) + \sum_{j \geq 1} \rho_j (r_1 \star E_{\nu_j})(t).$$

Proof. Coefficientwise. By $E_a \star E_\nu = (E_a - E_\nu)/(\nu - a)$ and the sum rules of Lemma 1.4(c), $\sum_j \rho_j (\varphi \star E_{\nu_j}) = \sum_k (-c_k \mu_k) E_{\mu_k} \sum_j \frac{\rho_j}{\nu_j - \mu_k} + \sum_j \rho_j E_{\nu_j} \sum_k \frac{-c_k \mu_k}{\mu_k - \nu_j} = -\pi_a \varphi + \sum_j \rho_j \hat{\varphi}(-\nu_j) E_{\nu_j}$, which is the first claim. Next $\varphi \star E_{\nu_0} = m_0 E_{\nu_0} - \frac{W}{1-\nu_0} E_1 + \tilde{J}$ (the level $\mu_k = 1$ has $\sum c_k = -W$), so $D(\varphi \star E_{\nu_0}) = \varphi - \nu_0(\varphi \star E_{\nu_0}) = \frac{W}{1-\nu_0} E_1 - r - \nu_0 m_0 E_{\nu_0} - \nu_0 \tilde{J}$. For $j \geq 1$, since $\varphi(0) = u_1/\mu = 0$, $D(\varphi \star E_{\nu_j}) = (D\varphi) \star E_{\nu_j} = (-WE_1 + r_1) \star E_{\nu_j} = -\frac{W}{\nu_j - 1} (E_1 - E_{\nu_j}) + r_1 \star E_{\nu_j}$. The total coefficient of WE_1 is $\frac{\rho_0}{1-\nu_0} - \sum_{j \geq 1} \frac{\rho_j}{\nu_j - 1} = \pi_a$, and $\pi_a D\varphi + \pi_a WE_1 = \pi_a r_1$. $\square$

7.2 Positivity and bounds for $q \leq \frac{1}{2}$

From now on $q \in (0, \frac{1}{2}]$. Then $\mu \mu_k \leq 2q \cdot \max_k \frac{1}{d} \sum_i \sin^2(k_i x) < 1$ and $\mu \nu_j < 1$ for all k, j , so all E 's are nonnegative and nonincreasing in t . Put $p_k := q a_k = d\mu \varepsilon_k \in (0, 1)$, $\beta_k := 1 - p_k \in (0, 1)$, and let $v_n := N(I - qA)^n(1, N) = \sum_k (-1)^k \gamma_k \beta_k^n$ (one-dimensional lazy chain).

Lemma 7.2 (one-dimensional structure). *$v_n - v_{n-1} = \mathbb{P}(H^D = n)$ for $n \geq 1$, where $H^D = G_1 + \dots + G_{N-1}$ with independent G_k geometric on $\{1, 2, \dots\}$ with success probability p_k . With $\psi(n) := (v_{n+1} - v_n)/p_1$ and π_2 the probability mass function of $G_2 + \dots + G_{N-1}$:*

$$\psi(n+1) - \beta_1 \psi(n) = \pi_2(n+1) \in [0, p_2 \gamma_2 (\varepsilon_2 - 1) \beta_2^n],$$

$$A\beta_1^n - \gamma_2 \varepsilon_2 \beta_2^n \leq \psi(n) \leq A\beta_1^n,$$

$$A\beta_1^n - \gamma_2 \beta_2^n \leq 1 - v_n \leq A\beta_1^n.$$

Moreover the sequences $(\psi(n))_{n \geq 0}$ and $(v_n)_{n \geq 0}$ are nonnegative, log-concave, with supports $\{n \geq N-2\}$ and $\{n \geq N-1\}$.

Proof. Lemma 1.7 says $\sum_k \frac{(-1)^k \gamma_k}{s + \varepsilon_k} = \frac{1}{s} \prod_{k \geq 1} \frac{\varepsilon_k}{s + \varepsilon_k}$ as rational functions. Substituting $s = \frac{1-z}{d\mu z}$ and multiplying by $\frac{1}{d\mu z}$: $\sum_n v_n z^n = \sum_k \frac{(-1)^k \gamma_k}{1 - \beta_k z} = \frac{1}{1-z} \prod_{k \geq 1} \frac{p_k z}{1 - \beta_k z}$, i.e. $(1-z) \sum_n v_n z^n = \mathbb{E} z^{H^D}$. Conditioning on G_1 : $p_1 \psi(n) = \mathbb{P}(H^D = n+1) = \sum_{m=1}^n p_1 \beta_1^{n-m} \pi_2(m)$, so $\psi(n) \beta_1^{-n} = \sum_{m \leq n} \beta_1^{-m} \pi_2(m)$ is nondecreasing with limit $\mathbb{E} \beta_1^{-(G_2 + \dots)} = \prod_{k \geq 2} \frac{p_k}{\beta_1 - \beta_k} = \prod_{k \geq 2} \frac{\varepsilon_k}{\varepsilon_k - 1} = A$ (Lemma 1.7; $\beta_1 - \beta_k = d\mu(\varepsilon_k - 1)$), and $\psi(n+1) - \beta_1 \psi(n) = \pi_2(n+1)$. In the same way $\pi_2(m) \beta_2^{-(m-1)} \uparrow p_2 \prod_{k \geq 3} \frac{\varepsilon_k}{\varepsilon_k - \varepsilon_2} = p_2 \gamma_2(\varepsilon_2 - 1)$. Then $A \beta_1^n - \psi(n) = \beta_1^n \sum_{m > n} \beta_1^{-m} \pi_2(m) \leq p_2 \gamma_2(\varepsilon_2 - 1) \frac{\beta_2^n}{\beta_1 - \beta_2} = \gamma_2 \varepsilon_2 \beta_2^n$, and summing $1 - v_n = p_1 \sum_{m \geq n} \psi(m)$ gives the last bounds.

Log-concavity: if $(\Pi(m))_{m \geq m_0}$ is positive and log-concave then so are its partial sums $F(n) = \sum_{m \leq n} \Pi(m)$: $F(n)^2 \geq F(n-1)F(n+1) \iff F(n)\Pi(n) \geq F(n-1)\Pi(n+1)$, and $\Pi(m)\Pi(n+1) \leq \Pi(m+1)\Pi(n)$ for $m \leq n-1$, so $F(n-1)\Pi(n+1) \leq \Pi(n) \sum_{m \leq n-1} \Pi(m+1) \leq \Pi(n)F(n)$. The mass function of $G_j + \dots + G_{N-1}$ is $p_j \beta_j^{n-1} \sum_{m \leq n-1} \beta_j^{-m} \pi_{j+1}(m)$, so by induction on j (starting from a geometric law) all these mass functions, in particular ψ , are log-concave on their supports, and so is v (partial sums of $p_1 \psi$). $\square$

Lemma 7.3 (multinomial representation; bounds). *Let $(n_1, \dots, n_d)$ be multinomial with t trials and equal cell probabilities. Then $u_t = \mathbb{E} \prod_i v_{n_i}$, $\varphi(t) = \mathbb{E} \sum_i \psi(n_i) \prod_{l \neq i} v_{n_l}$, and for numbers $b_1, \dots, b_d$: $\mathbb{E} \prod_i b_i^{n_i} = (\frac{1}{d} \sum_i b_i)^t$. Consequently, for $d \geq 2$ and all $t \in \mathbb{N}_0$, with $\tilde{q} := D\varphi + \varphi$:*

$$\begin{aligned} 0 \leq \varphi &\leq W E_1, & 0 \leq \tilde{q} &\leq C_1 E_2 + C_2(\varepsilon_2 - 1) E_{\varepsilon_2}, \\ W \mathbf{g} &\leq r \leq \bar{r} := C_1 E_2 + C_2 E_{\varepsilon_2}, \\ r &\leq r_1 = r + \tilde{q} \leq \bar{r}_1 := 2C_1 E_2 + C_2 \varepsilon_2 E_{\varepsilon_2}, \end{aligned}$$

where $\mathbf{g} := A E_2 - \gamma_2 E_{1+\varepsilon_2}$ (any $d \geq 2$) or, if $d = 3$, $\mathbf{g} := 2A E_2 - 2\gamma_2 E_{1+\varepsilon_2} - A^2 E_3 + 2A\gamma_2 E_{2+\varepsilon_2} - \gamma_2^2 E_{1+2\varepsilon_2}$. (These are the bounds of Lemma 1.9 with $e^{-k\tau}$ replaced by $E_k(t)$.)

Proof. $P = \frac{1}{d} \sum_i (I - q A_i)$, so one step of the walk chooses a coordinate uniformly and moves it by the one-dimensional lazy chain; hence $P^t(x_0, a) = \mathbb{E} \prod_i N^{-1} v_{n_i}$. Adding one trial in a uniform cell I : $u_{t+1} = \mathbb{E} \frac{1}{d} \sum_i v_{n_i+1} \prod_{l \neq i} v_{n_l}$, which gives the formula for $\varphi = (u_{t+1} - u_t)/\mu$ because $p_1 = d\mu$. The moment formula is the multinomial theorem; in particular $\mathbb{E} \beta_1^{n_i} = E_1$, $\mathbb{E} \beta_2^{n_i} = E_{\varepsilon_2}$, $\mathbb{E} \beta_1^{n_i + n_l} = E_2$, $\mathbb{E} \beta_1^{n_i} \beta_2^{n_l} = E_{1+\varepsilon_2}$, $\mathbb{E} \beta_1^{n_i + n_l + n_m} = E_3$, etc. (i, l, m distinct).

$\varphi \leq \mathbb{E} \sum_i A \beta_1^{n_i} = W E_1$. For $\tilde{q}$: writing $1 - \mu = \frac{1}{d}(\beta_1 + d - 1)$ and adding one trial,

$$\mu \tilde{q}(t) = \varphi(t+1) - (1-\mu)\varphi(t) = \mathbb{E} \frac{1}{d} \sum_i \left[(\psi(n_i+1) - \beta_1 \psi(n_i)) \prod_{l \neq i} v_{n_l} + \sum_{m \neq i} \psi(n_i) (v_{n_m+1} - v_{n_m}) \prod_{l \neq i, m} v_{n_l} \right],$$

which is ≥ 0 and, by Lemma 7.2, $\leq \mathbb{E} \frac{1}{d} \sum_i [p_2 \gamma_2(\varepsilon_2 - 1) \beta_2^{n_i} + \sum_{m \neq i} A \beta_1^{n_i} p_1 A \beta_1^{n_m}] = \mu [C_2(\varepsilon_2 - 1) E_{\varepsilon_2} + C_1 E_2]$. Next $r = W E_1 - \varphi = \mathbb{E} \sum_i (A \beta_1^{n_i} - \psi(n_i)) \prod_{l \neq i} v_{n_l} + \mathbb{E} \sum_i A \beta_1^{n_i} (1 - \prod_{l \neq i} v_{n_l})$. Upper bound: first term $\leq \mathbb{E} \sum_i \gamma_2 \varepsilon_2 \beta_2^{n_i} = C_2 E_{\varepsilon_2}$, second $\leq \mathbb{E} \sum_i A \beta_1^{n_i} \sum_{l \neq i} A \beta_1^{n_l} = C_1 E_2$. Lower bound: the first term is ≥ 0 ; with $\ell_n := A \beta_1^n - \gamma_2 \beta_2^n \in [0, 1 - v_n]$ we have $1 - \prod_{l \neq i} v_{n_l} \geq \ell_{n_i}$ for one $l \neq i$, and for $d = 3$, $1 - v_{n_l} v_{n_m} \geq \ell_{n_l} + \ell_{n_m} - \ell_{n_l} \ell_{n_m}$; taking expectations gives $W \mathbf{g}$. Finally $r_1 = -Dr = W E_1 + D\varphi = r + \tilde{q}$. $\square$

7.3 Unimodality for $q \leq \frac{1}{2}$

Theorem 7.4 (unimodality, discrete time, $q \leq \frac{1}{2}$). *Let $d \geq 2$, $N \geq 3$, $q \in (0, \frac{1}{2}]$. The sequence φ is log-concave with support $\{t \geq t_0\}$, $t_0 = d(N-1) - 1$; the function $B_D(t) := \mu(Dg)(t)/\varphi(t)$ is nonincreasing on $\{t \geq t_0\}$ and $g(t) = 0$ for $t < t_0$. Consequently $t \mapsto \mathbb{P}(T = t)$ is unimodal, and for integers $t \geq t_0$:*

$$\begin{aligned} (Dg)(t) < 0 &\implies \text{every maximiser } t^* \text{ of } \mathbb{P}(T = \cdot) \text{ satisfies } t^* \leq t + 1; \\ (Dg)(t) > 0 &\implies t^* \geq t + 2. \end{aligned}$$

Proof. By symmetry and Lemma 7.3, $\varphi(t) = d \cdot d^{-t} \sum_{n_1+\dots+n_d=t} \frac{t!}{n_1! \dots n_d!} \psi(n_1) v_{n_2} \dots v_{n_d}$, i.e.

$$d^{t-1} \varphi(t) = t! (\alpha * \beta^{*(d-1)})(t),$$

$$\alpha(n) := \frac{\psi(n)}{n!}, \quad \beta(n) := \frac{v_n}{n!},$$

where $*$ is the ordinary convolution of sequences on $\mathbb{N}_0$. The sequences α, β are nonnegative, not identically zero and summable (ψ and v are bounded), and $n! \alpha(n) = \psi(n)$, $n! \beta(n) = v_n$ are log-concave with supports that are half-lines (Lemma 7.2). A nonnegative summable sequence a with contiguous support for which $n \mapsto n! a(n)$ is log-concave is called *ultra log-concave* (of order ∞); equivalently $a(n)/\pi_\lambda(n)$ is log-concave for a Poisson mass function π_λ , and multiplying a by a positive constant (e.g. normalising it to a probability sequence) does not change the property. **The convolution of two ultra log-concave sequences is ultra log-concave:** D. W. Walkup, *Pólya sequences, binomial convolution and the union of random sets*, J. Appl. Probab. 13 (1976), 76–85, Theorem 1; T. M. Liggett, *Ultra logconcave sequences and negative dependence*, J. Combin. Theory Ser. A 79 (1997), 315–325, Theorem 2 (order ∞); a short proof is given by A. Marsiglietti and J. Melbourne, *Geometric and functional inequalities for log-concave probability sequences*, Discrete Comput. Geom. 71 (2024), 556–586, Corollary 3.4 (numbering of arXiv:2004.12005v4), whose definitions (Definition 2.1 and Example 2.5 there) are the ones used here. Applying this $d-1$ times, $\alpha * \beta^{*(d-1)}$ is ultra log-concave, i.e. $t \mapsto d^{t-1} \varphi(t)$ is log-concave with contiguous support; hence φ is log-concave, with support $\{t \geq (N-2) + (d-1)(N-1)\}$.

With $h(n) := \sum_j \rho_j E_{\nu_j}(n)$ (nonincreasing, as all $1 - \mu \nu_j \in (0, 1)$) and $\eta(n) := h(n-1) - h(n) \geq 0$,

$$g(t+1) - g(t) = \pi_a(\varphi(t+1) - \varphi(t)) + \mu \varphi(t) h(0) - \mu \sum_{n=1}^t \varphi(t-n) \eta(n).$$

For $t \geq t_0$ divide by $\varphi(t) > 0$: $\varphi(t+1)/\varphi(t)$ is nonincreasing in t and each $\varphi(t-n)/\varphi(t)$ ($= 0$ if $t-n < t_0$) is nondecreasing in t , so B_D is nonincreasing. For $t < t_0$, $g(t) = 0$. Therefore $\{t \geq t_0 : (Dg)(t) > 0\}$ is an initial segment and $\{(Dg)(t) < 0\}$ a final segment of $\{t \geq t_0\}$, g is nondecreasing on the first and strictly decreasing on the second, and the maximisers of g are $\geq t+1$ if $(Dg)(t) > 0$ and $\leq t$ if $(Dg)(t) < 0$. Finally $\mathbb{P}(T=t) = \mu g(t-1)$. $\square$

Remark 7.5. (i) The exact integer-arithmetic check `scripts/71_check_discrete_unimodal.py` confirms log-concavity of φ and the single sign change of $\mathbb{P}(T=t+1) - \mathbb{P}(T=t)$ for ten small cases ($q \in \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}\}$). (ii) On the citation: the theorem numbers for Walkup's and Liggett's papers are those given by O. Johnson, *Log-concavity and the maximum entropy property of the Poisson distribution*, Stochastic Process. Appl. 117 (2007), 791–802 (proof of Lemma 3.1: "Theorem 1 of Walkup" and "Theorem 2 of Liggett" for the closure of ultra log-concave laws under convolution); the two original papers were not consulted directly. The statement used, and its proof, can be checked in the openly available paper of Marsiglietti and Melbourne cited in the proof. (iii) Theorem 7.14 (with $q_{\max} = 0.8$) and Theorem 7.17 below cover $q \leq \frac{1}{2}$ without Theorem 7.4 for $N \geq 37$ ($d=2$) and $N \geq 12$ ($d=3$); Theorem 7.4 is needed only for the remaining range $11 \leq N \leq 36$ ($d=2$), $N \in \{10, 11\}$ ($d=3$) of Theorem 7.8 and for its better constants.

7.4 Window theorem and explicit bounds for $q \leq \frac{1}{2}$

Define $\kappa(t) := (r_1 \star E_{\nu_1})(t)/E_{\nu_1}(t)$, $E_-(t) := [\nu_0 \tilde{J}(t) + r(t)]/(\nu_0 m_0 E_{\nu_0}(t))$ and $P_{\text{rest}} := W \sum_{j \geq 2} \frac{\rho_j}{\nu_j - 1} E_{\nu_j} + \pi_a r_1 + \sum_{j \geq 2} \rho_j (r_1 \star E_{\nu_j})$, so that Proposition 7.1 reads $Dg = a_1 E_{\nu_1} [1 + \frac{\theta}{W} \kappa] - a_0 E_{\nu_0} [1 + E_-] + P_{\text{rest}}$. Put $\tau := \mu t$.

Lemma 7.6 (corrections, $q \leq \frac{1}{2}$). *For $t \geq 1$, with $\varrho_a := \frac{1-\mu a}{1-\mu \nu_1}$, $\mathbf{g} = \sum_i c_i E_{a_i}$ as in Lemma 7.3, and any $s_x = \mu S$, $S \in \mathbb{N}_{\geq 1}$:*

$$\begin{aligned}
(a) \quad \kappa_{\text{lo}}^D(t) &:= W - \frac{\bar{r}(t)}{E_{\nu_1}(t)} + \nu_1 W \sum_i c_i \frac{\varrho_{a_i} - \varrho_{a_i}^t}{a_i - \nu_1} \leq \kappa(t) \leq \bar{\kappa}_D := \frac{W}{1 - \mu\nu_1} + \\
&\nu_1 \left[W \frac{e^{\theta s_x/(1-\mu\nu_1)} - 1}{\theta} + \frac{C_1 e^{-(1-\theta)s_x}}{1-\theta} + \frac{C_2 e^{-(\varepsilon_2-1-\theta)s_x}}{\varepsilon_2-1-\theta} \right]; \\
(b) \quad \frac{W \mathbf{g}(t)}{\nu_0 m_0 E_{\nu_0}(t)} &\leq E_-(t) \leq \frac{1}{\nu_0 m_0 E_{\nu_0}(t)} \left[\frac{2C_1 E_2(t)}{2-\nu_0} + \frac{C_2 \varepsilon_2 E_{\varepsilon_2}(t)}{\varepsilon_2-\nu_0} \right]; \\
(c) \quad 0 &\leq P_{\text{rest}}(t) \leq E_2(t) \left[c_\beta \hat{\beta} + \sum_{i \in \mathcal{N}} \rho_i N_i^D(t) \right] + E_{\varepsilon_2}(t) C_2 \varepsilon_2 \left[\pi_a + \frac{\tau R_{\text{tot}}}{1 - \varepsilon_2 \mu} \right], \quad N_i^D := \\
&2C_1 \min\left(\frac{\tau}{1-2\mu}, \frac{1}{\nu_i-2}\right) + C_2 \varepsilon_2 \min\left(\frac{\tau}{1-\nu_i\mu}, \frac{1}{\varepsilon_2-\nu_i}\right); \\
(d) \quad \text{for } 0 \leq b < a \text{ with } a\mu < 1: &e^{-(a-b)\tau/(1-a\mu)} \leq E_a(t)/E_b(t) \leq e^{-(a-b)\tau}; \text{ and } \hat{\nu}_D := \frac{1}{\mu} \ln \frac{1-\mu\nu_0}{1-\mu\nu_1} \\
&\text{satisfies } \hat{\nu}_D - \hat{\nu} \in \mu \left[\frac{\nu_1^2}{2} - \frac{\nu_0^2}{2(1-\mu\nu_0)}, \frac{\nu_1^2}{2(1-\mu\nu_1)} - \frac{\nu_0^2}{2} \right].
\end{aligned}$$

Proof. (a) Summation by parts with $\mu r_1(s) = r(s) - r(s+1)$, $r(0) = W$ and $b := 1 - \mu\nu_1$: $\kappa(t) = \sum_{s < t} \frac{r(s) - r(s+1)}{b^{s+1}} = \frac{W}{b} - \frac{r(t)}{E_{\nu_1}(t)} + \mu\nu_1 \sum_{s=1}^{t-1} \frac{r(s)}{E_{\nu_1}(s+1)}$. Lower bound: $W/b \geq W$, $r(t) \leq \bar{r}(t)$, $r(s) \geq W \mathbf{g}(s)$ and $\mu \sum_{s=1}^{t-1} E_a(s)/E_{\nu_1}(s+1) = (\varrho_a - \varrho_a^t)/(a - \nu_1)$. Upper bound: κ is nondecreasing with limit $\frac{W}{b} + \mu\nu_1 \sum_{s \geq 1} r(s)/E_{\nu_1}(s+1)$; use $r \leq W E_1$ for $1 \leq s < S$: $\mu \sum_{s=1}^{S-1} E_1(s)/E_{\nu_1}(s+1) = (\varrho_1^S - \varrho_1)/\theta \leq (e^{\theta s_x/b} - 1)/\theta$ since $\varrho_1 = 1 + \mu\theta/b$; and $r \leq \bar{r}$ for $s \geq S$: $\mu \sum_{s \geq S} E_2(s)/E_{\nu_1}(s+1) = \varrho_2^S/(1-\theta) \leq e^{-(1-\theta)s_x}/(1-\theta)$, similarly for E_{ε_2} .

(b) $\tilde{J}(t) = \mu \sum_{s \geq t} r(s) E_{\nu_0}(t)/E_{\nu_0}(s+1) \in [0, \frac{C_1 E_2}{2-\nu_0} + \frac{C_2 E_{\varepsilon_2}}{\varepsilon_2-\nu_0}]$ by the geometric sum formula and $0 \leq r \leq \bar{r}$; add r/ν_0 as in Lemma 2.2(b).

(c) As in Lemma 2.2(c), using $E_{\nu_j} \leq E_2$ for $\nu_j \geq 2$ and, for $a < \nu$, $E_a \star E_\nu = E_a \frac{1-(E_\nu/E_a)}{\nu-a} \leq E_a \min(\frac{1}{\nu-a}, \frac{\tau}{1-\mu a})$ because $1 - \varrho^t \leq t(1 - \varrho)$ with $1 - \varrho = \frac{\mu(\nu-a)}{1-\mu a}$.

(d) $E_a/E_b = (1-z)^t$ with $z = \frac{\mu(a-b)}{1-b\mu}$ and $-\frac{z}{1-z} \leq \ln(1-z) \leq -z$, $\frac{z}{1-z} = \frac{\mu(a-b)}{1-a\mu}$. For $\hat{\nu}_D = \nu_1 \ell(\mu\nu_1) - \nu_0 \ell(\mu\nu_0)$ with $\ell(z) = -\ln(1-z)/z$ use $\frac{z}{2} \leq \ell(z) - 1 \leq \frac{z}{2(1-z)}$. $\square$

Theorem 7.7 (window theorem, discrete time, $q \leq \frac{1}{2}$). *For every integer $t \geq 1$,*

$$a_1 E_{\nu_1} \left[1 + \frac{\theta}{W} \kappa_{\text{lo}}^D \right] - a_0 E_{\nu_0} [1 + \bar{E}_-^D] \leq Dg \leq a_1 E_{\nu_1} \left[1 + \frac{\theta}{W} \bar{\kappa}_D \right] + \bar{P}_{\text{rest}}^D - a_0 E_{\nu_0} [1 + E_-^{\text{lo},D}]$$

(all functions evaluated at t ; the barred/lo quantities are those of Lemma 7.6). Hence $(Dg)(t) > 0$ as soon as $\ln \Lambda - \hat{\nu}_D \tau + \ln(1 + \frac{\theta}{W} \kappa_{\text{lo}}^D(t)) - \ln(1 + \bar{E}_-^D(t)) > 0$, and $(Dg)(t) < 0$ as soon as $\ln \Lambda - \hat{\nu}_D \tau + \ln(1 + \frac{\theta}{W} \bar{\kappa}_D + \frac{\bar{P}_{\text{rest}}^D(t)}{a_1 E_{\nu_1}(t)}) - \ln(1 + E_-^{\text{lo},D}(t)) < 0$.

Proof. Proposition 7.1 and Lemma 7.6; $E_{\nu_1}(t)/E_{\nu_0}(t) = e^{-\hat{\nu}_D \tau}$. $\square$

Theorem 7.8 (explicit bounds, discrete time, all $q \in (0, \frac{1}{2}]$; computer-assisted). *Let $d \in \{2, 3\}$, $q \in (0, \frac{1}{2}]$, $t_{\text{cf}} := \tau_{\text{cf}}/\mu = \text{MFPT} \cdot \ln(2dX)/(X+2d-1)$. For every $N \geq N_0$, every maximiser t^* of $t \mapsto \mathbb{P}(T=t)$ satisfies*

$$t_{\text{cf}} - \frac{K_{d,D}^-(N_0)}{X\mu} < t^* < t_{\text{cf}} + \frac{K_{d,D}^+(N_0)}{X\mu} + 2,$$

with the certified constants

$d = 2: N_0$	12	15	20	30	50	100	200	1000
$K_{2,D}^-$	8.546	6.119	4.479	3.131	2.099	1.332	0.770	-0.116
$K_{2,D}^+$	10.392	8.859	7.542	6.377	5.462	4.842	4.641	4.641
$d = 3: N_0$	10	15	20	30	50	100	200	1000
$K_{3,D}^-$	5.293	3.442	2.661	1.962	1.527	1.318	1.144	0.898
$K_{3,D}^+$	7.513	4.834	3.761	2.829	2.165	1.744	1.599	1.465

The constants are uniform in $q \in (0, \frac{1}{2}]$; $K_{d,D}^\pm(N_0)$ is the maximum of the block constants over all blocks that meet $[N_0, \infty)$, as in the proof of Theorem 5.1.

Proof. Fix a block and constants $c_\pm$. Let $t_+ := \lceil (\tau_{\text{cf}} + c_+/X)/\mu \rceil$ and $t_- := \lfloor (\tau_{\text{cf}} - c_-/X)/\mu \rfloor$; then $\mu t_+ = \tau_{\text{cf}} + c y'$ with $c \in [c_+, c_+ + \mu X]$, and $\mu t_- = \tau_{\text{cf}} + c y'$ with $c \in [-c_- - \mu X, -c_-]$. For $q \leq \frac{1}{2}$, $\mu \leq \frac{1}{d} \sin^2 x = \xi y/d$, hence $\mu X \leq \xi(1 + y\mathbb{E}U)/d$, $\mu\tau \leq \xi y\tau/d$ and $\mu\tau/y \leq \xi\tau/d$. The script `scripts/72_cert_discrete.py` evaluates, in ball arithmetic over the same blocks and enclosures as in Section 5 and with c replaced by these balls, an upper bound of the second sign function of Theorem 7.7 at t_+ and a lower bound of the first at t_- , divided by y . Compared with the continuous case (Lemma 5.2 applies verbatim to $\ln \Lambda - \hat{\nu}\tau$) the following changes are made, each justified by Lemma 7.6:

- the term $-(\hat{\nu}_D - \hat{\nu})\tau/y \in [-\frac{\xi\tau}{d}(\frac{\nu_1^2}{2(1-\mu\nu_1)} - \frac{\nu_0^2}{2}), 0]$ is added (for the unbounded block $\xi\tau$ is bounded using $\xi \ln \frac{1}{y} \leq \frac{\pi^2}{4N_a^2} M \ln M$, $M = 2\pi \ln N_a + 1.3$ resp. $c_3 N_a$, which is decreasing in N_a);
- upper side: $\bar{\kappa} \rightarrow \bar{\kappa}_D$ with $s_x \in [\ln((d-1)A), \ln((d-1)A) + \mu]$; $\tau \rightarrow \tau/(1-2\mu)$ in N_2 ; $R_{\text{tot}} \rightarrow R_{\text{tot}}/(1-\varepsilon_2\mu)$; the ratios E_2/E_{ν_1} , $E_{\varepsilon_2}/E_{\nu_1}$ are bounded by their continuous counterparts (Lemma 7.6(d)); in $E_-^{\text{lo},D}$ the positive terms of $W\mathbf{g}/E_{\nu_0}$ are multiplied by $1 - \frac{2(2-\nu_0)\mu\tau}{1-2\mu}$ (the term $2A\gamma_2 E_{2+\varepsilon_2}$ is dropped) and the negative ones are bounded by their continuous counterparts;
- lower side: in κ_{10}^D , $\frac{g_a - g_a^t}{a - \nu_1} \geq \frac{1 - e^{-(a-\nu_1)\tau}}{a - \nu_1} - \frac{\mu}{1 - \mu\nu_1}$ for the positive coefficients and $\leq \frac{1}{a - \nu_1}$ for the negative ones; $\bar{r}/E_{\nu_1}$ and $\bar{E}_-^D$ are bounded by their continuous counterparts.

For all blocks with $N_a \geq 11$ ($d = 2$) resp. $N_a \geq 10$ ($d = 3$) the script verifies the two strict inequalities (files `data/72_cert_discrete_d2.jsonl`, `data/72_cert_discrete_d3.jsonl`); the single block $d = 2$, $N = 10$ fails and is excluded. Hence $(Dg)(t_+) < 0$ and $(Dg)(t_-) > 0$, and Theorem 7.4 gives $t_- + 2 \leq t^* \leq t_+ + 1$, which implies the claim. $\square$

Corollary 7.9 (asymptotics in discrete time, $q \leq \frac{1}{2}$). *For $q \in (0, \frac{1}{2}]$ and $N \geq 20$ every maximiser t^* of the first-passage probability mass function satisfies*

$$d = 2 : \quad \frac{4N^2}{\pi^2 q} \left[\ln \ln N + \ln 8\pi - \frac{4.72 + 3 \ln(8\pi \ln N + 2.4)}{2\pi \ln N - 0.24} \right] < t^* < \frac{4N^2}{\pi^2 q} \left[\ln \ln N + \ln 8\pi \right] \left(1 + \frac{0.83}{N^2} \right) + 2,$$

$$d = 3 : \quad \frac{6N^2}{\pi^2 q} \left[\ln N + \ln(6c_3) - \frac{10.30 + 5 \ln(6c_3 N)}{c_3 N - 7.63} \right] < t^* < \frac{6N^2}{\pi^2 q} \left[\ln N + \ln(6c_3) \right] \left(1 + \frac{0.83}{N^2} \right) + 2.$$

In particular the laws $t^* \sim \frac{4}{\pi^2 q} N^2 \ln \ln N$ ($d = 2$) and $t^* \sim \frac{6}{\pi^2 q} N^2 \ln N$ ($d = 3$) of the computations of Sections 5–6 of the article are theorems for the discrete walk with $q \leq \frac{1}{2}$, with the same explicit rates as in Theorems 6.1–6.3.

Proof. Repeat the proofs of Theorems 6.1 and 6.2 with $(K^-, K^+) = (4.479, 7.542)$ resp. $(2.661, 3.761)$ from Theorem 7.8 ($N_0 = 20$): the upper bounds still hold since $0.62 + 7.542 - 11.09 < 0$ and $3.761 < 32$; in the lower bounds $0.24 + 4.479 \leq 4.72$ and $7.63 + 2.661 \leq 10.30$. $\square$

7.5 All $q < 1$: localisation without unimodality

For $q > \frac{1}{2}$ some of the numbers $\beta_k = 1 - 2q \sin^2(kx)$, $1 - \mu\mu_k$, $1 - \mu\nu_j$ are negative, the mass function need not be unimodal (for $q = 1$ it is not unimodal in general, because of parity effects), and Lemma 7.2 and Theorem 7.4 are not available. We show that the bounds of Lemma 7.3 survive when N is not too small, bound the poles with negative multiplier, and replace unimodality by a four-region argument. Throughout this subsection

$$q \in (0, 1), \quad N \geq 4, \quad \sin(2x) \leq \frac{1-q}{q}, \quad (\text{H})$$

and $\varrho_q := \max(2q - 1, 0)$. The third condition is void for $q \leq \frac{1}{2}$, and it holds if and only if $N \geq N_1(q) := \min\{N \geq 4 : \sin \frac{\pi}{N} \leq \frac{1-q}{q}\}$ ($\sin \frac{\pi}{N}$ decreases in N); $N_1(q) = 4$ for $q \leq \frac{1}{2}$ and $N_1(q) \approx \pi q / (1 - q)$ as $q \rightarrow 1$. Two consequences of (H) are used throughout: since $\mu = \frac{2q}{d} \sin^2 x$ and $q \sin 2x \leq 1 - q$,

$$\begin{aligned} \varepsilon_2 \mu &= \frac{2q}{d} \sin^2(2x) \leq \frac{2(1-q)}{d} \sin(2x) < 2(1-q), \\ \mu &\leq q \sin(2x) \leq 1 - q, \quad \text{i.e. } \varrho_q \leq 1 - 2\mu, \end{aligned} \quad (\text{H}')$$

where we used $\sin 2x < 1$ ($N \geq 4$) and $\sin x \leq d \cos x$.

Lemma 7.10 (pairing). *Assume (H). For $j \in \{1, 2, 3\}$ the power series $\prod_{k=j}^{N-1} (1 - \beta_k z)^{-1}$ has nonnegative coefficients; $\beta_1 \geq \beta_2 \geq 1 - q > 0$; and $|\beta_k| < \beta_j$ for $k > j$, $j \in \{1, 2\}$. Consequently all statements of Lemma 7.2 except log-concavity, and all statements of Lemma 7.3, remain true (with $\pi_j(m) := \lfloor \nu^m \rfloor \prod_{k \geq j} \frac{p_k z}{1 - \beta_k z} \geq 0$ in place of the mass functions); in particular $0 \leq v_n \leq 1$ is nondecreasing, $\varphi \geq 0$, $\tilde{q} \geq 0$ and*

$$\tilde{q} \geq q_{\text{lo}} := d(d-1)[A^2 E_2 - 2A\gamma_2 \varepsilon_2 E_{1+\varepsilon_2} - (d-2)A^3 E_3].$$

Proof. Fix j and pair k with $k' := N + j - 1 - k$, an involution of $\{j, \dots, N-1\}$. Since $k'x = \frac{\pi}{2} + (j-1)x - kx$, $\sin^2(kx) + \sin^2(k'x) = 1 + \sin^2(kx) - \sin^2(kx - (j-1)x) = 1 + \sin((2k-j+1)x) \sin((j-1)x) \leq 1 + \sin((j-1)x)$, hence $\beta_k + \beta_{k'} \geq 2 - 2q(1 + \sin((j-1)x)) \geq 0$ by (H) ($\sin((j-1)x) \leq \sin 2x$). For a pair $k \neq k'$ with, say, $\beta_k \geq \beta_{k'}$ we get $\beta_k \geq |\beta_{k'}|$, and the coefficients of $\frac{1}{(1-\beta_k z)(1-\beta_{k'} z)}$ are $\sum_{i=0}^n \beta_k^i \beta_{k'}^{n-i} = \frac{\beta_k^{n+1} - \beta_{k'}^{n+1}}{\beta_k - \beta_{k'}} \geq 0$ (or $(n+1)\beta_k^n$ if equal). A fixed point $k = k'$ has $\sin^2(kx) = \frac{1}{2}(1 + \sin((j-1)x))$, so $\beta_k = 1 - q(1 + \sin((j-1)x)) \geq 0$. A product of series with nonnegative coefficients has nonnegative coefficients.

$N \geq 4$ gives $\sin^2(2x) \leq \frac{1}{2}$, so $\beta_2 \geq 1 - q$. For $k > j$: $\beta_k < \beta_j$ and $\beta_k + \beta_j \geq 2 - 2q(\cos^2 x + \sin^2(jx)) = 2 - 2q(1 + \sin((j+1)x) \sin((j-1)x)) > 0$ for $j = 1, 2$ (for $j = 2$: $\sin 3x \sin x < \sin x \leq \sin 2x \leq \frac{1-q}{q}$).

Now repeat the proof of Lemma 7.2: $\pi_1(n+1) = \sum_{m=1}^n p_1 \beta_1^{n-m} \pi_2(m)$ with $\pi_2 \geq 0$, so $\psi(n) \beta_1^{-n} = \sum_{m \leq n} \beta_1^{-m} \pi_2(m)$ is nondecreasing; its limit is the value at $z = 1/\beta_1$ of $\prod_{k \geq 2} \frac{p_k z}{1 - \beta_k z}$, which is analytic there because $|\beta_k| < \beta_1$, i.e. $\prod_{k \geq 2} \frac{p_k}{\beta_1 - \beta_k} = A$. The same argument with $\pi_3 \geq 0$ and $|\beta_k| < \beta_2$ gives $0 \leq \pi_2(m) \leq p_2 \gamma_2 (\varepsilon_2 - 1) \beta_2^{m-1}$, and the remaining bounds of Lemma 7.2 follow verbatim; $v_n = \sum_{m \leq n} \pi_1(m) \in [0, 1]$ since $v_n \rightarrow 1$. The proof of Lemma 7.3 uses only these facts (all the numbers $E_k(t)$ occurring there have positive bases $\frac{1}{d}(\beta_{k_1} + \dots + \beta_{k_d})$ with $k_i \in \{0, 1, 2\}$).

Lower bound for $\tilde{q}$: in the formula for $\mu \tilde{q}$ in the proof of Lemma 7.3 drop the first (nonnegative) sum and use $v_{n_m+1} - v_{n_m} = p_1 \psi(n_m)$: $\mu \tilde{q} \geq \frac{p_1}{d} \mathbb{E} \sum_{i \neq m} \psi(n_i) \psi(n_m) \prod_{l \neq i, m} v_{n_l}$. With $L_n := A\beta_1^n - \gamma_2 \varepsilon_2 \beta_2^n \leq \psi(n) \leq A\beta_1^n$ and $\psi \geq 0$ one has $\psi(n) \psi(m) \geq A^2 \beta_1^{n+m} - A\gamma_2 \varepsilon_2 (\beta_1^n \beta_2^m + \beta_2^n \beta_1^m)$ (if $L_m \geq 0$: $\psi(n) \psi(m) \geq L_n L_m$; if $L_m < 0$ the right side equals $A\beta_1^n L_m - A\gamma_2 \varepsilon_2 \beta_2^n \beta_1^m \leq 0$); for $d = 3$, $v_{n_i} \geq 1 - A\beta_1^{n_i}$ and $\psi \psi \leq A^2 \beta_1^{n+m}$ give $\psi \psi v \geq \psi \psi - A^3 \beta_1^{n+m+l}$. Taking expectations ($p_1/d = \mu$, $d(d-1)$ ordered pairs) gives q_{lo} . $\square$

Call a pole *negative* if $\mu \nu_j \geq 1$ and *positive* otherwise; let $\mathcal{Q}$ be the set of negative poles (empty if $q \leq \frac{1}{2}$). The poles ν_0, ν_1 and the near poles are positive.

Lemma 7.11 (negative poles). *Assume (H). For $j \in \mathcal{Q}$: $|1 - \mu \nu_j| \leq \varrho_q \leq 1 - 2\mu$, and $\mu \sum_{j \in \mathcal{Q}} \rho_j \leq \mu R_{\text{tot}} = \frac{q}{2} \pi_a$. For all $t \in \mathbb{N}_0$:*

$$\begin{aligned} (a) \quad & \left| \sum_{j \in \mathcal{Q}} \rho_j \left[\frac{W}{\nu_j - 1} E_{\nu_j} + r_1 \star E_{\nu_j} \right](t) \right| \leq n_q(t) := \frac{q\pi_a}{2} \left[\left(\frac{W}{1-\mu} + \frac{2C_1}{2(1-q)-2\mu} \right) E_2(t) + \right. \\ & \left. \frac{C_2 \varepsilon_2}{2(1-q) - \varepsilon_2 \mu} E_{\varepsilon_2}(t) \right]; \\ (b) \quad & \left| \sum_{j \in \mathcal{Q}} \rho_j ((D\varphi) \star E_{\nu_j})(t) \right| \leq \omega_q \varphi(t) + \omega'_q(t), \quad \omega_q := \frac{q\pi_a}{2} \cdot \frac{1}{2(1-q)-\mu}, \quad \omega'_q(t) := \frac{q\pi_a}{2} \left[\frac{C_1 E_2(t)}{2(1-q)-2\mu} + \right. \\ & \left. \frac{C_2(\varepsilon_2-1)E_{\varepsilon_2}(t)}{2(1-q)-\varepsilon_2 \mu} \right]; \end{aligned}$$

$$(c) \left| \sum_{j \in \mathcal{Q}} \rho_j (\varphi \star E_{\nu_j})(t) \right| \leq \frac{q\pi_a}{4(1-q)}.$$

Proof. $\nu_j < m_L$ and $\mu m_L < 2q$, so $1 - \mu\nu_j \in (1 - 2q, 0]$ for $j \in \mathcal{Q}$; $\varrho_q \leq 1 - 2\mu$ is (H'). $\mu R_{\text{tot}} = \frac{2q}{d} \sin^2 x \cdot \frac{d}{4N^d \sin^2 x}$ (Lemma 1.4(c)). If $q \leq \frac{1}{2}$ then $\mu\nu_j < \mu m_L < 2q \leq 1$ for all j , so $\mathcal{Q} = \emptyset$, all sums over $\mathcal{Q}$ vanish and there is nothing to prove; let $q > \frac{1}{2}$, so $\varrho_q = 2q - 1$. For $0 \leq a \leq \varepsilon_2$, (H') gives $(1 - a\mu) - \varrho_q = 2(1 - q) - a\mu > 0$ and hence $\mu \sum_{s < t} E_a(s) \varrho_q^{t-1-s} = \mu \frac{E_a(t) - \varrho_q^t}{(1 - a\mu) - \varrho_q} \leq \frac{\mu E_a(t)}{2(1-q) - a\mu}$ (this inequality is false in general for $q < \frac{1}{2}$, where $\varrho_q = 0$, but it is not needed there). (a) $|E_{\nu_j}(t)| \leq \varrho_q^t \leq E_2(t)$, $\frac{1}{\nu_j - 1} \leq \frac{\mu}{1 - \mu}$, and $|r_1 \star E_{\nu_j}| \leq \mu \sum_{s < t} \bar{r}_1(s) \varrho_q^{t-1-s}$. (b) $D\varphi = \tilde{q} - \varphi$ with $\varphi, \tilde{q} \geq 0$, so $|D\varphi| \leq \varphi + \tilde{q}$; $\varphi(s) \leq \varphi(t) E_1(s)/E_1(t)$ for $s \leq t$ (as $\tilde{q} \geq 0$) gives $\mu \sum_{s < t} \varphi(s) \varrho_q^{t-1-s} \leq \frac{\mu \varphi(t)}{2(1-q) - \mu}$, and $\tilde{q} \leq C_1 E_2 + C_2(\varepsilon_2 - 1) E_{\varepsilon_2}$. (c) $\varphi = 1 - u - Q \leq 1$ with $Q := \mu \sum_{s \geq t} \tilde{q}(s) \geq 0$, so $|\varphi \star E_{\nu_j}| \leq \mu \sum_{s < t} \varrho_q^{t-1-s} \leq \frac{\mu}{2(1-q)}$. In each case multiply by $\sum_{\mathcal{Q}} \rho_j$. $\square$

Lemma 7.12 (the four regions). Assume (H), $d \in \{2, 3\}$ ($N \geq 4$). Let $\Phi_{\pm}^D$ denote the two bounds of Theorem 7.7 (with $\bar{P}_{\text{rest}}^D$ referring to the positive poles $j \geq 2$), $Q(t) := \mu \sum_{s \geq t} \tilde{q}(s) = 1 - u_t - \varphi(t)$, and

$$B_1(t) := \frac{1}{E_1(t)} \left[\frac{a_1 \nu_1}{W} E_{\nu_1}(t) - \frac{\rho_0 \nu_0}{1 - \nu_0} E_{\nu_0}(t) \right],$$

$$Q_{\text{lo}}(t) := d(d-1) \left[\frac{A^2 E_2(t)}{2} - \frac{2A\gamma_2 \varepsilon_2 E_{1+\varepsilon_2}(t)}{1 + \varepsilon_2} - \frac{(d-2)A^3 E_3(t)}{3} \right].$$

(A) For $t \geq 1$: $\Phi_{-}^D(t) - n_q(t) \leq (Dg)(t) \leq \Phi_{+}^D(t) + n_q(t)$. Moreover $\mathcal{A}(t) := [a_1 E_{\nu_1}(1 + \frac{\theta}{W} \bar{\kappa}_D) + \bar{P}_{\text{rest}}^D + n_q](t)/(a_0 E_{\nu_0}(t))$ is nonincreasing on $\{t : \mu t \geq 1\}$ and $(Dg)(t) \leq a_0 E_{\nu_0}(t) [\mathcal{A}(t) - 1]$.

(B) For $t \geq 1$: $(Dg)(t) \geq \varphi(t) [B_1(t) - \omega_q] - \omega'_q(t)$; B_1 is nonincreasing, φ/E_1 is nondecreasing with $\varphi/E_1 \geq W - \bar{r}/E_1$, and ω'_q/E_1 is nonincreasing.

(C) For all t : $\rho_0 E_{\nu_0}(t) (1 - W E_1(t)) - \frac{q\pi_a}{4(1-q)} \leq g(t) \leq \rho_0 \left[1 - Q(t) + \frac{\nu_0 \varphi(t)}{1 - \nu_0} \right] + \frac{q\pi_a}{4(1-q)}$; Q is nonincreasing, $Q \geq Q_{\text{lo}}$ and $\varphi \leq 1$.

Proof. (A) In Proposition 7.1 split the sums over $j \geq 2$ into positive and negative poles. Lemma 7.6 (whose proof uses only Lemma 7.3, $0 < 1 - \mu\nu_j < 1$ for the poles concerned, and the convergence of the geometric series defining $\tilde{J}$, which holds because $|1 - \mu\mu_k| \leq \max(1 - 2\mu, \varrho_q) < 1 - \mu\nu_0$) bounds the positive part, Lemma 7.11(a) the negative part. For the monotonicity write $\mathcal{A} = \frac{E_1}{E_{\nu_0}} \cdot \frac{1}{a_0} [a_1(1 + \frac{\theta}{W} \bar{\kappa}_D) \frac{E_{\nu_1}}{E_1} + \frac{\bar{P}_{\text{rest}}^D + n_q}{E_1}]$: E_1/E_{ν_0} and E_{ν_1}/E_1 , E_2/E_1 , E_{ε_2}/E_1 are nonincreasing, and $t \mapsto t(E_2/E_1)(t)$, $t(E_{\varepsilon_2}/E_1)(t)$ are nonincreasing for $\mu t \geq 1$ (the ratio of consecutive values of $t\varrho^t$ is $(1 + \frac{1}{t})\varrho \leq 1$ iff $t \geq \frac{\varrho}{1-\varrho}$, and $\frac{\varrho}{1-\varrho} = \frac{1-2\mu}{\mu}$ resp. $\frac{1-\varepsilon_2\mu}{(\varepsilon_2-1)\mu}$); hence each term $\min(c\tau, c') E_2/E_1$ and $\tau E_{\varepsilon_2}/E_1$ of $\bar{P}_{\text{rest}}^D/E_1$ is nonincreasing. The last inequality is $Dg \leq \Phi_{+}^D + n_q$ with $E_{-}^{\text{lo}, D} \geq 0$ dropped.

(B) $Dg = \pi_a D\varphi + \sum_j \rho_j (D\varphi) \star E_{\nu_j}$ (as $\varphi(0) = 0$) and $(D\varphi) \star E_{\nu} = \varphi - \nu(\varphi \star E_{\nu})$. For a positive pole, $\varphi(s) \leq \varphi(t) E_1(s)/E_1(t)$ ($s \leq t$) gives $(\varphi \star E_{\nu})(t) \leq \varphi(t) \frac{(E_1 \star E_{\nu})(t)}{E_1(t)}$, i.e.

$$(D\varphi) \star E_{\nu_j} \geq \varphi \left[-\frac{1}{\nu_j - 1} + \frac{\nu_j}{\nu_j - 1} \frac{E_{\nu_j}}{E_1} \right] \quad (j \geq 1),$$

$$(D\varphi) \star E_{\nu_0} \geq \varphi \left[\frac{1}{1 - \nu_0} - \frac{\nu_0}{1 - \nu_0} \frac{E_{\nu_0}}{E_1} \right],$$

and $\pi_a D\varphi \geq -\pi_a \varphi$. Summing over the positive poles and using $\frac{\rho_0}{1-\nu_0} - \pi_a - \sum_{j \geq 1, j \notin \mathcal{Q}} \frac{\rho_j}{\nu_j - 1} = \sum_{j \in \mathcal{Q}} \frac{\rho_j}{\nu_j - 1} \geq 0$ (Lemma 1.4(c)) and dropping the nonnegative terms with $j \geq 2$: the positive poles contribute at least φB_1 . The negative poles are bounded by Lemma 7.11(b). Monotonicity: E_{ν_1}/E_1 decreases, E_{ν_0}/E_1 increases; $(\varphi/E_1)(t+1) - (\varphi/E_1)(t) = \mu \tilde{q}(t)/E_1(t+1) \geq 0$; $r \leq \bar{r}$.

(C) $g = \pi_a \varphi + \sum_j \rho_j (\varphi \star E_{\nu_j})$. Upper bound: $(\varphi \star E_{\nu_0})(t) \leq \mu \sum_{s < t} \varphi(s) = u_t$; for positive $j \geq 1$, $(\varphi \star E_{\nu_j}) \leq \varphi/(\nu_j - 1)$ and $\sum_{j \geq 1, j \notin \mathcal{Q}} \frac{\rho_j}{\nu_j - 1} \leq \frac{\rho_0}{1-\nu_0} - \pi_a$; so the positive part is $\leq \rho_0 [u + \frac{\varphi}{1-\nu_0}]$, and

$u + \varphi = 1 - Q$ (sum $D\varphi + \varphi = \tilde{q}$ over $s \geq t$). Lower bound: the positive part is $\geq \rho_0(\varphi \star E_{\nu_0})(t) \geq \rho_0 E_{\nu_0}(t) u_t$ and $1 - u_t = \mu \sum_{s \geq t} \varphi(s) \leq W E_1(t)$. The negative part is bounded by Lemma 7.11(c). $Q \geq Q_{\text{lo}}$ follows from Lemma 7.10 and $\mu \sum_{s \geq t} E_k(s) = E_k(t)/k$. $\square$

Theorem 7.13 (localisation without unimodality). *Assume (H), $d \in \{2, 3\}$, and let $t_E \leq t_B \leq t_- < t_+$ be integers with $\mu t_+ \geq 1$ such that*

- (A) $\mathcal{A}(t_+) < 1$;
- (B) $\Phi_-^D(t) - n_q(t) > 0$ for every integer $t \in [t_B, t_-]$;
- (C) $c_E := W - \bar{r}(t_E)/E_1(t_E) > 0$, $B_1(t_B) > \omega_q$ and $c_E(B_1(t_B) - \omega_q) > \omega'_q(t_E)/E_1(t_E)$;
- (D) $1 - Q_{\text{lo}}(t_E) + \frac{\nu_0}{1 - \nu_0} + \frac{q\pi_a}{2(1 - q)\rho_0} < E_{\nu_0}(t_-)(1 - W E_1(t_-))$.

Then every maximiser t^ of $t \mapsto \mathbb{P}(T = t)$ satisfies $t_- + 2 \leq t^* \leq t_+ + 1$.*

Proof. By (A) and Lemma 7.12(A), $\mathcal{A}(t) \leq \mathcal{A}(t_+) < 1$ and so $(Dg)(t) < 0$ for all $t \geq t_+$: no maximiser of g exceeds t_+ . By (C) and Lemma 7.12(B), for $t_E \leq t \leq t_B$: $(Dg)(t)/E_1(t) \geq c_E(B_1(t_B) - \omega_q) - \omega'_q(t_E)/E_1(t_E) > 0$; by (B) and Lemma 7.12(A), $(Dg)(t) > 0$ for $t_B \leq t \leq t_-$. Hence g is strictly increasing on $[t_E, t_- + 1]$: no maximiser in $[t_E, t_-]$. By (D) and Lemma 7.12(C), for $t \leq t_E$: $g(t) \leq \rho_0[1 - Q_{\text{lo}}(t_E) + \frac{\nu_0}{1 - \nu_0}] + \frac{q\pi_a}{4(1 - q)} < \rho_0 E_{\nu_0}(t_-)(1 - W E_1(t_-)) - \frac{q\pi_a}{4(1 - q)} \leq g(t_-)$: no maximiser in $[0, t_E]$. So the maximisers of g lie in $[t_- + 1, t_+]$, and $\mathbb{P}(T = t) = \mu g(t - 1)$. $\square$

Theorem 7.14 (explicit bounds for $q \leq q_{\text{max}} < 1$; computer-assisted). *Let $d \in \{2, 3\}$, $q_{\text{max}} \in \{0.8, 0.9, 0.95, 0.99\}$ and $q \in (0, q_{\text{max}}]$. For every $N \geq N_0$ every maximiser t^* of $t \mapsto \mathbb{P}(T = t)$ satisfies*

$$t_{\text{cf}} - \frac{K^-}{X\mu} < t^* < t_{\text{cf}} + \frac{K^+}{X\mu} + 2,$$

with the following certified constants (K^-, K^+) ; the first N_0 in each row is the smallest for which the certificate succeeds for all $N \geq N_0$:

q_{max}	d	$(N_0; K^-, K^+)$
0.8	2	(37; 2.850, 6.613), (50; 2.255, 6.152), (100; 1.448, 5.560), (200; 0.875, 5.145), (1000; -0.016, 5.135)
0.8	3	(13; 5.189, 6.634), (20; 3.459, 4.729), (30; 2.527, 3.715), (50; 1.851, 2.973), (100; 1.568, 2.488), (200; 1.334, 2.305), (1000; 1.021, 2.141)
0.9	2	(46; 2.456, 6.303), (50; 2.300, 6.171), (100; 1.460, 5.566), (200; 0.878, 5.147), (1000; -0.015, 5.135)
0.9	3	(29; 2.762, 3.804), (50; 1.946, 2.980), (100; 1.619, 2.489), (1000; 1.028, 2.141)
0.95	2	(60; 2.065, 6.052), (100; 1.479, 5.578), (200; 0.884, 5.151), (1000; -0.015, 5.135)
0.95	3	(60; 1.971, 2.807), (100; 1.647, 2.492), (1000; 1.032, 2.141)
0.99	2	(312; 0.550, 5.135), (1000; 0.003, 5.135)
0.99	3	(312; 1.236, 2.231), (1000; 1.033, 2.141)

As in Theorem 5.1, each entry is the maximum of the block constants over all blocks that meet $[N_0, \infty)$. In particular, for the value $q = \frac{4}{5}$ the closed form satisfies $|t^ - t_{\text{cf}}| < \frac{6.613}{X\mu} + 2$ for all $N \geq 37$ ($d = 2$) and $|t^* - t_{\text{cf}}| < \frac{6.634}{X\mu} + 2$ for all $N \geq 13$ ($d = 3$). The finitely many smaller N are not covered by this theorem (Remark 7.19).*

Proof. We verify the hypotheses of Theorem 7.13 with $t_+ := \lceil (\tau_{\text{cf}} + c_+/X)/\mu \rceil$, $t_- := \lfloor (\tau_{\text{cf}} - c_-/X)/\mu \rfloor$, $t_B := \lfloor (\tau_{\text{cf}} - c_-/X - b_B)/\mu \rfloor$, $t_E := \lceil \tau_E/\mu \rceil$, where $(\tau_E, b_B) = (1.2, 2.4)$ for $d = 2$ and $(1.8, 3.0)$ for $d = 3$. The script `scripts/76_cert_q_large.py` works with the blocks and enclosures of Section 5, restricted to $N \geq N_1(q_{\text{max}})$ (which gives the third condition of (H) for all $q \leq q_{\text{max}}$), and with $\mu \leq 2q_{\text{max}}\xi y/d$; the constants $n_q, \omega_q, \omega'_q, \frac{q\pi_a}{1-q}$ are increasing in q and are evaluated at q_{max} , using $\pi_a/y = N^{-d}X_s \leq N_a^{-d}\bar{X}^{(d)}(N_b)$. In ball arithmetic it checks, for each block:

- $\varepsilon_2\mu < 2(1 - q_{\max})$ and $2q_{\max} - 1 < 1 - 2\mu$ (these follow from (H') and are checked only as a safeguard), and $c_+ + c_- > 0$, $\tau_{\text{cf}} + c_+y' \geq 1$ (so $t_- < t_+$ and $\mu t_+ \geq 1$);
- (A): the upper sign function of Theorem 7.8 with $E_-^{\text{lo},D}$ replaced by 0 and $n_q/(a_1 E_{\nu_1})$ added inside the logarithm is negative for $\tau \in [\tau_{\text{cf}} + c_+y', \tau_{\text{cf}} + c_+y' + \mu]$ (this is $\mathcal{A}(t_+) < 1$);
- (B): the lower sign function with $n_q/(a_1 E_{\nu_1})$ subtracted inside the logarithm is positive for all $\tau \in [\tau_{\text{cf}} - c_-y' - \mu - b_B, \tau_{\text{cf}} - c_-y']$; this interval is covered by the shifts $b \in [b_i, b_{i+1}]$ of a mesh $0 = b_0 < b_1 < \dots = b_B$ (first step 0.01, ratio 1.25, maximal step 0.06), each treated as a ball; the term $b\hat{\nu}/y$ of S/y is bounded below by $b_i \inf \hat{\nu} / \sup y$;
- (C): with the bounds $E_{\nu_1}/E_1 \geq e^{-\theta\tau/(1-\nu_1\mu)}$, $E_{\nu_0}/E_1 \leq e^{(1-\nu_0)\tau/(1-\mu)}$ (Lemma 7.6(d)) at $\tau_B \in [\tau_{\text{cf}} - c_-y' - b_B - \mu, \tau_{\text{cf}} - c_-y' - b_B]$, $B_1(t_B)/y \geq \alpha_1 \frac{\nu_1}{W} e^{-\theta\tau_B/(1-\nu_1\mu)} - \frac{(1-s)^2 W}{1-\nu_0} e^{\delta_B - \nu_0\tau_B + (1-\nu_0)\mu\tau_B/(1-\mu)}$, $c_E \geq W - C_1 e^{-\tau_E} - C_2 e^{-(\varepsilon_2-1)\tau_E}$, $\omega'_q(t_E)/E_1(t_E) \leq \frac{q\pi_a}{2} \left[\frac{C_1 e^{-\tau_E}}{2(1-q)-2\mu} + \frac{C_2(\varepsilon_2-1)e^{-(\varepsilon_2-1)\tau_E}}{2(1-q)-\varepsilon_2\mu} \right]$, and $\tau_E + \mu < \tau_{\text{cf}} - c_-y' - b_B$ (so $t_E \leq t_B$);
- (D): with $E_2(t_E) \geq e^{-2(\tau_E+\mu)/(1-2\mu)}$, $E_{1+\varepsilon_2}(t_E) \leq e^{-(1+\varepsilon_2)\tau_E}$, $E_3(t_E) \leq e^{-3\tau_E}$, $\rho_0 \geq y(1-s)/(1+\nu_0^2 Z_d/(1-\nu_0)^2)$, $E_{\nu_0}(t_-) \geq e^{-\nu_0\tau_-/(1-\nu_0\mu)}$ and $WE_1(t_-) \leq ye^{-\delta_-+\mu}$.

The constant c_- is the smallest value (to 10^{-3}) for which the lower sign condition is verified, increased by 0.1. The certificate files `data/76_cert_q0p8_d2.jsonl`, ... list all blocks with the five Boolean results; for every block that meets $[N_0, \infty)$, with N_0 as in the table, all are **true**. Theorem 7.13 then gives $t_- + 2 \leq t^* \leq t_+ + 1$. $\square$

Corollary 7.15 (asymptotic laws for the discrete walk, $q \leq 0.99$). *Let $q \in (0, 0.99]$ and $N \geq N_0(q)$, where $N_0 = 100$ for $q \leq 0.95$ and $N_0 = 312$ for $0.95 < q \leq 0.99$. Then every maximiser t^* of the first-passage probability mass function satisfies*

$$d = 2 : \quad \frac{4N^2}{\pi^2 q} \left[\ln \ln N + \ln 8\pi - \frac{1.72 + 3 \ln(8\pi \ln N + 2.4)}{2\pi \ln N - 0.24} \right] < t^* < \frac{4N^2}{\pi^2 q} \left[\ln \ln N + \ln 8\pi \right] \left(1 + \frac{0.83}{N^2} \right) + 2,$$

$$d = 3 : \quad \frac{6N^2}{\pi^2 q} \left[\ln N + \ln(6c_3) - \frac{9.28 + 5 \ln(6c_3 N)}{c_3 N - 7.63} \right] < t^* < \frac{6N^2}{\pi^2 q} \left[\ln N + \ln(6c_3) \right] \left(1 + \frac{0.83}{N^2} \right) + 2.$$

Proof. For these N_0 , Theorem 7.14 gives $(K^-, K^+) \leq (1.479, 5.578)$ for $d = 2$ (the row $q_{\max} = 0.95$, $N_0 = 100$; the block $\{99, \dots, 106\}$ is included) and $\leq (1.647, 2.492)$ for $d = 3$. Repeat the proofs of Theorems 6.1 and 6.2: the upper bounds hold since $0.62 + 5.578 < 12.88 \leq 3 \ln(4X) \frac{X}{X+3}$ and $2.492 < 41.4 \leq 5 \ln(6X) \frac{X}{X+5}$ for $N \geq 100$ (the right-hand sides increase in X ; use $X \geq 2\pi \ln N - 0.24$ resp. $X \geq c_3 N - 7.63$); in the lower bounds $0.24 + 1.479 \leq 1.72$ and $7.63 + 1.647 \leq 9.28$. $\square$

7.6 All $q < 1$ at once: constants uniform in q

The constants n_q , ω_q , ω'_q and $\frac{q\pi_a}{1-q}$ of Lemma 7.11 and Theorem 7.13 blow up as $q \rightarrow 1$ for fixed N , but under (H) they are bounded in terms of N alone. This removes the restriction $q \leq q_{\max}$ of Theorem 7.14.

Lemma 7.16 (constants uniform in q). *Let $d \in \{2, 3\}$, $N \geq 10$, $x = \frac{\pi}{2N}$, and let $q \in (0, 1)$ satisfy (H). Put*

$$D_1(x) := 2 \sin 2x - \frac{2}{d} \sin^2 x, \quad D_2(x) := 2 \sin 2x - \frac{4}{d} \sin^2 x, \\ D_{\varepsilon_2}(x) := \sin 2x \left(2 - \frac{2}{d} \sin 2x \right).$$

(a) $\mu \leq \frac{2}{d} \sin^2 x$, and (H') holds.

(b) For $a \in \{1, 2, \varepsilon_2\}$: $2(1-q) - a\mu \geq q D_a(x) > 0$, hence $\frac{q}{2(1-q) - a\mu} \leq \frac{1}{D_a(x)}$; and $\frac{q}{1-q} \leq \frac{1}{\sin 2x}$.

(c) $D_1, D_2, D_{\varepsilon_2}$ are increasing on $(0, \frac{\pi}{20}]$, and $N \mapsto N D_a(\frac{\pi}{2N})$ ($a \in \{1, 2, \varepsilon_2\}$) and $N \mapsto N \sin \frac{\pi}{N}$ are increasing on $N \geq 10$.

Consequently, for all t ,

$$n_q(t) \leq \bar{n}(t) := \frac{\pi_a}{2} \left[\left(\frac{W}{1-\mu} + \frac{2C_1}{D_2(x)} \right) E_2(t) + \frac{C_2 \varepsilon_2}{D_{\varepsilon_2}(x)} E_{\varepsilon_2}(t) \right],$$

$$\omega_q \leq \bar{\omega} := \frac{\pi_a}{2D_1(x)},$$

$$\omega'_q(t) \leq \bar{\omega}'(t) := \frac{\pi_a}{2} \left[\frac{C_1 E_2(t)}{D_2(x)} + \frac{C_2(\varepsilon_2 - 1) E_{\varepsilon_2}(t)}{D_{\varepsilon_2}(x)} \right],$$

$$\frac{q\pi_a}{1-q} \leq \frac{\pi_a}{\sin 2x},$$

and the right-hand sides do not depend on q .

Proof. (a) $\mu = \frac{2q}{d} \sin^2 x \leq \frac{2}{d} \sin^2 x$; (H') was derived from (H) above. (b) By (H), $2(1-q) \geq 2q \sin 2x$, and $a\mu = q \cdot \frac{2a}{d} \sin^2 x$; for $a = \varepsilon_2 = 4 \cos^2 x$ one has $\frac{2\varepsilon_2}{d} \sin^2 x = \frac{2}{d} \sin^2 2x$. Hence $2(1-q) - a\mu \geq q D_a(x)$, and $D_a(x) > 0$ because $2 \sin 2x = 4 \sin x \cos x > \frac{2a}{d} \sin^2 x$ for $a \leq 2$, $x \leq \frac{\pi}{20}$, and $\sin 2x < 1 \leq \frac{d}{2}$. The bound for $\frac{q}{1-q}$ is (H). (c) For $a \in \{1, 2\}$, $D'_a(x) = 4 \cos 2x - \frac{2a}{d} \sin 2x \geq 4 \cos \frac{\pi}{10} - 2 \sin \frac{\pi}{10} > 0$; $D_{\varepsilon_2} = h(\sin 2x)$ with $h(s) = 2s - \frac{2}{d} s^2$ increasing for $s < \frac{d}{2}$. Next $N \sin \frac{\pi}{N} = \pi \frac{\sin u}{u} \Big|_{u=\pi/N}$ increases with N , and $N \sin^2 \frac{\pi}{2N} = \frac{\pi}{2} \sin v \cdot \frac{\sin v}{v} \Big|_{v=\pi/(2N)}$ decreases with N ; so $N D_a(\frac{\pi}{2N}) = 2N \sin \frac{\pi}{N} - \frac{2a}{d} N \sin^2 \frac{\pi}{2N}$ ($a = 1, 2$) and $N D_{\varepsilon_2}(\frac{\pi}{2N}) = N \sin \frac{\pi}{N} (2 - \frac{2}{d} \sin \frac{\pi}{N})$ increase with N . The final bounds follow by inserting (b) and $q < 1$ into the definitions of n_q, ω_q, ω'_q in Lemma 7.11. $\square$

Theorem 7.17 (window for the discrete walk, uniform in $q < 1$; computer-assisted). *Let $d \in \{2, 3\}$ and $N_* := 59$ for $d = 2$, $N_* := 12$ for $d = 3$. For every $q \in (0, 1)$ and every integer $N \geq \max(N_*, N_1(q))$, where $N_1(q) = \min\{N \geq 4 : \sin \frac{\pi}{N} \leq \frac{1-q}{q}\}$, every maximiser t^* of $t \mapsto \mathbb{P}(T = t)$ satisfies*

$$t_{\text{cf}} - \frac{K_{d,u}^-(N_0)}{X\mu} < t^* < t_{\text{cf}} + \frac{K_{d,u}^+(N_0)}{X\mu} + 2$$

whenever $N \geq \max(N_0, N_1(q))$, $N_0 \geq N_*$,

with the following certified constants $(N_0; K_{d,u}^-, K_{d,u}^+)$, the same for all q :

d	$(N_0; K_{d,u}^-, K_{d,u}^+)$
2	(59; 2.082, 6.061), (100; 1.508, 5.594), (200; 0.908, 5.167), (312; 0.578, 5.135), (1000; -0.009, 5.135)
3	(12; 6.379, 7.183), (15; 5.094, 5.907), (20; 3.923, 4.766), (30; 2.847, 3.741), (50; 2.047, 2.989), (100; 1.675, 2.495), (200; 1.398, 2.309), (312; 1.243, 2.234), (1000; 1.036, 2.142)

In particular, for every fixed $q \in (0.99, 1)$ the window holds for all $N \geq N_1(q)$ (here $N_1(q) \geq 312 > N_*$, and $N_1(q) \approx \pi q / (1 - q)$ as $q \rightarrow 1$).

Proof. Lemma 7.12 and Theorem 7.13 remain true when n_q, ω_q, ω'_q and $\frac{q\pi_a}{1-q}$ are replaced by the larger quantities $\bar{n}, \bar{\omega}, \bar{\omega}'$, $\frac{\pi_a}{\sin 2x}$ of Lemma 7.16: their proofs use only upper bounds for these quantities and, in Lemma 7.12(A), the fact that $\bar{n}/E_1$, like n_q/E_1 , is a combination with nonnegative coefficients of E_2/E_1 and E_{ε_2}/E_1 , both nonincreasing. Fix a block $B = \{N_a, \dots, N_b\}$ of Section

5.3 with $N_a \geq 10$. For $N \in B$ and q with (H): $\mu \leq \frac{2}{d}\xi y$ (Lemma 7.16(a)); $\pi_a X_s/D_a(x) \leq N_a^{-d} \bar{X}^{(d)}(N_b)/D_a(\frac{\pi}{2N_b})$ and $\pi_a X_s/\sin 2x \leq N_a^{-d} \bar{X}^{(d)}(N_b)/\sin \frac{\pi}{N_b}$ if $N_b < \infty$ (Lemma 7.16(c), (E2)); and, if $N_b = \infty$, $\pi_a X_s/D_a(x) = N^{1-d} X_s/(ND_a(x)) \leq N_a^{1-d} M(N_a)/(N_a D_a(\frac{\pi}{2N_a}))$ with $M(N) = 2\pi \ln N + 1.3$ ($d = 2$), $c_3 N$ ($d = 3$), because $X_s \leq M(N)$, $N^{1-d} M(N)$ decreases and $ND_a(\frac{\pi}{2N})$ increases (Corollary 4.7, Lemma 7.16(c)); likewise with $\sin 2x$ in place of D_a . Here $\pi_a/y = \pi_a X_s$. The script `scripts/78_cert_allq.py` (function `quni_bounds` in `scripts/certlib.py`) verifies, in ball arithmetic and for each block, the hypotheses (A)–(D) of Theorem 7.13 exactly as in the proof of Theorem 7.14, with $q_{\max}$ replaced by 1 in $\mu \leq \frac{2q_{\max}}{d}\xi y$ and with $\bar{n}, \bar{\omega}, \bar{\omega}', \frac{\pi_a}{\sin 2x}$ in place of the q -dependent constants; it also verifies $c_+ + c_- > 0$ and $\tau_{\text{cf}} + c_+ y' \geq 1$. All checks succeed for every block with $N_a \geq N_*$ (files `data/78_cert_allq_d2.jsonl`, `data/78_cert_allq_d3.jsonl`; 171 resp. 387 blocks). Theorem 7.13 gives $t_- + 2 \leq t^* \leq t_+ + 1$ for every N in such a block and every q with (H). The table lists, for each N_0 , the maximum of $c_{\pm}$ over the blocks that meet $[N_0, \infty)$. $\square$

Corollary 7.18 (asymptotic laws for the discrete walk, every $q < 1$). *Let $q \in (0, 1)$ and $N \geq \max(N_*, N_1(q))$. Then every maximiser t^* of the first-passage probability mass function satisfies*

$$d = 2: \quad \frac{4N^2}{\pi^2 q} \left[\ln \ln N + \ln 8\pi - \frac{2.33 + 3 \ln(8\pi \ln N + 2.4)}{2\pi \ln N - 0.24} \right] < t^* < \frac{4N^2}{\pi^2 q} \left[\ln \ln N + \ln 8\pi \right] \left(1 + \frac{0.83}{N^2} \right) + 2,$$

$$d = 3: \quad \frac{6N^2}{\pi^2 q} \left[\ln N + \ln(6c_3) - \frac{14.01 + 5 \ln(6c_3 N)}{c_3 N - 7.63} \right] < t^* < \frac{6N^2}{\pi^2 q} \left[\ln N + \ln(6c_3) \right] \left(1 + \frac{0.83}{N^2} \right) + 2,$$

and for $N \geq \max(100, N_1(q))$ the same holds with the numerators 1.75 ($d = 2$) and 9.31 ($d = 3$) in place of 2.33 and 14.01.

Proof. Repeat the proofs of Theorems 6.1 and 6.2 (which use Corollary 4.7, valid for $N \geq 10$, and, for $d = 2$, $X \leq 1.032L$, valid for $N \geq 20$) with the constants of Theorem 7.17 at $N_0 = N_*$ resp. $N_0 = 100$: (2.082, 6.061) and (1.508, 5.594) for $d = 2$, (6.379, 7.183) and (1.675, 2.495) for $d = 3$. Upper bounds: $0.62 + 6.061 < 12.39 \leq 3 \ln(4X) \frac{X}{X+3}$ for $N \geq 59$ and $0.62 + 5.594 < 12.88$ for $N \geq 100$; $7.183 < 28.8 \leq 5 \ln(6X) \frac{X}{X+5}$ for $N \geq 12$ (the right-hand sides increase in X). Lower bounds: $0.24 + 2.082 \leq 2.33$, $0.24 + 1.508 \leq 1.75$, $7.63 + 6.379 \leq 14.01$, $7.63 + 1.675 \leq 9.31$. $\square$

7.7 What remains open in discrete time

Remark 7.19. (i) *Uncovered parameters.* For $q \leq \frac{1}{2}$ the window is proved for $N \geq 12$ ($d = 2$; the certificate also covers $N = 11$, with the weaker constants $(K^-, K^+) = (10.476, 11.191)$) and $N \geq 10$ ($d = 3$) (Theorem 7.8). For $\frac{1}{2} < q < 1$ it is proved for $N \geq \max(N_*, N_1(q))$ (Theorem 7.17) and, for $q \leq q_{\max} \in \{0.8, 0.9\}$, also for $N \geq 37$ resp. 46 when $d = 2$ (Theorem 7.14). Not covered are: all $N < N_1(q)$, where (H) fails (for q close to 1 this is the range $N \lesssim \pi q/(1 - q)$); for $d = 2$ and $\frac{1}{2} < q < 1$ the N with $N_1(q) \leq N \leq 58$ that are not covered by Theorem 7.14; for $d = 3$ and $\frac{1}{2} < q < 1$ the values $N \in \{10, 11\}$ with $N \geq N_1(q)$; and $N < 10$. In the uniform certificate the blocks below N_* fail only in the value inequality (D) for $36 \leq N \leq 58$ ($d = 2$) and for $N \in \{10, 11\}$ ($d = 3$); for $d = 2$ they fail also in (B) when $32 \leq N \leq 35$, in (B) and (C) when $12 \leq N \leq 31$, and already in the lower sign condition when $N \in \{10, 11\}$. (ii) *The simple walk.* For $q = 1$, condition (H) fails for every N , and the mass function is not unimodal in general: for $d = 2$, $N = 9$, exact integer arithmetic gives $\mathbb{P}(T = 108) = 0.0020122644 \dots > \mathbb{P}(T = 109) = 0.0020121967 \dots < \mathbb{P}(T = 110) = 0.0020122182 \dots$ (two local maxima two steps apart; the same pattern occurs for $N = 11$, but not for $N \in \{8, 10\}$). The present method gives nothing for $q = 1$. We conjecture that the global maximiser still satisfies the laws of Corollary 7.18 (the observed values of $X(\mu t^* - \tau_{\text{cf}})$ for $q = 1$ lie in the ranges quoted in Numerical observation 7.20).

Numerical observation 7.20. (a) `scripts/77_validate_q_large.py`: all exact discrete modes of the computations of Sections 5–6 of the article with $q \in \{0.8, 0.9\}$ covered by the certificates lie in the windows of Theorem 7.14, and for $(d, N, q) \in \{(2, 37, 0.8), (2, 44, 0.8), (3, 13, 0.8), (3, 14, 0.8), (2, 40, 0.5)\}$

a dense eigen-decomposition confirms the four region statements of Theorem 7.13 with the certified constants. (b) `scripts/75_check_q_large.py` checks Lemmas 7.10–7.12 by brute force for nine cases with q between 0.5 and 0.9 (0 violations; the one-dimensional part in 40-digit arithmetic). (c) `scripts/74_observe_q_large.py`: for $q \in \{0.8, 0.9, 1\}$ the observed $X(\mu^* - \tau_{\text{cf}})$ ranges over $[0.24, 1.67]$ ($d = 2$, $12 \leq N \leq 401$) and $[-1.09, -0.35]$ ($d = 3$, $13 \leq N \leq 100$). (d) `scripts/79_validate_allq.py`: for ten cases (d, N, q) , eight of them with q the largest value allowed by (H) (rounded down to six digits; e.g. $(2, 320, 0.990278)$, $(2, 200, 0.984535)$, $(3, 40, 0.927248)$, $(3, 12, 0.794395)$, none of them covered by Theorem 7.14) and two with $q \leq \frac{1}{2}$, the maximiser of the mass function computed by direct iteration lies in the window of Theorem 7.17; a spectral tail bound (Cauchy interlacing for the killed transition matrix) shows that it is the global maximiser, up to floating-point rounding. The observed $X(\mu^* - \tau_{\text{cf}})$ lies in $[0.99, 1.61]$ ($d = 2$) and $[-0.93, -0.46]$ ($d = 3$).

8 What is not proved (exact remaining gaps)

1. **Discrete time, q close to 1.** Theorem 7.17 covers every $q \in (0, 1)$, but only for $N \geq \max(N_*, N_1(q))$, and $N_1(q) \approx \pi q / (1 - q) \rightarrow \infty$ as $q \rightarrow 1$: for $N < N_1(q)$ the pairing lemma (Lemma 7.10) and hypothesis (H) fail and nothing is proved. Between $N_1(q)$ and N_* part of the range is covered by Theorems 7.8 and 7.14; the rest is listed in Remark 7.19(i).
2. **The simple random walk ($q = 1$):** the PMF is not unimodal in general (parity; Remark 7.19(ii) gives an exact example), condition (H) fails for every N , and the present method does not apply; only Propositions 1.5 and 7.1 hold. That the laws of Corollary 7.18 hold for $q = 1$ is a conjecture.
3. **Sharpness of the constants.** $X(\tau^* - \tau_{\text{cf}})$ is enclosed in an interval a few units wide; the true value appears to converge to $K_2^\infty \approx 3.74$, $K_3^\infty \approx -0.22$ (Remark 5.3(iii)). Proving the limit would require two-sided bounds for S_ν , $\mathbb{E}U$ and $\int_0^\infty re^s ds$ with $o(1)$ errors; the present enclosures have $O(1)$ slack (T_1 : tail bound $\approx 10.4/K$ in $d = 3$; κ : $[\kappa_{\text{lo}}, \bar{\kappa}]$). Consequently Theorem 5.1 proves a relative error of the closed form of at most $K_d / (X \ln(2dX)) \cdot (1 + o(1))$, e.g. $\leq 10.4\%$ at $N = 20$ and $\leq 2.2\%$ at $N = 1000$ in $d = 2$, whereas the observed error is about 1%; in $d = 3$ the proven bound is 0.42% at $N = 20$ and 0.03% at $N = 100$ (observed: 0.016% and 0.0015%).
4. **Small N .** Theorem 5.1 starts at $N_0 = 12$ ($d = 2$; at $N = 10, 11$ the certificate gives the weaker constants $(12.89, 12.14)$, $(9.03, 11.07)$) and $N_0 = 10$ ($d = 3$); smaller N are finite computations that are not done here.
5. **Other geometries** (corner–centre etc.) are not treated; the decomposition $T \stackrel{d}{=} U + T_\pi$ holds for any pair, but u' is log-concave and the alternating bounds of Lemma 1.8 hold only when the one-dimensional factors are end-to-end passages.
6. **Cited results.** Theorem 7.4 (hence Theorem 7.8, but not Theorems 7.13, 7.14 and 7.17, which cover $q \leq \frac{1}{2}$ for $N \geq 37$ ($d = 2$) resp. $N \geq 12$ ($d = 3$)) uses the closure of ultra log-concave sequences under convolution (Walkup 1976, Theorem 1; Liggett 1997, Theorem 2; short proof: Marsiglietti–Melbourne 2024, Corollary 3.4 of arXiv:2004.12005v4). The theorem numbers for the two older papers are taken from Johnson (2007), who cites them for this statement; those two papers were not consulted directly (Remark 7.5(ii)). Everything else is self-contained except the Laplace-transform uniqueness theorem (Feller) and the Bessel-function identity used to evaluate κ numerically (DLMF 10.32.3).
7. **Computer assistance.** Propositions 4.5, 4.6, Theorem 5.1, Theorems 6.1–6.3, Theorems 7.8, 7.14, 7.17 and Corollaries 7.9, 7.15, 7.18 depend on ball-arithmetic computations (Arb via python-flint 0.9.0): the constants of `50_cert_constants.py` and the block certificates

of 52_cert_theoremB.py, 72_cert_discrete.py, 76_cert_q_large.py, 78_cert_allq.py (all re-run with scripts/run_all_certs.sh). The trusted base is Arb, python-flint and the transcription of the formulas into certlib.py; the transcription is cross-checked by 51_validate_enclosures.py and 53_check_decomposition.py, which compare every enclosure with separately computed floating-point values, and the windows are compared with brute-force modes in 54, 73, 77 and 79. The certificate files of 52, 72 and 76 were also re-checked block by block with separately written code using mpmath interval arithmetic (with the T_1 enclosures recomputed in floating point), within the same project and on the same machine (code not released); this re-check is not part of the proof, and it does not cover 78.

A Files

In the repository, scripts/ is code/msc_rigorous_asymptotics/ and data/ is data/msc_rigorous_asymptotics/; the run logs cited in the % src comments are in data/msc_rigorous_asymptotics/logs/, other logs and working notes are not released.

file	role
scripts/certlib.py	ball-arithmetic library: X_s bounds (Props 4.5, 4.6), T_1 enclosures (Lemma 4.8), pole enclosures (Section 3), sign function (Lemma 5.2, Thm 7.7)
scripts/50_cert_constants.py → data/50_cert_constants.json	certified constants $Z_2, Z_3, c_{\square}, J, c_3, c_{\kappa}, T_d(1)$
scripts/52_cert_theoremB.py → data/52_cert_theoremB_d{2,3}.jsonl	certificate for Theorem 5.1 (220 + 389 blocks)
scripts/72_cert_discrete.py → data/72_cert_discrete_d{2,3}.jsonl	certificate for Theorem 7.8
scripts/76_cert_q_large.py → data/76_cert_q{0p8,0p9,0p95,0p99}_d{2,3}.jsonl	certificate for Theorem 7.14 (four regions, no unimodality)
scripts/78_cert_allq.py → data/78_cert_allq_d{2,3}.jsonl	certificate for Theorem 7.17 (four regions, constants uniform in $q < 1$, Lemma 7.16)
scripts/80_tables.py → data/80_tables.json, logs/80_tables.log	tables of certified constants (maximum over all blocks meeting $[N_0, \infty)$)
(commands in reproduce.md)	serial re-run of all block certificates and of the tables
scripts/83_compare_cert_runs.py → logs/83_compare_*.log	block-by-block comparison of certificate re-runs (0 differences)
scripts/r4lib.py	float library for sanity checks
scripts/40_check_thmA.py, 41_windows_exact.py	checks of Sections 1–2 (dense eigen-decomposition; exact modes)
scripts/51_validate_enclosures.py, 53_check_decomposition.py, 54_bruteforce_theorem51.py	checks of Sections 3–5 (and of Theorem 7.8 for small N)
scripts/60_check_asymptotics.py	checks of Section 6 against exact modes up to $N = 1.7 \cdot 10^7$ ($d = 2$), 4096 ($d = 3$)
scripts/70_check_discrete.py, 71_check_discrete_unimodal.py, 73_validate_discrete.py, 74_observe_q_large.py, 75_check_q_large.py, 77_validate_q_large.py, 79_validate_allq.py, 82_q1_parity_example.py	checks of Section 7
scripts/81_check_text_constants.py → logs/81_check_text_constants.log	interval checks of the numerical claims in the text (Corollaries 4.7, 7.15, 7.18, display (6.1), Lemma 7.16)
data/	outputs of all scripts